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Magnetically induced force and torque on temporary epicardial leads

Experimental Device Development and Clinical testing

by

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A Thesis

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Abstract

Patients with retained epicardial pacing leads following cardiac surgery are restricted MRI access due to risks of static-field translational forces and torques, despite frequent diagnostic need for MRI in this patient population. This thesis developed and clinically tested three custom non-magnetic fixtures to quantify these interactions at micro-Newton scales: a PVC platform for magnetic field characterization, a pendulum for displacement force, and a strain-gauge sensor and fixture for torque.

Each fixture was validated in a controlled laboratory environment using an air-core solenoid and Helmholtz/neodymium fields (≤180 G), then tested on a 3 T Siemens Magnetom™ scanner at CHI Health CUMC–Bergan Mercy. Laboratory field mapping confirmed the expected field spatial decay, establishing the $B\nabla B$ product needed for force predictions. Trial pendulum measurements of a ferromagnetic object yielded $\chi_m/\rho = (2.01 \pm 0.05) \times 10^{-3}$ m^3/kg ($\tan\alpha = 0.002$–0.012), validating the linear model before clinical deployment. Torque sensor calibration produced consistent linear voltage-to-force slopes across all orientations ($\tau_{\min} = 1.24 \times 10^{-5} N \cdot m$), and laboratory measurements confirmed $\tau \propto B^2$ dependence ($k = 379 \pm 9$ N/m, 2.3%).

Clinical field mapping yielded $|B\nabla B|$ between 12.68 and 22.95 T^2/m at the bore entrance. Translational-force ratios (FR $= \tan\alpha$) remained below the ASTM F2052-15 threshold of 1.0 for all tested materials. The Medtronic STREAMLINE™ 6492 unipolar epicardial lead reached the highest deflection ($\alpha = 9.5°$, FR = 0.166), substantially higher than titanium references (FR = 0.006) yet well within the safety criterion. Extrapolation yielded allowable gradients of 107 T/m at 1.5 T and 54 T/m at 3.0 T for the unipolar lead, both far exceeding the scanner's physical maximum ($\leq$ 19 T/m). Clinical torque runs at $B \approx 1.5$ T produced no measurable deflection, requiring further investigation.

The results indicate that micro-Newton-scale magnetic-field mechanical effects on the tested epicardial lead can be measured under 3 T conditions. Though the findings are limited to a single configuration and do not support general safety conclusions, it established a methodology that can be extended to evaluate a broader range of lead designs and conditions.

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Chapter 1

Introduction

Medical diagnostics heavily rely on imaging tools such as Magnetic Resonance Imaging (MRI), a powerful and evolving application capable of showing high contrast sensitivity to soft tissue while not depending on ionizing radiation. Instead, MRI uses strong magnetic fields and Radiofrequency (RF) waves to excite hydrogen protons in the body, then detects the signals these excited protons emit as they return to their original state to create detailed images of internal tissues [1].

The magnetic nature of MRI makes it a hazardous environment where surgical equipment is limited. Conductive temporary epicardial pacing leads are frequently placed as part of Cardiac Implantable Electronic Device (CIED) systems to regulate cardiac rhythm during open-heart surgery [2]. Furthermore, a recognized practice is to cut the epicardial lead wire (if significant resistance is encountered), intentionally leaving a small segment inside the patient to prevent immediate complications [3, 4, 5]. These cases introduce long-term repercussions, most notably the loss of access to MRI scanners.

In the United States alone, more than 300,000 individuals receive new CIED implants every year, as the Heart Rhythm Society (HRS) reported in 2024 [6], and there is a growing interest in the interactions between the MRI environment and various metallic or conductive objects and medical devices. Key standards from the ASTM International F04 committee address these concerns through standardized test methods for evaluating safety parameters such as magnetically induced displacement force (ASTM F2052 [7]), torque (ASTM F2213 [8]), and radiofrequency (RF)-induced heating (ASTM F2182 [9]), as well as uniform labeling practices for MRI safety (ASTM F2503 [10]).

According to ASTM F2503, a device labeled as MRI safe is expected to be non-conductive, non-electrical, non-magnetic, and to present no known hazards in the MRI environment [10]. However, this classification does not apply to all epicardial leads, as they are rarely physically characterized after removal, nor properly labeled by vendors. By the time they are cut, these leads have fulfilled their clinical purpose, and systematic safety evaluations are seldom performed.

For instance, conductive items such as Electrocardigram (ECG) leads, even if nonfunctional and relatively short, are generally considered unsafe or require extreme caution in the MR environment [11, 12, 13]. Main concerns include magnetically induced translational forces and torques, RF heating, or inadvertent cardiac stimulation [14, 15, 16]. First, the strong magnetic field and its spatial gradient can exert translational forces and torques on the lead, resulting in mechanical displacement along the direction of the magnetic field. Second, the lead can absorb energy from the MRI's (RF) field, potentially causing localized heating that poses a thermal risk to the patient. Third, the RF field may induce alternating currents within the lead, which can electrically stimulate nearby cardiac tissue, potentially leading to dangerous physiological effects [17]. The risk of injury effectively prevents patients with CIEDs or cardiac surgery history, who are most likely to need a MRI scan during their lifetime, to not be able to enter MRI room [11, 18].

There is still a dearth of focused studies evaluating the precise dangers associated with retained epicardial leads, despite the frequent use of such leads and their repercussions [11, 19, 20]. Existing clinical recommendations tend to be too cautious and frequently disqualify these individuals from MRI eligibility even when there is little evidence to support it. Recent studies regarding this problem have shown both no image quality degradation and, most importantly, no serious safety events such as, device malfunction, myocardial injury or arrhythmias, associated with MRI examinations in patients with CIEDs, such as abandoned epicardial leads [19, 21]. While these studies have evaluated, to some extent, the resulting consequences of RF-induced heating, mechanical forces and conductivity, additional study is needed.

Specifically, focused quantitative investigations are required to address key unanswered questions regarding the physical forces experienced by lightweight devices such as retained epicardial pacing leads in the MR environment. While substantial research has quantified the projectile hazards posed by large, strongly ferromagnetic hospital objects in the MR environment (e.g., oxygen tanks, scissors, or monitors), far less attention has been given to the much smaller forces experienced by thin, low-mass leads with weak magnetic susceptibility, such as retained epicardial leads [18, 22, 23]. The importance of accurately assessing such low-magnitude forces is underscored by limitations in standard test methods [7, 8], which often require adaptations for ultra-lightweight or needle-like devices where conventional measurements struggle to capture micronewton-level effects [24]. To properly situate this work, the following section reviews pertinent literature on MRI risks, safety principles, and previous studies to establish current state of the art regarding epicardial leads on MRI environment.

1.1 Literature Review

From a safety standpoint, the static magnetic field (B_0) represents the main hazard and must be contained properly. A vast majority of diagnostic clinical scans are completed daily without incidents indicating that magnetic interactions with normal tissues are within the bounds of safety up to the highest fields now in use for MRI [17]. However, there have been well-documented incidents associated with ferromagnetic objects either implanted in the patient or inappropriately brought into the scanner environment (Zone IV) [23, 25]. Recent cases continue to underscore that such accidents most commonly result from failures in pre-entry screening and access control protocols, rather than from the technology itself [26].

MRI safety protocols are grounded in a system of absolute and relative contraindications. Absolute contraindications include non–MR-conditional cardiac implantable electronic devices (CIEDs), metallic intraocular and foreign bodies, among others. Relative contraindications consider implants such as stents, surgical clips, and programmable shunts that may be scanned under controlled conditions [27]. The safety classification of each device is guided by testing standards and certification data, with CIEDs representing a critical category due to their susceptibility to electromagnetic interactions [17, 28].

The American College of Radiology (ACR) Manual on MR Safety (2024) [29] outlines the current regulatory and procedural framework for MRI safety. Recommendations include structured zoned access control and interprofessional collaboration through clearly defined positions (MR Medical Director, MR Safety Officer, and MR Safety Expert) to prevent unscreened individuals or ferromagnetic objects from inadvertently entering the magnet room. Pre-scan screening, implant model and status verification, and continuous patient monitoring are still necessary for this safety infrastructure [27].

In addition to current safety guidelines, studies have centered on qualitative findings. For example, a large observational study by Schaller *et al.* (2021) evaluated 200 MRI examinations in 139 patients with a total of 243 abandoned leads [30]. Their objective was to assess the safety of 1.5-T MRI in patients with at least one abandoned CIED lead and to evaluate any potentially harmful effects on the active CIED leads.

Before the scans, safety was a priority. Each case was discussed by radiologists and the ordering healthcare professionals in order to balance the advantages of alternative tests against the dangers associated with MRI. Next, baseline battery voltage, sensing, pacing thresholds, and lead impedances were recorded. During the MRI, patients were monitored continuously via electrocardiography telemetry and pulse oximetry and the scans were performed in a normal specific absorption rate (SAR) mode, ensuring they did not exceed 2.0 Watts per kg for the whole body. In some cases involving the brain, a transmit-receive radiofrequency coil was used to avoid the central location of the leads relative to the coil.

Immediately after the MRI, the CIED evaluation was repeated to check for changes in capture threshold, sensing, and lead impedance. The research found a low rate of adverse events (3%), with no serious complications such as sustained arrhythmias or device resets. The observed events were primarily transient, non-life-threatening changes in lead sensing parameters that resolved spontaneously. Notably, this safety profile held true even for thoracic and cardiac MRI scans, where the theoretical risk is highest.

Complementing this, a retrospective study by Vuorinen *et al.* (2021) focused on 26 MRI scans in 17 patients with epicardial leads [21]. With a similar objective, on a 1.5-T MRI, researchers distinguished between "old" epicardial leads (implanted before 2000) and "modern" leads (implanted in 2000 or later) to compare safety outcomes. They followed a dedicated institutional protocol, previously implemented, which included measuring the length of the leads using chest x-rays prior to MRI scan to determine if the lead reached lengths of 24 cm, which might increase heating risks. Similarly, researchers documented CIED generator and lead models and parameters with implantation dates.

The study's key finding was that MRIs performed on patients with "modern" epicardial leads had no detectable adverse events. However, one definite and one possible adverse event were noted in patients with "old" leads, suggesting that lead age and technology may be a factor in the risk profile.

These observations rely on clinical monitoring and empirical outcomes rather than direct quantification of physical interactions with the static magnetic field. To bridge this gap and enable more precise risk evaluation, standardized quantitative protocols are essential for measuring the magnetically induced forces on such objects.

1.1.1 Quantification of Magnetically Induced Displacement Force and Torque

ASTM F2052-21 offers a standardized technique for assessing magnetically induced displacement force on medical equipment in order to set safety standards for such situations. The methodology includes a non-magnetic pendulum apparatus with a protractor that allows for the suspended object to be pulled by the magnetic field. The characterization of the static magnetic field and its spatial gradient is required. For this purpose, Ferreira *et al.* (2017), developed a custom-designed fixture with an AlphaLab Vector/Magnitude Gaussmeter (Model VGM) probe to map the spatial gradients inside a Siemens Magnetom Trio ™3T MRI scanner [31]. Measurements of the magnetic field vector components (X, Y, and Z) were collected along the Z-axis in 5 cm increments, correctly capturing the static magnetic field and ensuring repeatability and showing a steady spatial gradient.

To quantify the magnetically induced displacement force, Woods *et al.* (2021), conducted a similar experiment following ASTM F2052-21 on eleven metal alloys commonly used in medical implants [32]. For deflection force testing, nine specimens (in three nominal mass categories: 2, 20, and 200 g) of each alloy were tested in a

Siemens MAGNETOM Trio™ 3T MR system with the objective to establish evidence for MR conditional device labeling.

The main results regarding magnetically induced deflection force showed that for seven of the eleven tested alloys, the magnetic force was found to be less than the device weight in any current clinical 1.5 T, 3.0 T, and 7.0 T MR system. These safe alloys included various grades of CP titanium, Ti-6Al-7Nb, Ti-6Al-4V ELI, Ti-15Mo, Biodur 108, and Co-28Cr-6M. However, for the other alloys, including 316L stainless steel and 22Cr-13Ni-5Mn, the magnetically induced force was calculated to be greater than the device weight in certain high-field systems, such as 3.0 T and 7.0 T clinical scanners.

Following a series of torque measurements, Woods *et al.* (2021) reported that for medical instruments made from their list of alloys the magnetically induced torque will always be less than the worst case torque produced by gravity in a 3.0 T static magnetic field [32]. Notably, no measurable movement was observed in any of the torque specimens tested at 3.0 T. However, the ASTM conservative safety criterion requires that magnetically induced torque do not exceed its gravitational equivalent, $\tau < Lmg$. Using this criterion, the authors established an upper bound on torque by measuring the coefficient of static friction between specimens and the test surface, finding that magnetic torque remained below roughly 30% of the gravitational torque across all tested alloys. This result suggests that under typical clinical MRI circumstances, torque is not a limiting safety concern for implant-grade versions of these materials. This finding demonstrates that torque effects are secondary to displacement forces. Consequently, device designers can prioritize addressing displacement force effects while being confident that torque is generally well-controlled under typical clinical field strengths.

However, the study was probably done using ASTM F2213-17 “The Low Friction Surface Method” which estimates magnetically induced torque by placing the device on a low-friction, non-metallic, non-conductive surface near the MRI isocenter. The device is rotated step-wise, and any spontaneous alignment or rotation toward the static magnetic field is observed. If no movement occurs, the maximum possible torque is estimated from the device's weight and the surface's friction coefficient outside the MR environment. ASTM F2213-17 states that, if rotation is observed, more precise tests should be used instead which means this method relies on macroscopic motion and may not be able to detect mN forces.

In this regard ASTM F2213-17 define other methodologies, mainly the torsional spring or pulley method. A study by Seo *et al.* (2021), utilized "The pulley method" for the safety assessment of magnetically induced torques on Active Implantable Medical Device (AIMD) in 1.5 T and 3.0 T magnetic resonance imaging (MRI) environments [33]. They constructed a cylindrical shaped apparatus entirely from acrylic, including moving parts like the bearings, to minimize magnetic field resistance.

This methodology determines the maximum magnetically induced torque by fixing the AIMD onto a rotating plate placed at the MRI scanner isocenter. A thread attached to the plate is connected to a force gauge located far enough to avoid interference with the magnetic field. As the plate rotates to align with field, the thread tension is measured.

Prior to MRI scanner deployment, the apparatus was validated using calibrated masses applied at known radii to generate controlled torques. Across a range of applied loads, the measured torque agreed with theoretical values to within an average error of approximately 1.5%. The preliminary experiment was configured to replace the magnetic field's effect with known gravitational forces (weights). Weights ranging from 100–500 g were placed on the rotating plate at the point where the AIMD would be located, which was 2.9 cm from the center pillar. A force gauge, connected via a thread, was placed 4.9 cm from the center and used to pull the plate in the opposite direction of the weights. The force gauge used a load cell with a resolution of 0.01 N.

Each test was performed ten times for each weight (100 g, 200 g, 300 g, 400 g, and 500 g), and measurement was performed only when the equipment was stable.

Additionally, Seo *et al.* (2021) addressed the friction issue not by eliminating it entirely, but by quantifying and compensating for it mathematically within their final torque calculation [33]. The friction of the rotating plate (F_f) was determined in a prior experiment. Then, the friction value was explicitly used in the formula for magnetically induced torque (τ) calculations, which follows the principles of their pulley method. They explained that the torque (τ) is obtained by multiplying the distance from the center of rotation (r) to the AIMD's position by the difference between the measured force (F_m) and the friction of the rotating plate (F_f), $\tau = r(F_m - F_f)$.

In the clinic, they tested two types of AIMDs. First, they tested a bone conduction implant, a device that captures sound via an external microphone and transmits it wirelessly through the skull to the cochlea using an internal magnet [33]. Second, they tested nonmagnetic devices, such as pacemakers and Implantable Cardioverter-Defibrillator (ICD) which are generally not significantly affected by magnetic fields because their enclosures are typically made of MRI-compatible metallic materials like titanium or zirconium.

The study showed that the bone conduction implant consistently exhibited the highest average torque (9.460 $\text{N} \cdot \text{cm}$ at 1.5 T and 14.618 $\text{N} \cdot \text{cm}$ at 3.0 T), while the pacemaker showed the lowest (0.255 $\text{N} \cdot \text{cm}$ at 1.5 T and 0.668 $\text{N} \cdot \text{cm}$ at 3.0 T). AIMD containing magnets, such as the cochlear and bone conduction implants, generally showed significantly higher torques than devices without magnets like ICD and pacemakers.

Similarly, Stoianovici *et al.* (2024) showcases a variation of the ASTM F2213 Torsional Spring measurement technique, referred as the 2-string method [24]. This method addresses the difficulty of measuring very low magnetically induced torques on lightweight, slender medical devices, such as needles. This approach utilizes two strings to suspend the device by its ends and creates a torsional spring mechanism that is frictionless.

The needle rotates due to magnetic torque, the torque is calculated analytically based on the measured rotation angle and compared to the gravitational reference torque of the object. The study showed that the method successfully measured torques between 10^{-4}N$\cdot$ m and 1.8×10^{-3} N$\cdot$m in the MRI environment. The maximum magnetically induced torque deflection angle remained below the 25° limit required by the ASTM F2213.

1.1.2 Quantification of Magnetically Induced Heating

Similarly, the ASTM F2182 guideline offers a standardized test procedure for detecting RF-induced heating on or near passive implants in tissue-mimicking phantoms in order to quantify the heating effects of MRI scans on AIMD. Similar methods were used in earlier research by Aboyewa *et al.* (2021) to examine the interaction of leads with the RF field, assessing the energy absorbed and the ensuing heat generation [34].

The experiment was performed on a 3T Siemens Tim Trio™ MRI scanner using a turbo spin echo pulse sequence. The pulse sequence was set to produce a whole-body Specific Absorption Rate (SAR) of approximately 2 W/kg and fluoroptic probes where used to measure the temperature rise. Several factors on induced heating were tested, including lead length (such as 6.33 cm, 12.65 cm, and 18.98 cm), lead type (bipolar vs. unipolar), orientation, and shape.

The study showed that lead length, type, and orientation significantly influence heating behavior while short leads ($<$ 13 cm) posed no significant thermal hazard during MRI at a specific absorption rate (SAR) of 2 W/kg, the maximum temperature rise was observed at the lead tips. For both lead types investigated, the heat induced increased with increasing lead length. Notably, bipolar leads showed slightly higher heating than unipolar leads and the straight lead configuration produced the highest maximum temperature rise [34].

1.1.3 Relevant Studies Significance Summary

The overall significance of these studies is in establishing reliable experimental and computational frameworks for quantifying MRI-induced mechanical effects, particularly where conventional ASTM approaches fall short. Stoianovici *et al.* (2024) solves the technical challenge of measuring very low magnetically induced torques on slender, lightweight medical devices, such as needles. Similarly, Seo *et al.* (2021) validates the pulley methodology for AIMD. The modeling of clinically relevant alloys done by Woods *et al.* (2021) shows experimental basis to rely on the listed alloys but requires further testing in terms of torque. While the modeling of the static magnetic field done by Ferreira *et al.* (2017) provided a practical way to accurately map the magnetic field and calculate its spacial gradient. Finally, measurements by Aboyewa *et al.* (2021) focused on the RF-heating rather than mechanical effects. They employed the same MRI system and testing framework we utilize here, reinforcing the feasibility of our experimental setup and situating our work within established practices.

1.2 Scope of Research

This study focused primarily on measuring magnetically induced translational displacement forces and torques acting on retained epicardial pacing leads near and inside the MRI scanner bore, and to map how those forces evolve with distance from the isocenter. This study investigated whether retained epicardial pacing leads are subject to torque or translational magnetic forces when interacting with the scanner's main field at clinically significant strength levels. We provided experimental evidence to address a gap in the literature.

The core scientific aim was the development and implementation of adapted methodologies capable of reliably measuring forces at the micro-Newton scale. First, in a controlled laboratory environment and subsequently in the clinical 3T MRI setting. The methodology was capable of testing micro-Newton forces with precision for clinical application clarifying whether the static magnetic field in MRI poses a real risk to patients with retained leads.

Additionally, radiologists frequently request guidance on challenging cases involving patients with retained cardiac leads during MRI safety evaluations. One key objective of this work was to create practical educational resources that help clinicians confidently address these difficult clinical scenarios. Specifically, we aimed to develop a straightforward set of decision-support tools and teaching materials that enable radiologists, particularly radiology residents, to systematically evaluate the risks of static magnetic field–induced translational forces and torques on retained leads, improving safety decision-making and resident training in real-world practice.

To achieve these aims, the work proceeded in two main phases. In the laboratory phase, the work employed a Polyvinyl Chloride (PVC) pendulum-based measurement setup following ASTM F2052 for the deflection angle and a custom-made solenoid simulating a magnetic field similar to that found on a MRI scanner bore entrance at scale. Angular deflection of a ferromagnetic test object was recorded across a range of field strengths to extract the mass magnetic susceptibility χ_m/ρ and validate the linear force model prior to clinical deployment.

A second setup was specifically developed to quantify magnetically induced torque, which is particularly challenging for slender, lightweight structures where conventional methods may suffer from friction, low sensitivity, or inadequate resolution for micro-scale effects. While our torque apparatus does not strictly adhere to any single ASTM F2213 method, it draws conceptual similarities from established low-friction techniques. For example, to further reduce potential errors from string-based systems, our design elevates the test sample to be positioned at a isocenter using non-magnetic supports for near-frictionless suspension. Torque is opposed and measured via a sideways lever arm and a thin, non-magnetic rod attached to the fixture, which bends slightly under the counter-force from induced rotation. This deflection induces strain, detected by bonded strain gauges integrated into a custom circuit for direct, quantitative readout of torque magnitude under controlled orientations.

These methods allowed for high sensitivity in detecting small magnetic forces in a controlled environment, without the complications of patient variability or direct MRI scanner use. Measurements were interpreted in the context of magnetic susceptibility theory and compared to their gravitational equivalents, which is a common normalization approach in MRI safety literature to assess clinical significance [7, 35].

Building on this foundation, a limited clinical phase in a 3T MRI environment followed. Given the challenges associated with clinical access for experimental testing, such as MRI scanner weekly availability and short measurement periods, the clinical experiments were therefore designed with two complementary goals. First, to verify device performance under real conditions. And second, to identify workflow bottlenecks, operator errors, or opportunities for optimization in the measurement chain.

The techniques developed in this document will help with more thorough evaluations of MRI safety in patients with retained temporary epicardial leads and set the stage for a future clinical adaptation of the protocol employing a 3T MRI scanner.

1.3 Summary

Epicardial leads have the problem of being broadly labeled MRI-unsafe under ASTM F2503. This classification originates from theoretical concerns over static magnetic field interactions (translational force and torque), RF-induced heating, and inadvertent electrical stimulation, yet it restricts MRI access for cardiac patients most in need of diagnostic imaging.

Hundreds of MRI tests using abandoned leads, including contemporary epicardial types, have shown no significant adverse events, indicating good safety profiles in large observational studies. There is still a significant gap in the accurate, micro-Newton-scale characterization of mechanical forces on thin, lightweight epicardial leads, despite the fact that standardized testing (ASTM F2052, F2213 [7],[8]) and earlier research on implant alloys confirm low magnetic interactions in many materials. The methodology on this project follows that of the literature reviewed in this chapter. Featuring a static field mapping fixture, a pendulum for deflection angle determination, and a newly developed torque movement device that resembles a suspension method but uses a low friction bearing to enable rotation and a highly sensitive sensor to measure mN forces.

Building on this foundation, Chapter 2 will provide essential background on the underlying physics of MRI scanners. It includes the generation and spatial distribution of static field, along with a broad introduction to gradient, and RF fields. Alongside physics, a description of epicardial lead design, material composition, and implantation configurations is provided. Key concepts from electromagnetism (magnetic susceptibility, force, and torque on conductive elements) and mechanics (deflection, rotation, and tissue anchoring) will form the thematic backbone, enabling a rigorous analysis of lead-MRI interactions and informing the interpretation of experimental results in subsequent chapters.

Chapter 2

MRI Fundamentals and Safety Concerns regarding Epicardial Leads

This chapter provides a review of the most important topics to fully understand this work. It includes an introduction to MRI scanners and the physics behind them, including the possible physical interactions of the system with matter, such as epicardial leads. For example, active implantable medical devices (AIMDs), such as epicardial leads, are particularly prone to safety risks, as even trace amounts of ferromagnetic materials in the device components can experience mechanical forces and torques within the scanner environment. To understand these risks we focus our attention on the static magnetic field that the MRI scanner continuously produces, and the effect this field has on the specialized alloys used in medical devices. A brief discussion on lead design is included followed by implications for the clinic and the lab setting. Most of this chapter content is based on physics and imaging from well known textbooks, including Griffith's Introduction to Electrodynamics [36], and Bushberg's Physics of Medical Imaging [1].

2.1 Magnetic Resonance Imaging

MRI or Magnetic Resonance Imaging is a medical imaging modality that uses magnetic fields and radiofrequency (RF) energy, rather than ionizing radiation like X-rays, to produce images. These images are made by processing the signals produced when the net magnetization vector returns to equilibrium following an RF excitation pulse. In fact, MRI has exceptional utility in clinical settings, particularly for neurological and musculoskeletal imaging, because it produces images with high sensitivity to anatomical variations and superior soft tissue contrast [1].

This contrast comes from the proton density and how freely their water molecules move within different tissue types such as fat, gray matter, white matter, and cerebral spinal fluid. When placed in a strong static magnetic field, these hydrogen-rich tissues develop a net equilibrium magnetization aligned with the field. An RF pulse temporarily disturbs this equilibrium, and each tissue returns to it with characteristic times of relaxation that define image contrast. Importantly, generating and sustaining this equilibrium magnetization requires an exceptionally strong magnetic field.

To fully characterize the safety considerations relevant to epicardial leads, we must first understand the components of the system. An MRI scanner has by five main components: the main magnet, the gradient coils, the RF coils, the control electronics, and the control console [37]. A real example of this system can be found in Chapter 3.

Figure 2.1 shows an exploded view of three cylindrical components arranged concentrically around the patient couch at the center. These cylinders represent the primary magnetic field–generating structures, which must be isolated from the external environment. The outermost and largest cylinder corresponds to the main magnet, which houses superconducting windings capable of generating high magnetic field strength. The magnetic field strength, also known as magnetic flux density (B), is measured in Tesla (T). An alternative unit is the Gauss (G), where 1 T = 10,000

G. For perspective, a clinical MRI scanner has a range from 0.5 T to 3 T, and up to 9 T in some specialized research systems, which is orders of magnitude stronger than Earth's magnetic field (0.5 G)[1, 37, 38].

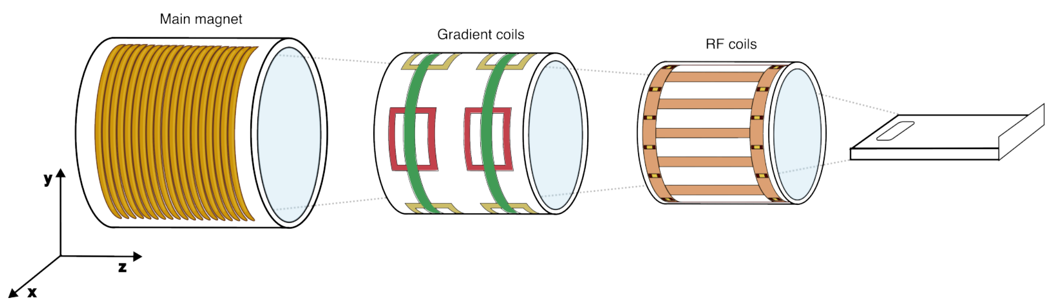


Figure 2.1: Exploded view of an MRI scanner: the main magnet, gradient coils, RF system and patient couch.

The magnetic field strength required for clinical imaging is set up this way to interact with materials normally considered non-magnetic in everyday context, like human tissue. The image formation process exploits the microscopic magnetic behavior of atomic nuclei within this field. Hydrogen (specifically its proton, ^1H) is the most utilized element in clinical MRI because it has the largest magnetic moment and is highly abundant in biological tissues, primarily in water and fat, which constitute the majority of soft tissue composition.

Normally, these hydrogen atoms have a random orientation of their nuclear magnetic moments (protons) due to thermal energy. When placed in a strong static magnetic field (B_0), these nuclear magnetic moments weakly align in parallel (same direction of B_0, low-energy state) and antiparallel (opposite direction of B_0, high-energy state) directions.

In addition to aligning with B_0, the protons also experience precession, much like the Earth's rotation around its tilted axis or a spinning top under gravity. This precession occurs at an angular frequency (ω_0) known as the Larmor frequency, which is directly proportional to the magnetic field strength (B_0), as described by the Larmor equation $\omega_0 = \gamma B_0$. This frequency also corresponds to the separation between the proton's magnetic energy levels ($\hbar\omega_0$), and it is the resonance condition in which an RF pulse can be applied to make the transition between these states.

In Figure 2.1, the innermost cylinder represents a birdcage resonator or body coil used to transmit RF into the patient. Once this RF pulse (B_1) is applied, synchronized to the Larmor frequency of protons (about 42.58 MHz per Tesla), the protons excite and the net magnetization (M) tips away from the longitudinal direction (z) becoming anti-aligned. In doing so, it causes the magnetic moments of many protons to become synchronized, or phase-coherent. They start to precess together in the transverse plane, creating transverse magnetization (M_{xy}).

The instant the RF pulse is turned off, the relaxation process starts, these precessing protons start to lose that synchronization due to slight variations in their local magnetic environments. This causes the transverse magnetization signal to gradually fade over time. The decaying oscillations of this signal are detected by the receiver coil, which potentially can be the same that sent them. The time at which the plane (M_{xy}) decays is described as transverse relaxation (T2). Independently, the process in which the nuclear spins return to alignment along B_0 occurs with characteristic time T1 (longitudinal relaxation). The final MRI image's gray-scale depiction of a tissue is determined by the RF signal produced during these relaxation processes taken together.

However, this covers a single signal from a particular coordinate. Spatial localization of the signal requires knowing the origin of each measurement across all three axes. This is achieved using small, precisely controlled variations in the magnetic field known as gradient fields. The gradient fields are superimposed on B_0 and make the magnetic field slightly stronger or weaker depending on position, causing the Larmor frequency of the protons to vary with location.

In practice, three gradient coils systems work together to map space. Their configuration is depicted on Figure 2.1 color-coded. First, a Slice Select Gradient (SSG) (green) selects, typically along the z-axis, a particular axial "slice" of the body that is being excited by the RF pulse. This defines the plane of imaging. Then, a Phase Encoding Gradient (PEG) in yellow, briefly changes the precession speed of protons along an orthogonal axis, i.e. the y-axis, giving each row of tissue a unique phase "signature". Finally, a Frequency Encoding Gradient (FEG) is applied along the x-axis, shown as red, so that protons precess at slightly different frequencies during signal readout.

Each of these three subsystem is associated with distinct physical interactions that become relevant when conductive or magnetic objects, such as surgical hardware, are present in the scanner environment. Notably, the static magnetic field spatial gradient exerts translational forces in the fringe field [39]. Similarly, the rapid switching gradient, which is required to accomplish spatial encoding can also induce peripheral nerve stimulation and acoustic noise [40]. Meanwhile, RF coils deposit energy that may lead to tissue heating [41]. These safety concerns are further explored in Section 2.3.

Continuing on to the image acquisition steps, the resulting signals are stored in a frequency domain matrix called k-space. Each line in k-space represents one measurement with a specific gradient configuration. When all necessary lines are filled, the Fourier Transform reconstructs the data into a spatial image, where each pixel corresponds to a small volume element (voxel) of tissue.

The control and computing subsystems coordinate every stage of image acquisition and reconstruction. The host computer and operator console provide the interface through which the technologist selects and supervises scanning protocols, while the pulse programming and measurement control systems synchronize gradient and RF activity [1]. Supporting units such as the digitizer, image processor, and compute engine translate the raw analog signals into interpretable digital images through Fourier-based reconstruction algorithms [37]. Finally, the dedicated power supplies, patient table mechanics, and electromagnetic shielding ensure stable operation, patient positioning, and noise minimization. Of these elements, the static field of the main magnet is the most significant to comprehend for this study as it establishes the imaging environment and is the major source of mechanical risk for implanted devices.

2.1.1 Physics of the Static Field

Electric current flowing through a wire coil generates a magnetic field, and is the fundamental physical principle of the MR system. Field strength, temporal stability, and homogeneity are used to evaluate system performance. Clinical MRI main magnets are typically made of superconducting wire-wrapped cylinders. These wires are made from niobium-titanium alloys, show no resistance to electric current when maintained at extremely low temperatures, and align the resulting magnetic field along the patient's cranial-caudal axis [37].

The amplitude of the current I and n, the turn density (turns per unit length), determine the magnetic field strength magnitude for an ideal solenoid $B = \mu_0 nI$. The field is directed along the solenoid's long axis. A good estimate of the magnetic field at position z along the bore axis, measured from the solenoid center, within a finite solenoid of length L and radius R can be calculated from Biot–Savart's law

$$B(z) = \frac{\mu_0 n I}{2}\left[\frac{z+\frac{L}{2}}{\sqrt{R^2+\left(z+\frac{L}{2}\right)^2}} - \frac{z-\frac{L}{2}}{\sqrt{R^2+\left(z-\frac{L}{2}\right)^2}}\right], \tag{2.1}$$

where μ_0 is the permeability of free space [36]. This formula provides the axial field along the solenoid, with the maximum occurring at the center ($z = 0$) and decreasing smoothly toward the ends. A simulation of this model for a 1.4 m length by 0.6 m diameter ideal solenoid was used, to approximate a real MRI system, such as the

Siemens Magnetom Trio™, used at CHI Health Creighton University Medical Center–Bergan Mercy (CUMC–Bergan Mercy).

Figure 2.2 showcases this model in two parts. First, the magnetic field (B_0) along the z-axis, where $z = 0$ corresponds to isocenter with dashed vertical lines and light blue shading marking the solenoid boundaries at z = ± 0.704 m away from isocenter. The field is normalized to a maximum of $B_0 = 3$ T, shown as a dashed horizontal reference line, with no shielding or distortion applied.

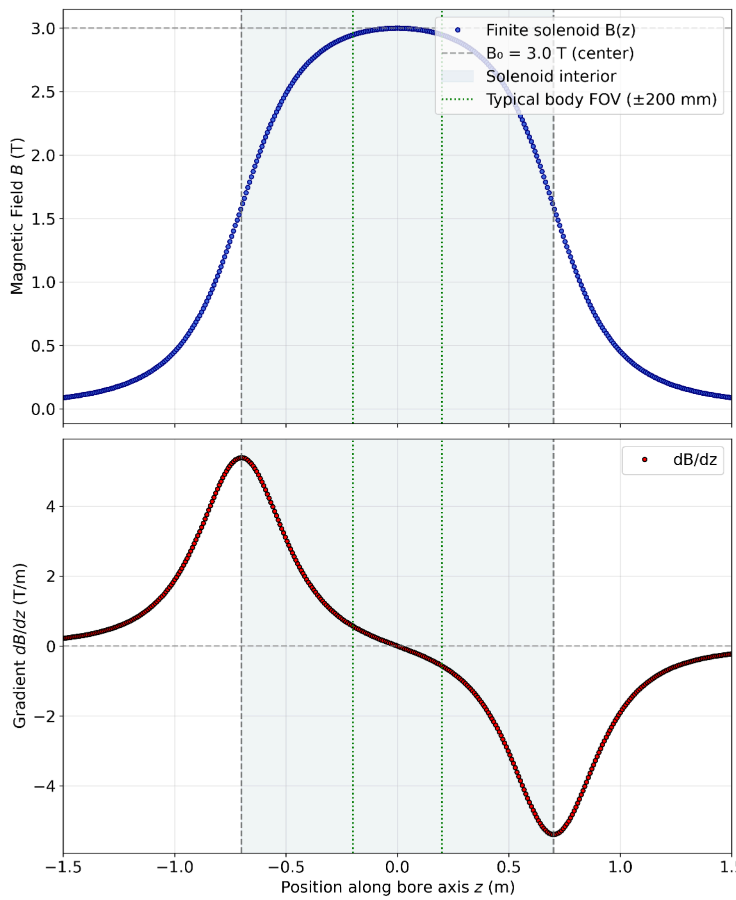


Figure 2.2: Axial magnetic flux density B_z and spatial gradient for an idealized 3 T MRI solenoid (L = 1.4 m, R = 0.3 m). Top: The magnetic field peaks at 3T at the isocenter (z = 0 m) and decreases symmetrically toward both ends. Grey dashed lines indicate the solenoid edges (bore entrances). Green dotted lines represent the typical clinical imaging field of view. Bottom: The spatial gradient dB_z/dz reaches its maximum magnitude at the bore entrances (z = ±0.704 m).

Second, the axial field $B(z)$ from the finite solenoid model yields a spatial gradient dB_z/dz, its maximum value is confirmed to be at the bore entrance position [7, 31] ($z \approx \pm 0.704$ m for this 1.4 m length example), as shown in the bottom panel of Figure 2.2. These strong gradients (often several T/m near entrances) exert attractive forces on paramagnetic/ferromagnetic objects (pulling toward higher $|B|$) and repulsive on diamagnetic ones (pushing away), posing projectile hazards [1, 7, 31]. This gradient-driven behavior supports ASTM displacement force testing and motivates shielding designs in modern systems. High-susceptibility materials experience the strongest pull toward the isocenter, with the maximum risk at bore entrances.

Having established the spatial distribution of the static field, we now turn to how this field interacts with matter. These interactions are governed by fundamental electromagnetic theory and provide the foundation for classifying materials by magnetic susceptibility which quantifies MRI safety risks for implanted devices according to ASTM standards [7, 8].

2.1.2 The Static Field Interaction with Matter

Any material placed in a magnetic field develops a magnetization, $\mathbf{M}$, defined as the net magnetic dipole moment per unit volume, $\mathbf{M} \equiv \frac{\boldsymbol{m}}{V}$ [36]. In a way this $\mathbf{M}$ represents a "smoothed" magnetization of an average field effect over many atoms. At the atomic level, the magnetic field inside a material fluctuates rapidly due to particles in motion. This phenomena forms the basis for MRI image generation. However, for the purpose of this thesis, we focus on the macroscopic behavior mostly. To simplify the macroscopic analysis of materials in magnetic fields, the auxiliary magnetic field $\mathbf{H} \equiv \frac{1}{\mu_0}\mathbf{B} - \mathbf{M}$, where $\mathbf{B}$ is the macroscopic magnetic flux density and μ_0 is the permeability of free space, accounts for the contribution to the magnetic field from free currents, separating it from the contribution of bound currents induced by magnetization [36]. This distinction becomes useful when analyzing mechanical effects on magnetic dipoles.

Different materials respond differently to an applied field. For linear materials magnetization is also directly proportional to the field strength $\mathbf{M} = \chi_m \mathbf{H}$, the degree to which a material becomes magnetized is quantified by its magnetic susceptibility, χ_m (dimensionless), which is typically around -10^{-5} to 0 for biological tissues. It is convenient to define the material permeability as $\mu \equiv \mu_0(1 + \chi_m)$, so that the magnetic flux density takes the compact form

$$\mathbf{B} = \mu_0(1 + \chi_m)\mathbf{H} \quad \text{or} \tag{2.2a}$$

$$\mathbf{B} \approx \mu \mathbf{H}. \tag{2.2b}$$

Diamagnetic materials, such as water, calcium, and most organic compounds, develop a magnetization opposite to the applied B-field due to paired electrons. The susceptibilities of most human tissues are very small, in the range from -7$\times 10^{-6}$ to -11.0$\times 10^{-6}$. Paramagnetic substances, like molecular oxygen, deoxyhemoglobin, or gadolinium-based MRI contrast agents, align weakly with the field, enhancing the local magnetic flux with χ_m values from 0 to $+10^{-2}$ [17]. Contrast agents, such as gadolinium or iron-oxide, represent a separate class of regulated pharmaceuticals administered only in carefully controlled doses [42].

For these material types which are relevant to the MR environment, the susceptibility satisfies ($|\chi_m| \ll 1$), and Eq. 2.2b reduces to $B = \mu_0 H$. Ferromagnetic materials, by contrast, exhibit strong self-sustained magnetization, about 10^2 to 10^5, and are not allowed in MRI suites due to safety hazards [43]. This classification directly determines how the components of an implanted device such as an epicardial pacing lead responds mechanically to the MRI static field, which is the primary topic of this work.

2.2 Epicardial Leads

CIED systems refer to cardiac implantable electronic devices, including pacemakers, ICDs, Cardiac Resynchronization Therapy (CRT) devices, and implanted rhythm monitors. These are advanced medical devices used to monitor and regulate heart rhythm disorders. Out of these components, cardiac leads are a type of very thin, ∼4 to 5 French gauge (Fr, where 1 Fr = 0.33 mm), electrical wire that connects a pulse generator to the heart enabling two-way communication between them. Their main functions include, monitoring, pacing, and defibrillation.

Cardiac lead types are differentiated by three main things, placement in the heart, polarity, and their fixation mechanism. The later is the method used to secure the lead tip to the heart tissue to prevent movement. In practice, Figure 2.3 shows the two types of leads that are used, epicardial leads which are implanted directly on the epicardium (the outer surface of the heart) and transvenous leads, inserted through the veins and positioned inside the heart's chambers, making contact with the endocardium (the inner lining of the heart). Both are prone to complications such as fracture, infection, and venous obstruction (transvenous only) requiring complex extraction procedures [44, 45].

The position of leads is an important distinction. Clinically, these leads are indicated for different populations and diagnostics. The lead position carries a variety of management decisions (leaving vs. extracting) and different complications (percutaneous lead extraction or transvenous lead extraction) [19, 47]. Physically, the type of tissue contact differs between lead placements and carries distinct mechanical

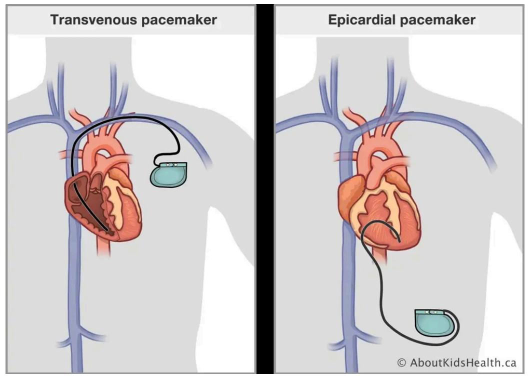


Figure 2.3: Comparison of transvenous vs epicardial lead placement in pacemakers. Reproduced in accordance with the AboutKidsHealth Image Use Policy [46], © 2004–2026 AboutKidsHealth.

and thermal consequences. For instance, transvenous leads are surrounded by blood, which can cool the lead and reduce its effective heating capacity.

Beyond thermal effects, any induced displacement force or vibration on transvenous leads may be modified by surrounding blood flow, vessel-wall contact, and tissue fixation, which can constrain lead motion and alter force transmission. Epicardial leads, by contrast, are mechanically constrained differently. They are exposed on the outer surface of the heart, anchored by suture or screw fixation directly onto the epicardium, so any mechanically induced force acts at that fixation interface [2]. The fixation mechanisms for each lead type are shown in Figure 2.4.

Epicardial leads are most commonly fixed by suturing directly onto the tissue, Figure 2.4b shows a Medtronic STREAMLINE™ 6492 unipolar epicardial lead with one end a swaged-on suture needle. Other mechanisms include screwing an active-fixation helix into the myocardium, shown in Figure 2.4a bottom panel. Transvenous leads are placed internally and secured using either passive-fixation tines that grip the heart's trabecular muscle bundles or an active-fixation helix, as shown in Figure 2.4a [2].

Within epicardial leads, design may vary, including lead polarity and body structure. Figure 2.5 shows the distinction between unipolar and bipolar configurations in terms of electrode placement and the resulting current pathways during cardiac stimulation. Unipolar leads contain a single conductor and one electrode (the cathode) is in contact with the heart. Bipolar leads contain two conductors that connect to two electrodes on the lead itself, a tip electrode (cathode) and a ring electrode (anode).

Moreover, bipolar leads exhibit further structural variation in their body design. As shown in Fig. 2.6, coaxial design features an inner coil conductor nested within

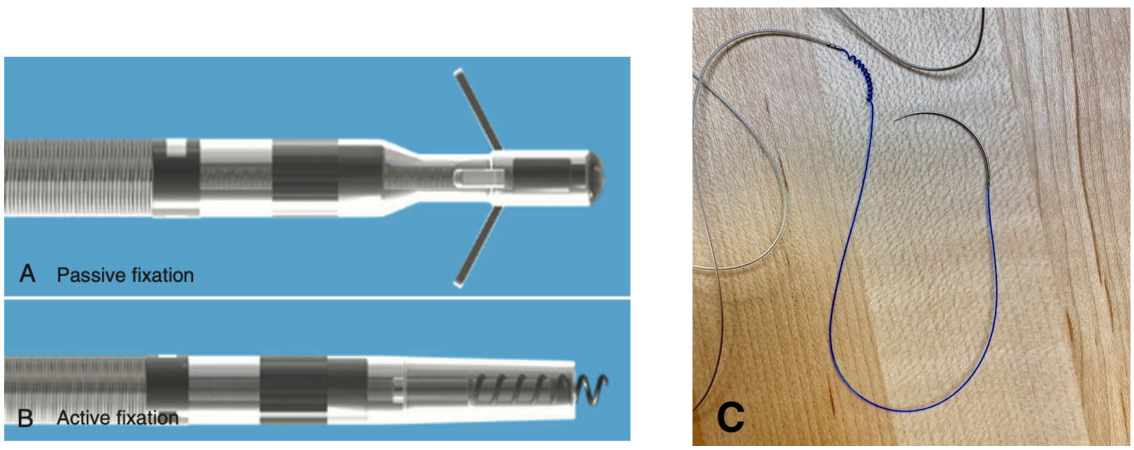


Figure 2.4: Lead fixation variations in pacing leads. (A) Passive-fixation leads rely on flexible tines that anchor within myocardial trabeculae. (B) Active-fixation leads employ a retractable helix that screws into the myocardium. Reprinted from *Clinical Cardiac Pacing, Defibrillation, and Resynchronization Therapy*, Haqqani et al., p. 133, Copyright (2011), with permission from Elsevier [2]. (C) Medtronic STREAMLINE™ 6492 unipolar epicardial lead with suture needle at the proximal end followed by a blue fixation coil.

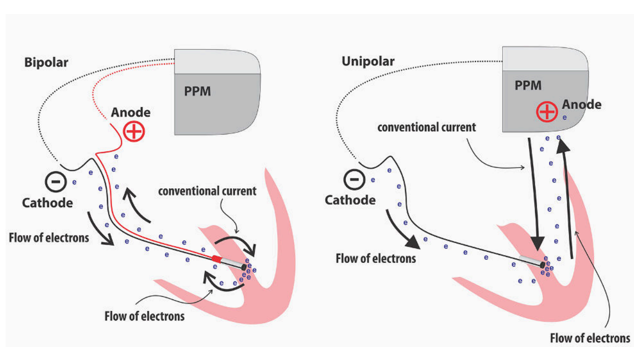


Figure 2.5: Comparison of unipolar and bipolar epicardial pacing leads using a schematic of the heart's ventricle. The current flows in different pathways on each case, Reprinted from [48] under CC BY-NC-ND 4.0.

an outer coil conductor, each wrapped in a separate insulation. Co-radial designs consist of a single coil containing two parallel, intertwined conductor strands, each individually insulated [2].

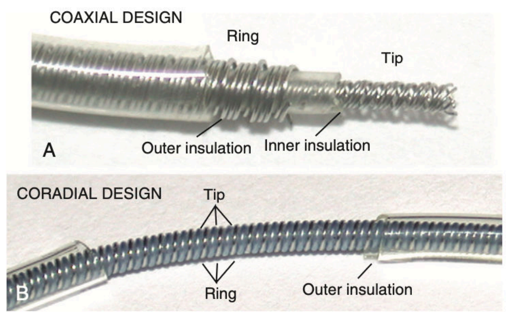


Figure 2.6: Body design variations in pacing bipolar leads. Coaxial designs use two nested coil conductors with separate insulation layers, whereas co-radial designs incorporate parallel, intertwined conductors within a shared lumen. Reprinted from *Clinical Cardiac Pacing, Defibrillation, and Resynchronization Therapy*, Haqqani et al., p. 133, Copyright (2011), with permission from Elsevier [2].

Another important distinction is the clinical use of the epicardial lead, i.e. permanent vs. temporary leads. Permanent leads are relatively uncommon, implanted by surgeons, and primarily indicated in pediatric patients with complex congenital heart disease [49]. In contrast, temporary epicardial leads are routinely placed after open-heart surgery to manage arrhythmia that might arise in the postoperative period [5, 49]. Once the patient is rhythmically stable, these wires are manually removed. However, if resistance is encountered during extraction, they may be clipped at the skin and allowed to retract, resulting in retained wire fragments within the subcutaneous tissues [3, 49].

The several components of an epicardial lead are made of different materials chosen for their biocompatibility, durability, and electrical properties. Table 2.1 summarizes the materials found in a epicardial lead and relates to its magnetic susceptibility. Electrodes are made from inert materials to minimize corrosion and inflammatory reactions over the long term. These include titanium, platinum-iridium alloys, Elgiloy (an alloy of cobalt, iron, chromium, and other metals), and vitreous carbon

[2, 50]. Similarly, conductors which carry the electrical signals between the pulse generator and the electrode are commonly made out of MP-35N (alloy of nickel, cobalt, chromium, and molybdenum) [2, 51]. Insulation, a protective outer coating of the lead that prevents short-circuiting is made of different polymers like Silicone Rubber (Polysiloxane), Polyurethane, Fluoropolymers [2].

Table 2.1: Qualitative magnetic susceptibility of common epicardial lead materials and their relevance for MRI safety. [9, 10, 50, 51]

Component	**Material**	**Magnetic susceptibility category**	**Relevance for MRI safety**
Electrode	Titanium	Paramagnetic	Low; minimal torque/force in MRI
Electrode	Platinum–Iridium alloy	Paramagnetic	Low; generally MRI safe
Electrode	Elgiloy (Co–Cr–Fe–Ni alloy)	Paramagnetic	Moderate; may experience slight translational force
Electrode	Vitreous carbon	Diamagnetic	Minimal; very low MRI interaction
Conductor	MP-35N	Weakly paramagnetic	Low; minimal contribution to MRI forces
Insulation	Silicone rubber (polysiloxane)	Diamagnetic	Negligible; non-conductive and non-magnetic
Insulation	Polyurethane	Diamagnetic	Negligible; non-magnetic
Insulation	Fluoropolymer (PTFE)	Diamagnetic	Negligible; MRI inert

Because epicardial leads are composed of conductive and paramagnetic materials, an abandoned lead may pose a potential risk in the MR environment. Moreover, its physical dimensions influence the nature of these interactions: greater material volume increases the extent to which the lead can interact with the magnetic field.

An abandoned epicardial lead could measure full-length or partial fragments after being left behind [21, 30]. Some fragments are described as long as 5 cm, being 2 cm an average length. Further, longer leads are reported up to 30 cm, while multiple fragments together can add up to approximately 21 cm. The diameter may be similar to standard transvenous pacing leads, typically ranging from 0.5 mm to 1.5 mm, and thus narrower than standard transvenous leads ($\sim$5 F, $\simeq$ 1.7 mm outer diameter) [21]. For instance, lead length governs RF-induced heating through the antenna effect. Longer leads couple more strongly with the radiofrequency field. Maximum heating is expected near the resonant length, defined as half the RF wavelength, which is $\approx$ 12 cm for a 3 T system [18, 34]. For translational force and torque, the total material volume is relevant as well, as both interactions scale with volume and are evaluated relative to gravitational effects. These relationships are discussed further in Section 2.3.

Finally, current institutional recommendations regarding MRI in patients with retained epicardial leads remain conservative despite evolving evidence. Major guideline bodies, including the Heart Rhythm Society (HRS), the European Society of Cardiology (ESC), and the Centers for Medicare & Medicaid Services (CMS), allow MRI in patients with MRI-conditional or certain legacy cardiac implantable electronic devices (CIEDs), but continue to restrict or discourage imaging in patients with abandoned or epicardial leads [21].

At the time of the HRS 2017 consensus there was insufficient data to comment on the safety of MRI with an abandoned cardiac lead. Afterwards, the 2021 ESC guidelines noted that while MRI is generally permitted for many CIED patients, it is not recommended for those with epicardial leads due to theoretical risks like heating. If a patient had an MRI-conditional system along with an abandoned lead, lead extender, lead remnants, fractured lead(s), or surgically implanted epicardial leads, the system was considered non–MRI-conditional leading to a case-by-case consideration for whether to image or not and in which modality [30]. CMS similarly excludes abandoned or epicardial leads from coverage due to perceived insufficient safety data, even though large observational studies indicate that modern epicardial leads (implanted after 2000) may undergo MRI safely under carefully monitored conditions [21].

This cautious institutional stance demonstrates the need for experimental characterization of mechanical interactions between retained epicardial leads and MRI fields. This supports the rationale for the present study. By quantifying these forces under controlled laboratory conditions, the work shown in this document directly addresses whether current institutional restrictions are justified and lays the groundwork for evidence-based updates to clinical safety policies.

2.3 Safety Concerns

In section 2.2 the main object of study on this work was reviewed and in 2.1 the components of an MRI scanner were introduced, illustrated in Figure 2.1. These components are associated with various risks and safety concerns. The three main sources of hazard are the gradient coils, the RF coils, and the main magnet, the latter presenting the most critical safety issues due to its powerful static magnetic field.

2.3.1 Hazards Attributed to the Encoding Gradient Fields

Encoding gradients fields must switch rapidly and with large current flows. The typical slew rate ranges from 5 to 250 mT/m/ms, with currents on the order of 100–200 A [1, 37], to accomplish encoding. That quick shifting of magnetic gradients introduce hazard mechanisms beyond magnetic field strength. For example, when the currents are repeatedly turned on and off in the gradient coils, the resulting Lorentz forces, cause significant vibration. That, is the source of the loud "knocking" sounds during MR scans. Noise levels in modern MRI systems can easily exceed 130 dB, which is roughly a million times more intense in sound pressure than normal conversation or similar to an airplane taking off about 100 meters away, leading to temporary hearing loss as reported in early studies [18]. Consequently, hearing protection is now considered mandatory during MRI procedures.

Moreover, the rapid temporal variation of MRI gradient fields induces electric fields within conductive tissues via Faraday's law. The effect scales with both the rate of change of the magnetic field and the spatial extent of the conductive path through tissue, so even though the absolute magnetic field perturbation from gradients (10–60 mT/m, or less than 1% of B_0) is small, the rapid temporal ramping can induce voltages large enough to reach neural activation thresholds [52]. Typical clinical systems, limited by FDA guidelines to $\sim$ 40 T/s, remain below the threshold for cardiac excitation, but research systems capable of 300 mT/m gradients and higher slew rates approach that regime, posing theoretical cardiac stimulation risks [1, 18, 37]. Although gradient fields induce significantly less tissue heating than RF fields, the time-varying electric potentials they generate can depolarize excitable membranes, producing Peripheral Nerve Stimulation (PNS) or, in extreme cases, involuntary muscle twitching [18, 22].

2.3.2 Heating as a Consequence of Radio Frequency Pulses

Regarding the heating concern, the RF field produced by an MRI scanner consists of both magnetic and electric components. The time-varying magnetic field induces electrical currents within conducting loops or materials, as described by Faraday's law. Because there is a significant difference in electrical conductivity between the metallic implant and the surrounding human tissue, these currents experience Joule losses [18, 34]. This energy is dissipated as heat directly into the adjacent tissue, causing a localized rise in temperature.

For straight metallic implants like epicardial leads, the antenna effect is a dominant mechanism for heating. The tangential component of the RF electric field that runs parallel to the lead causes free charges on the conductor's surface to migrate towards the tip of the lead. There they accumulate, creating a scattered electric field resulting in a much higher total electric field at the lead tips [34, 53]. The antenna effect describes the broad mechanism by which the lead couples with the RF field, inducing currents that concentrate at the tip and heat surrounding tissue. This heating is further amplified if the physical length of the lead is approximately half the wavelength of the RF pulse in the body. Here the lead can act as a resonator, creating a standing wave of current along its length and absorbing energy with peak efficiency. For a 3 T MRI system, this resonant length is approximately 12–13 cm [34].

Finally, the extent of the heating action is determined by several variables, Specific Absorption Rate (SAR), lead configuration, and lead structure. SAR is used to quantify the power deposited per unit mass in the tissue, with higher SAR levels leading to higher temperature rises [18, 34]. The shape, orientation, and termination significantly modify how the RF field couples with the lead [18, 34].

2.3.3 Risks Associated with the Static Magnetic Field

In terms of the static magnetic field, the safety concerns are visible and straight forward. The main magnet is the greatest source of risk. The immense field strength can cause everyday objects to behave in non-intuitive and dangerous ways, most notably turning ferromagnetic items into high-velocity projectiles [18].

A ferromagnetic object experiences a translational magnetic force that pulls it toward the scanner [17, 18]. This force is strongest at the entrance to the bore, as determined by the spatial gradient of the B_0 field. This aligns with theoretical insight of ASTM F2052-15 [7]. Similarly, objects with asymmetric shapes, such as cylinders or needles, will experience torque. Torque is the force that twists an object so that its long axis aligns with the magnetic field lines and is dependent on this field.

In linear media, the magnetic flux density is directly proportional to the applied field strength. For magnetic dipoles $\boldsymbol{m}$ (or objects well-approximated as dipoles, such as small/thin cylindrical shapes), the force and torque due to the magnetic field B_0, are

$$F_z = m_0 \frac{dB_0}{dz} \quad \text{and} \tag{2.3}$$

$$\tau = m_0 B_0 \sin\theta. \tag{2.4}$$

These equations hold universally for induced or permanent dipoles in the low-susceptibility regime relevant to MRI safety testing. The torque expression (Equation 2.4), in particular, is independent of any specific measurement protocol. However, its practical evaluation in the laboratory, such as the determination of $\boldsymbol{m}$ or the orientation relative to the field, depends on the experimental setup and will be detailed further in the context of our measurements. For weakly paramagnetic materials, the induced magnetic moment can be estimated as

$$m_0 = \frac{1}{\mu_0}\chi_m B_0 V, \tag{2.5}$$

where where χ is the magnetic susceptibility, V is the volume of the material, and μ_0 is the permeability of free space. With these foundational expressions, we can now develop the specific physical framework for the translational force relevant to our experimental setup.

2.3.3.1 Translational Forces Induced by the Static Field

In the MRI environment, the dominant safety concern for non-ferromagnetic objects arises not at isocenter, where the static field B_0 is uniform and $\frac{dB_0}{dz} \approx 0$, but in the fringe field region outside the bore, where strong spatial gradients exist.

Consider a simple pendulum with a metallic object at its end. This object is free to swing along the z-axis of an MRI. A static magnetic field along that same axis is then applied. The pendulum would be pulled towards the magnet with a force F_m until it reaches equilibrium with its weight mg and tension T_s. Figure 2.7 shows a complete free body diagram.

From Newton's laws at equilibrium point, the magnetic force can be found by measuring the angular deflection α such that

$$F_m = mg\tan(\alpha). \tag{2.6}$$

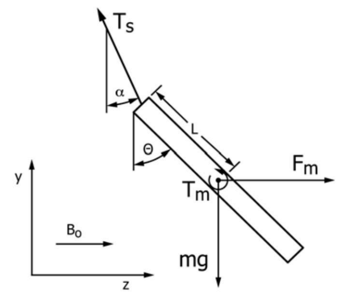


Figure 2.7: Free-body diagram of device in magnetic field (adapted from ASTM F2052-15 [7], Appendix X2).

Noting that this solution is independent of the point of attachment of the string, as the derivation relies solely on force equilibrium, and holds if the center of magnetic force does not coincide with the center of mass [7]. On the other hand, by using Equations (2.3), and (2.5), the magnetic force also is

$$F_m = \frac{\chi_m V}{\mu_0} B_0 \frac{dB_0}{dz}. \tag{2.7}$$

The safety condition is found by comparing the magnetic force to the gravitational force ($F_g = mg$). If the device deflects less than $\alpha = 45°$, then the magnetically induced deflection force is less than the force on the device due to its weight [7]. This formulation explains why the measured deflection angle α (and thus F_m) depends not on material properties or field conditions independently, but on their product.

Although the permanent magnets and solenoids available in our laboratory generate small magnetic fields (only about 30 to 300 Gauss), we can use samples with proportionally larger χ_m/ρ to replicate clinically relevant force-to-weight ratio or Force Ratio (FR) ($\tan(\alpha)$). An estimation for the expected χ_m/ρ of a material in the clinic to produce a set deflection of 1 cm can be found on appendix B.1 and summarized in Table 2.2.

The measurable magnetic deflection depends on the ratio between the mechanical term and the magnetic force term. To produce a deflection of 1 cm at 3 T given known laboratory conditions, the product of χ_m/ρ and $B \cdot \nabla B$ must be preserved. Scaling the field down by a factor of 10^3 requires increasing the sample's effective susceptibility by roughly 10^5 to maintain the same mechanical response.

Table 2.2: Minimum χ_m/ρ and corresponding χ_m required to produce a measurable deflection ($\tan\alpha = 0.01$, i.e. 1 cm over a 1 m string) under reduced field conditions, computed from the on-axis solenoid model at each field strength. The 3 T row uses the ASTM F2052 cited upper bound [7]. Laboratory rows use a bench solenoid geometry ($R = 5\,\mathrm{cm}$, $L = 20\,\mathrm{cm}$), evaluated at the maximum-sensitivity location near the coil exit. Reference density $\rho = 7900\,\mathrm{kg/m^3}$ for common steel.

Condition	$B_0\,\nabla B_0$ ($\mathrm{T^2/m}$)	χ_m/ρ ($\mathrm{m^3/kg}$)	χ_m
3 T clinical (ASTM F2052)	4.80×10^{1}	2.57×10^{-9}	$\approx 1.8 \times 10^{-5}$
3 T model (ideal solenoid)	9.76×10^{0}	1.26×10^{-8}	$\approx 8.8 \times 10^{-5}$
300 G lab (bench solenoid)	6.09×10^{-3}	2.02×10^{-5}	≈ 0.16
30 G lab (bench solenoid)	6.09×10^{-5}	2.02×10^{-3}	≈ 15.98

For typical solenoids producing 30–300 G, the factor of χ_m/ρ spans from approximately ten-thousandth to one-thousandth of that found in ferromagnetic materials. Consequently, to appropriately mimic translational forces under these laboratory conditions, a realistic value of χ_m for a substitute should lie between one and ten. For example, a nonmagnetic stainless steel wire ($\chi_m \approx 10^{-4}$) might show a magnetic force of 30% of the object's weight, whereas ferromagnetic metals like nickel can exceed two million percent [18, 32, 34]. That is more than enough to become dangerous projectiles unless restrained .

On the other hand, the torque induced by the static field depends only on field strength and not its gradient, meaning it will always be smaller than the translational force under the same conditions.

2.3.3.2 Torque Induced by the Static Field

Consider a metallic rigid cylinder shaped object (a wire or rod) free to rotate on a low friction surface, and aligned to a static magnetic field (B_0). The cylinder experiences a torque and has a magnetic dipole moment according to Equation (2.4).

For an idealized thin, linear paramagnetic material, the induced magnetic moment depends on its orientation relative to the static magnetic field. It can be expressed as

$$\mathbf{m} = m_0 \cos(\theta), \tag{2.8}$$

where m_0 is the maximum induced dipole moment, which occurs when the material's axis is aligned parallel to the B_0 field ($\theta = 0°$). In practice, the moment is reduced but does not fully vanish at B_0 ($\theta = 90°$) due to finite geometry. However, this approximation is sufficient for slender devices where the transverse contribution is small. This follows from the projection of the dipole moment vector onto the magnetic field axis.

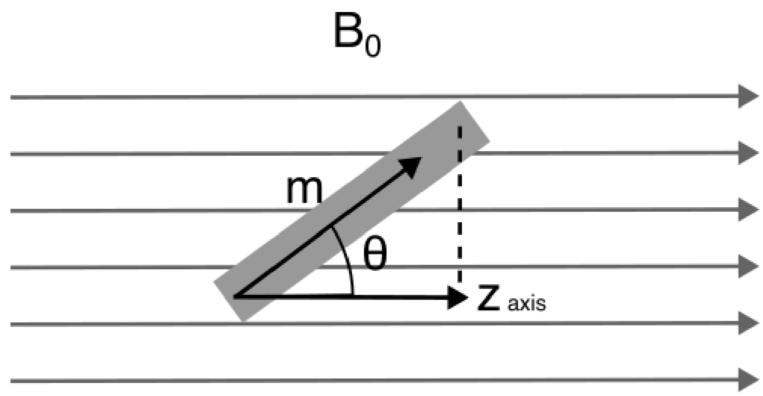


Figure 2.8: Vector diagram of a magnetic dipole **m** in a uniform magnetic field **B** directed along the z-axis.

In Figure 2.8, the dipole of magnitude m, experiences a torque that follows Equation 2.4. From the geometric projection of the dipole moment onto the magnetic field direction (Equation 2.8), the torque experienced by the dipole is

$$|\boldsymbol{\tau}| = m_0 B_0 \sin(\theta) \cos(\theta) = m_0 B_0 \frac{\sin(2\theta)}{2}.$$

Using equation 2.5 for the dipole moment,

$$\tau(\theta) = \tau_{max} \sin(2\theta), \quad \text{with} \tag{2.9a}$$

$$\tau_{max} = \frac{\chi_m B_0^2 V}{2\mu_0}. \tag{2.9b}$$

Once the torque is found, a direct comparison to the object's weight can be made. According to ASTM F2213 [8, 24], the criterion (c_τ) for evaluating magnetically induced torque on medical devices in a magnetic resonance environment is

$$c_\tau = \frac{\tau_{max}}{mgl}, \tag{2.10}$$

where τ_{max} is the maximum torque measured or obtained by equation 2.9b, thus the maximum induced torque is comparable to the product of the test object's longest dimension (l) and its weight (mg). Under the ASTM gravity-based acceptability criterion, a device is considered acceptable when $c_\tau < 1$.

With the equations developed in this chapter, primarily equations (2.7,2.9b), we aim to estimate the magnetic susceptibility and quantify the force and torque of objects as they interact with Gauss field magnets. We also project these forces and torques onto materials with much lower susceptibility and correspondingly, higher magnetic fields for clinical applications, keeping in mind the proportionality shown in Table 2.2.

2.4 Concluding Remarks

Safety hazards associated with the three primary field components were systematically examined. Gradient fields produce acoustic noise and possible peripheral nerve stimulation, radiofrequency pulses can induce localized heating via the antenna effect, and the static field generates both translational forces, maximized at bore entrances where spatial gradients are steepest, and torques on conductive implants [17, 18].

This chapter has reviewed the essential theoretical foundations for evaluating MRI safety in patients with retained epicardial leads. MRI fundamentals, static field (B_0), RF excitation, gradient encoding, and relaxation (T_1, T_2), were provided to explain the scanner function. Furthermore, the interaction of matter with the static field and the physics that describe such interactions were explored seeking to characterize the force and torque acting on objects with the MRI. Translational forces and torques can be induced on conductive implants. Magnetic susceptibility (χ_m) is used to classify materials as diamagnetic, paramagnetic, or ferromagnetic, with most human tissues and the primary constituents of modern epicardial leads (MP-35N conductors, platinum-iridium or titanium electrodes, and polymer insulation) occupying the low-to-moderate paramagnetic range.

Temporary epicardial leads, as parts of CIEDs, differ from transvenous leads in placement, design (unipolar vs. bipolar, coaxial vs. coradial), usage, and more importantly, fixation (suturing vs. screwing). Their direct attachment to the epicardium and occasional abandonment (often as 5–30 cm fragments, 0.5–1.5 mm diameter) after surgery raise concern for mechanical interactions with MRI scanner fields, despite low-to-moderate magnetic susceptibilities that suggest minimal risk [17].

Current guidelines (HRS 2017, ESC 2021, CMS) remain conservative, considering systems with abandoned or epicardial leads non–MRI-conditional due to insufficient data, despite evidence that post-2000 leads may be safe under monitoring [21, 30].

The material presented here serves as a physical framework needed to move from theoretical prediction to empirical validation. Quantifying static-field forces and torques on epicardial lead under controlled conditions in the next experimental chapter addresses the evidence gap that currently restricts MRI access for a growing patient population.

Chapter 3

Materials and Methods

This chapter details the experimental setups and measurement procedures developed to quantify MRI-induced translational forces and torque acting on post-surgical epicardial pacing lead when exposed to its static magnetic field. Three non-magnetic test fixtures were constructed, specifically designed to measure these forces with the precision required for clinical applications, in accordance to ASTM guidelines [7, 8, 9, 10]. First, a spatial field mapping platform was fabricated to characterize B_0 and ∇B_0. Alongside but independently, a pendulum-based deflection apparatus was made to assess translational force through angular displacement. Finally, a strain sensor with a low-friction pole and lever arm measurement system were designed, cut, and assembled for rotational force (torque) assessments.

All three fixtures were developed to be tested in a laboratory framework first. As such, this chapter is divided into the three experiments listed above. Each, constituted of three parts, a technical description of the fixtures developed, the in-lab testing, and an initial clinical methodology. We also outline the design rationale, key modifications, and resulting improvements in sensitivity and precision to support reproducibility.

In the laboratory, the complete suite of instruments and associated control/acquisition software was constructed and tested in a controlled low-field environment (in range of 30 - 300 G). Although this field strength is two to three orders of magnitude below the 3 T clinical scanner, it was deliberately chosen to enable precise evaluation of the fixtures' accuracy, sensitivity, resolution, and repeatability without requiring scanner time. The laboratory setup also allowed direct observation of the fundamental physical phenomena described in Sections 2.3.3.1 and 2.3.3.2 on representative test samples.

Importantly, both translational force and torque scale predictably with the magnetic field strength, $F \propto B_0 \frac{dB_0}{dz}$ and $\tau \propto B_0^2$. We reproduced the same normalized force and torque signatures that will be encountered in the MRI suite by scaling the effective susceptibility χ_m and density ρ of the test objects (see Appendix B.1) for use under small magnetic fields available in our laboratory. The laboratory results are therefore directly valuable. They confirmed that the instruments function as intended and that the physics is captured correctly before any clinical measurements were attempted.

As outlined in Section 1.2, the several constraints of clinical MRI access necessitated thorough laboratory validation of the instruments and protocols beforehand. Through institutional collaboration, CUMC–Bergan Mercy accommodated a dedicated measurement session on their Siemens 3T Magnetom Trio™ MRI scanner. Figure 3.1 depicts the scanner in which measurements were conducted under supervision to ensure safety protocols were followed. Accordingly, the clinical session pursued the two complementary goals previously defined, namely to verify device performance and identify workflow issues. Moreover, a clinical session provided our first dataset on the MRI-induced translational forces and torques acting on post-surgical epicardial pacing leads, thereby closing the loop between bench validation and in-vivo applicability.

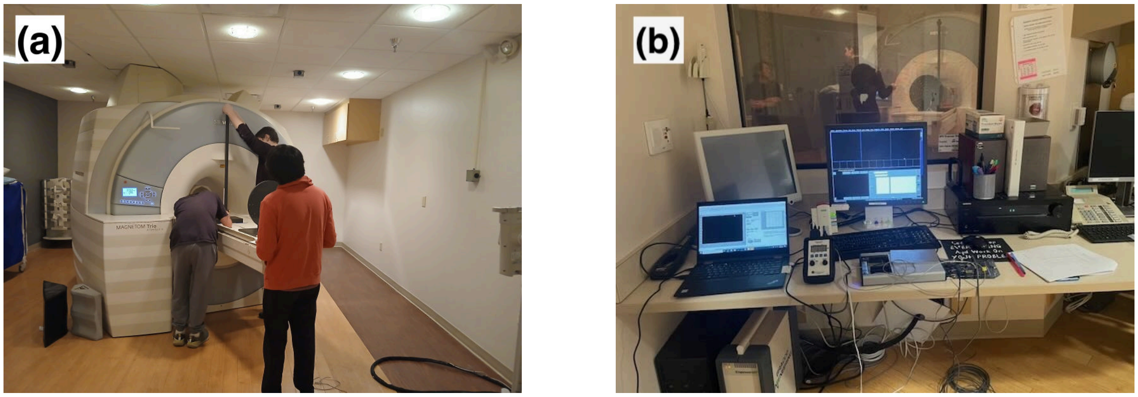


Figure 3.1: (a) Siemens 3T Magnetom Trio™ system at CUMC–Bergan Mercy, where experiments were conducted. (b) Control room showing the main control computer in stand-by and research instruments on the desk.

The first challenge in estimating the induced magnetic force in an object is determining the spatial behavior of the magnetic field around the object experiencing the force given by equation 2.7. The next section describes the method used for the first experiment, both in the lab and at the clinic.

3.1 Mapping the Magnetic Field

Magnetic field strength (B_0) and gradient (∇B_0) measurements were performed using a commercial Gaussmeter (Model GM2, AlphaLab Inc., Salt Lake City, UT). The Gaussmeter used has an accuracy of 1% of the DC reading in the 16°C to 29°C range [54].

The first step in accurately mapping the static magnetic field of an MRI scanner is to know the precise position of the measurement probe at all times relative to the scanner bore entrance or isocenter. To achieve this spatial accuracy and ensure reproducible field measurements in a clinical environment, a rigid non-magnetic fixture constructed from PVC was developed [35].

3.1.1 Mapping Fixture

The fixture consists of a rectangular base (300(W) × 420(L) × 40(H) mm) and a vertical circular plate of 200 mm diameter. It has a total weight of about 9 kg and height of 350 mm, with its center at 250 mm above the surface to align it with the MRI scanner isocenter. The plate is perforated with a polar grid of probe-access holes, with 12 radial lines spaced at 30°. Each line contains five holes at uniform of 20 mm radial spacing, all sharing a central hole as shown in Figure 3.2, resulting in 49 measurement positions. Each access point was numerically indexed to facilitate 2D field mapping in the X-Y plane.

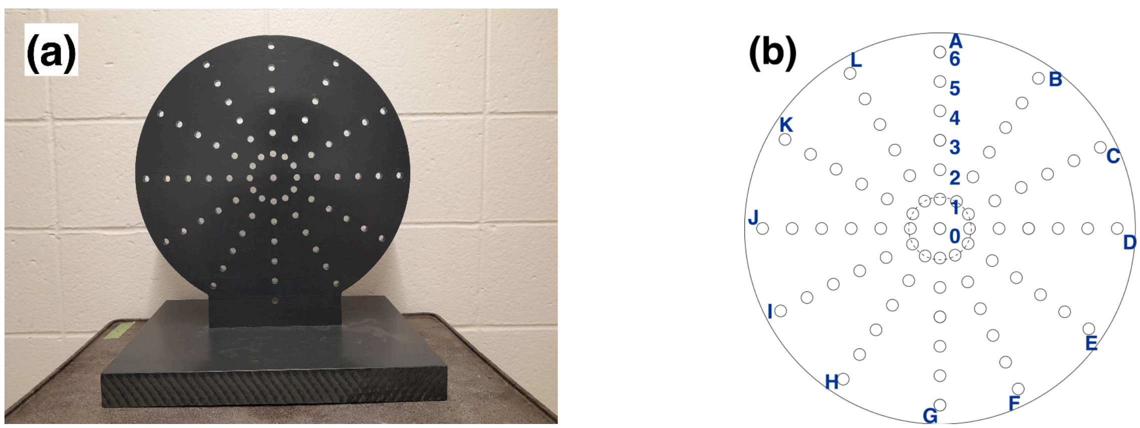


Figure 3.2: PVC-based magnetic field mapping platform and coordinated circular grid layout labeled from A to L angular and from 0 to 6 radial. This platform was used for spatial B_0 and ∇B_0 measurement, based on design from Ferreira, et al. (2017) [31, 35].

These holes in the fixture's plate were used to insert the gaussmeter probe for data capture using LabVIEW, chosen for its flexibility and precision. The manufacturer's default software, AlphaApp, provides basic data-logging, recording, live screen streaming, and plotting of measurements from USB-enabled AlphaLab meters [55]. However, AlphaApp's control over sampling frequency is limited by design, therefore limiting fine temporal resolution when the probe is moved into steep field gradients. In contrast, using LabVIEW Virtual Instrument (VI) interface allows for configurable sampling intervals, buffer control, and precise timing protocols, enabling higher accuracy as the probe approaches and crosses high gradient regions within the MRI fringe. We developed that VI by leveraging the AlphaLab Data Acquisition Communication Protocol [56] to command the device directly via a USB series port, providing control over timing and sample rate.

The entire fixture was shifted axially along the Z-direction of a horizontal bore MRI scanner using controlled couch increments (e.g., 10 cm or continuously). This translation allowed for volumetric sampling of B_0 over several planes. The spatial magnetic field gradient ∇B_0 was estimated numerically using finite difference calculations between adjacent measurement points on a field strength map along the z-axis

$$\frac{dB}{dz} \approx \frac{B_2 - B_1}{z_2 - z_1}, \tag{3.1}$$

where B_1 and B_2 represent the measured magnetic flux densities at positions displaced symmetrically from the location of interest. This type of fixture has previously been applied to MRI mapping in the work by Ferreira [31].

3.1.2 Laboratory Methodology - Solenoid and Neodymium Magnets

Seeking to achieve the most reliable approximation of clinical MRI conditions, magnetic fields in the lab were generated using either neodymium permanent magnets or a copper-wound solenoid. Each approach presents advantages and limitations.

For example, a neodymium magnet can produce a very strong magnetic field (B_0 up to 1–1.4 T near its surface), but its strength at the sample/probe position depends heavily on distance from the magnet (falling off approximately as $1/r^3$) and results in a steep spatial gradient. These properties make neodymium magnets ideal for generating strong localized torque on small samples, where the sample can be positioned very close to the magnet to exploit the peak intensity without requiring power supplies or cooling. The fixed nature of the field, however, limits fine adjustments without physical repositioning.

In contrast, the magnetic field of a solenoid depends directly on the current flowing through its wire, allowing precise, continuous tuning of the field without mechanical repositioning. With appropriate coil design, solenoids can also provide better field uniformity over a defined region, making solenoids better suited for controlled translational force assessments, where variable field strength and gradient are needed to measure forces consistently across distances.

The approach used in this section was the latter. A custom-built air-core solenoid of approximately 15 cm in length and 10 cm in outer diameter was densely wound with copper wire ($\sim$1 mm diameter, winding density of $50,000$ turns per meter), forming an inner core of approximately 5 cm. A Pasco Scientific SF-9584 DC power supply was used to deliver currents in the range of 0–4 A, corresponding to magnetic field strengths from 1 to 18.9 Gauss at about 8 cm from the aperture along its long axis (z-axis) for this solenoid.

Figure 3.3 shows the experimental setup used to test the probe and obtain a reference map of the magnetic field of our solenoid. Constant current of 4 A was applied to the solenoid. At the solenoid entrance, three positions were selected to place the Gaussmeter probe to check for homogeneity, at the center (r=0 cm,θ=0°), southwest (r=10 cm, θ=225°), and east (r=10 cm, θ=0°). This was achieved by using a translational stage positioning the probe 10 cm away from the solenoid's entrance.

The stage uses three Thorlabs Z812 servomotors, one for each cartesian axis, to accurately position the probe with changes of less than 0.1 mm, driven via the Kinesis software interface [57, 58]. While Kinesis-controlled stages support sub-micrometer incremental motion and repeatability on the order of microns [57], our experiment only required positional accuracy on the order of 0.5 mm. Thus the stage system offered

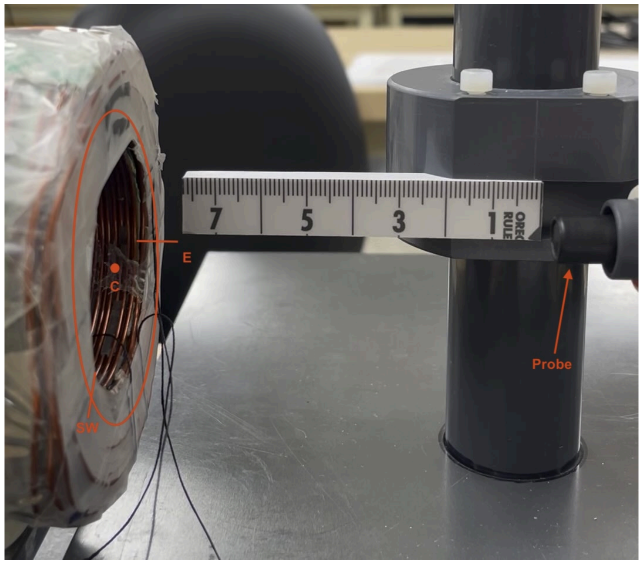


Figure 3.3: Probe mapping magnetic field generated from solenoid. The field was measured at three positions around the solenoid's face: E (East), SW (Southwest) and C (Center). The probe was manually stepped 20 cm away from the solenoid face along the axial direction.

more than sufficient precision and stability for repeatable setup between trials. The translation speed was set up using Kinesis for 0.3 m/s and the maximum displacement of each actuators was 11 mm. This setup was also used for the experiment in Section 3.2.

To make a quick check across a wider range, a ruler was fixed at the same three positions described and the probe was steadily translated about 20 cm to the solenoid entrance and back. Data was recorded using the manufacturer's default software, AlphaApp [55] with a sampling time of 1 reading per second, for about 2 min. Video recording was used to map time, and distance enabling a rough map of the B field and its gradient.

3.1.3 In Clinic Measurements - Magnetic Strength and Gradient

To characterize the magnetic field gradient along the scanner's longitudinal (z) axis, the circular grid fixture was mounted on the patient couch, as shown in Figure 3.4. Couch motion controls were first calibrated, allowing precise positioning in millimeters via the four-digit display. Fixture-to-bore clearance was verified, and the fixture proved marginally over-sized for full insertion, constraining the maximum travel depth. The gaussmeter probe was routed through the fixture plate into the MR suite with a dedicated acquisition computer, located in the control room. Measurement range was defined from 1 cm away from the bore entrance to approximately 1 m away.

Figure 3.4: The mapping fixture was mounted near the bore entrance, researcher calibrated and noted initial position, bore entrance clearance and proper alignment to laser locators.

Prior to each acquisition series, the fixture was aligned to scanner isocenter using the built-in laser crosshair and a calibrated ruler. Its position was permanently marked on the table surface at (8.5 cm from table side edge) for reproducibility. Grid coordinates examined were the central position (00) and off-axis locations A3, A6, G6, and G3 (vertical plane), with the physical heights of G6 and G3 recorded to ensure correspondence with translational couch motion. Uniformity across the grid was additionally confirmed by measurements at right-side coordinate D3.

At each grid coordinate, the couch was translated continuously while the gaussmeter logged data in paired passes, one run advancing into the bore and one run retracting, producing two independent datasets for hysteresis assessment. Continuous translation was chosen over a step-and-shoot approach because the patient couch controllers do not support positioning to a specific coordinate or velocity. This way, alignment was performed once and the couch was driven manually through the full range, with position tracked from both the logged data and the motion controller screen.

In total, 15 raw datasets were acquired, four at the central position (00), two at each off-axis lettered coordinate (A3/A6, G3/G6, D3), and one repeated file following a LabVIEW crash. All files were saved as Comma Separated Values (CSV) with the structure illustrated in Appendix C.1.

Post-acquisition, the CSV files were imported into Python using the `Pandas` library. Column headers were standardized, and data integrity checks (duplicate removal, timestamp validation) were performed. To better illustrate this process, the next set of figures show examples for a single coordinate point (G6 from Figure 3.2b) which is representative of all the other coordinates datasets.

First, Figure 3.5 depicts the raw data in time for both runs, backwards on the left, forward on the right. Two outliers were observed in all datasets. These outliers were removed with a quick exponential fit to use as baseline. Outliers were removed where the value was greater than 3σ. Excess measurements were taken at the beginning and end of each measuring run, most visible at small times on the left panel, where a sudden plateau appears. This was cut as well, done by removing the leading/trailing near-constant flux checking with nearest neighbors within a tolerance of 1 Gauss.

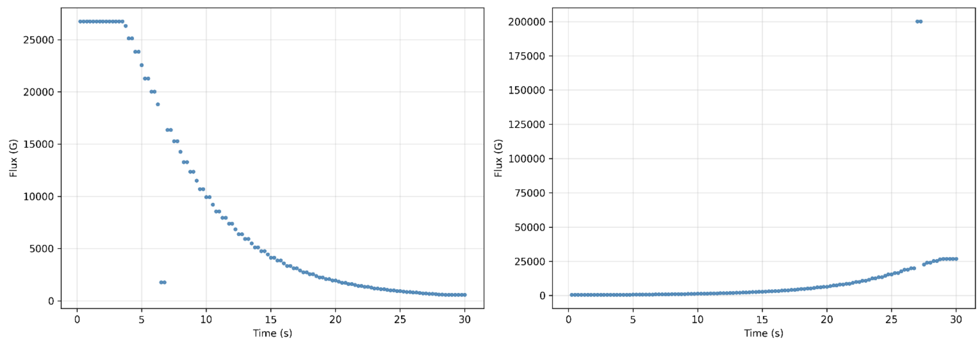


Figure 3.5: Raw magnetic field measurements at coordinate G6 as a function of time. The left panel corresponds to the backward run and the right panel to the forward run. Two clear outliers are visible and appear consistently across all datasets.

Once both of these problems were removed, the velocity of the patient couch with our fixture was calculated assuming constant speed by averaging the total time with the total distance (1252.0 mm) on each run. This allowed for conversion of the time axis to position along the couch, Figure 3.6 top rows in blue shows these plots. However, notice that some of the points on the plot are repeated making the plot stepped. This in turn distorts the gradient calculations as central difference no longer works, as seen in Figure 3.6 bottom rows in red. This repetition arises from the resolution factor of our gaussmeter probe. The gaussmeter probe was set to sample at 4 Hz (one sample every 0.25 s). At this field strength, the instrument became saturated, hence the repeated readings. To address this, the datasets were separated into two interleaved sequences (sub-run A and sub-run B), effectively reducing the sampling rate to 2 Hz (0.50 s per sample) and eliminating redundant values for subsequent gradient calculations. Singles were kept in sub-run A.

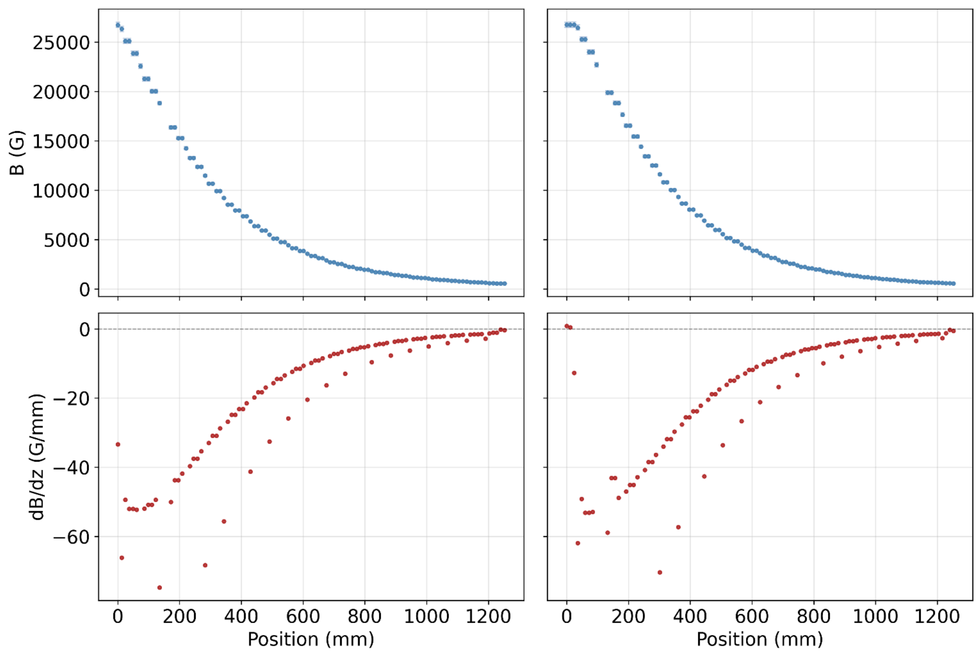


Figure 3.6: Magnetic field (top row) and corresponding gradient (bottom row) along the measurement path after converting time to position but prior to separating repeated measurements. Cleaned up data can be found in Chapter 4.1.2

Afterwards, the flux column was subsequently converted to physical units, and the longitudinal gradient was extracted by numerical differentiation to determine the optimal representation of the field profile.

With the spatial behavior of B and ∇B explored, we move to the test object itself and how it physically responds to that field. In the next section, we present the fixture used to measure the actual displacement a small test object experiences (experiment two), record the resulting deflection angle, and use that angle to compare the magnetic force directly against the object's weight. At the same time, we take the measured force and work backwards using the field values to determine (or confirm) the material properties, such as effective susceptibility. This cross-check confirms that the underlying physics holds as expected. In the clinic, we assessed deflection angle independently from magnetic field behavior providing first insight onto positive control materials and epicardial lead configurations.

3.2 Translational Force Measurement via Angular Deflection

To assess MRI-induced translational forces, our lab developed a pendulum-style apparatus or fixture following the geometry and physics outlined in ASTM F2052-15 [7]. This apparatus enables measurement of small deflection angles to derive magnetic forces via gravitational force balance. The following subsections describe the fixture design and features, the test objects used (including positive controls and epicardial lead holders), the smallest measurable force, and the measurement protocols used in both the laboratory (low-field solenoid) and in the clinical 3 T MRI environment.

3.2.1 Measurement Device and Test Objects

Figure 3.7 shows the experimental apparatus. Figure 3.7a presents CAD renders of the pendulum fixture created in CATIA to document the design intent and associated engineering drawings (full set in Appendix A.1). The fixture was constructed entirely from PVC components, including a vertical post of about 1.5 meters and a rigid base of dimensions 30(L)×20(W)×4.5(H) cm for stabilization, as shown in Figure 3.7b. The vertical post includes a set of two rings fixed with nylon screws, for height adjustment. A default suspension height of $L = 1$ meter between these two rings was used for consistency across trials, maintaining the geometry and proportions shown in equation (3.2). These distances were measured from the last contact point of the string at the rubber interface on the top ring (see Figure 3.7c) to the freely hanging sample aligned with the ruler scale. The lower ring, shown in Figure 3.7d, features a slit where a custom made Kydex™ ruler is secured with smaller nylon screws. This ruler is detachable, allowing for easier transportation and interchangeable ruler sizes.

After describing the physical implementation of the pendulum fixture, we now evaluate its measurement capabilities. In particular, we determine the smallest de-

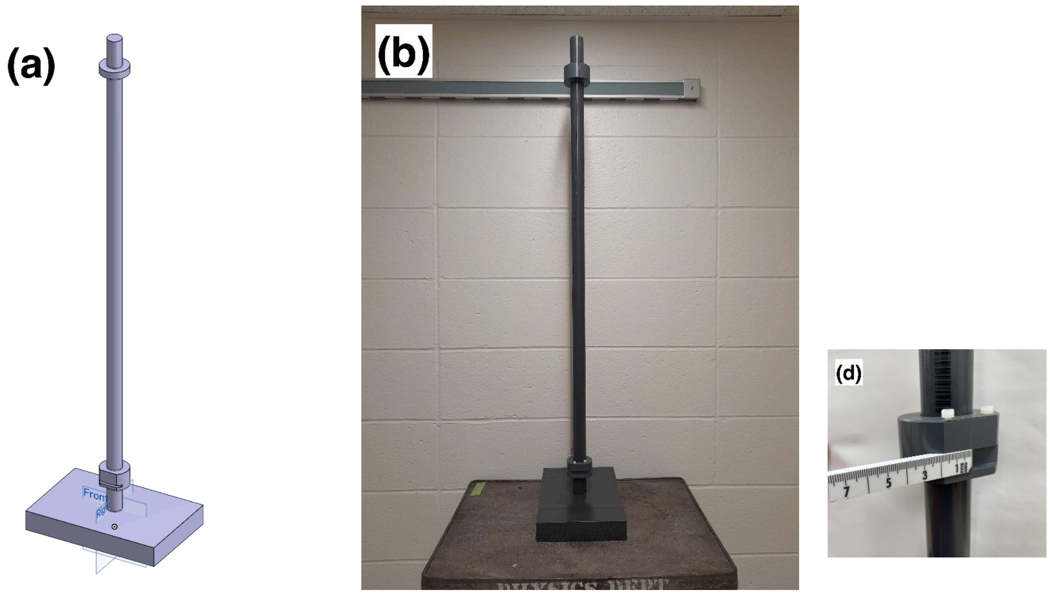


Figure 3.7: Full and detailed views of the pendulum fixture. (a) Full ensemble as CAD model. (b) Real-size pendulum fixture made of PVC. (c) The upper ring designed to secure the pendulum string with clamp and rubber attachment. (d)The lower ring containing the slit used to align a 1cm-wide by 20 cm ruler perpendicular to the string.

tectable translational force along the z-axis by analyzing the geometry of the deflected pendulum at equilibrium. Whether forces on the order of mN, comparable to the weight of 1 g, can be resolved depends on the finite precision of the ruler-based deflection measurement and the pendulum's length.

3.2.1.1 Smallest Measurable Translational Force

The smallest measurable translational force can be obtained by considering the geometry of the device. The pendulum from Figure 3.7b forms a right triangle where the vertical line is the pole *"L"*, the horizontal gives the position on the pendulum *"z"* and the hypotenuse is the pendulum's string. Both of these measurements have a 1 mm precision, yielding a $L = 1$ m and $z = 1$ mm triangle (Figure 3.8). Then, the smallest deflection angle capable of measurement is given by $\alpha = \tan(z/L) \sim z/L$ for small angle which is about of a 10^{-3} rad (~ 0.057 deg).

Since the angular deflection corresponds to the ratio of magnetic to gravitational forces the smallest measurable magnetic force is

$$F_{m,min} = mg\tan(\alpha_{min}) = mg(\frac{z}{L}). \tag{3.2}$$

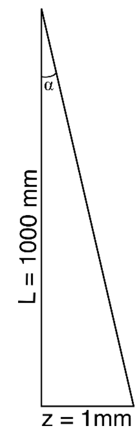


Figure 3.8: Right-angled triangle formed by the pendulum geometry at equilibrium. The length of the string L and the pendulum's position z define the deflection angle α that a test object experiences.

For a 0.1 g test mass, this corresponds to a detection threshold of $F_{m,min} \sim 10^{-6}N$ ($1\mu N$). Thus, the pendulum fixture can resolve magnetic force as small as one-thousandth of the object's weight. This resolution defines the lower limit of measurable magnetic susceptibility using the pendulum method. For nonmagnetic or weakly paramagnetic materials, such sensitivity ensures that even subtle translational forces can be quantified, making the setup suitable for assessing device safety under MRI static field gradients. Sample calculations on this can be found in appendix B.2.

3.2.1.2 Test Objects and Baskets

In the laboratory setting, a solid ferromagnetic cylinder was used as sample. While testing for the clinical environment used materials that are expected to exhibit minimal translational force under clinical MRI conditions. Positive controls were prepared from two common biocompatible, non-ferromagnetic metals, titanium and 316L stainless steel wire, both sourced from TEMCo Industrial (USA) and supplied on their standard spools [59]. The physical characteristics of all test objects used in this study are summarized in Table 3.1, while Figure 3.10 shows the samples in real size.

These positive controls and the epicardial lead were mounted in a custom 3D-printed Polylactic Acid (PLA) basket, with the added benefit of easily change test objects while maintaining MRI compatibility. Figure 3.9 shows a 3D render with the intended proportions. The corresponding engineering design is provided in Appendix A.2, with STL files available in this Github repository [60]. The mesh's lightweight design aimed to minimize its contribution to the total gravitational force. Its bounding volume would be 224 mm^3. However due to the hexagonal lattice structure, the effective solid volume is 95.2 mm^3, resulting in a mass of at least 118 mg based on PLA material properties. Once printed, it had a final weight of 130 mg.

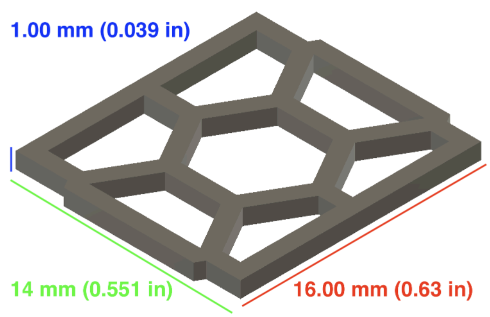


Figure 3.9: The design of the lead holder was made as a rectangular lightweight hexagonal mesh (16 (L) × 14 (W) × 1 (H) mm) to wrap the epicardial lead around. It is held by the pendulum string

Table 3.1: Summary of test objects and their physical properties used in laboratory and clinical measurements.

Object	Material	Diameter (mm)	Length (mm)	Mass (g)
Ferromagnetic sample	Steel	3.00	6	0.3648
Titanium wire	Ti (Grade 1)	1.02	23	0.1549
316L SS wire	Stainless steel	0.76	22	0.1690
Titanium coil	Ti	–	–	1.7964
Epicardial lead	–	0.46	378	0.1386
Hexagonal Basket	PLA	16 × 14 × 1 mm (Figure 3.9)		0.130

So far, we have described the materials used in both the lab and the clinic, from the solenoid (in Section 3.1.2) to the translational fixture and test objects for both environments. The following two sections explain the methodology followed in each of the two environments.

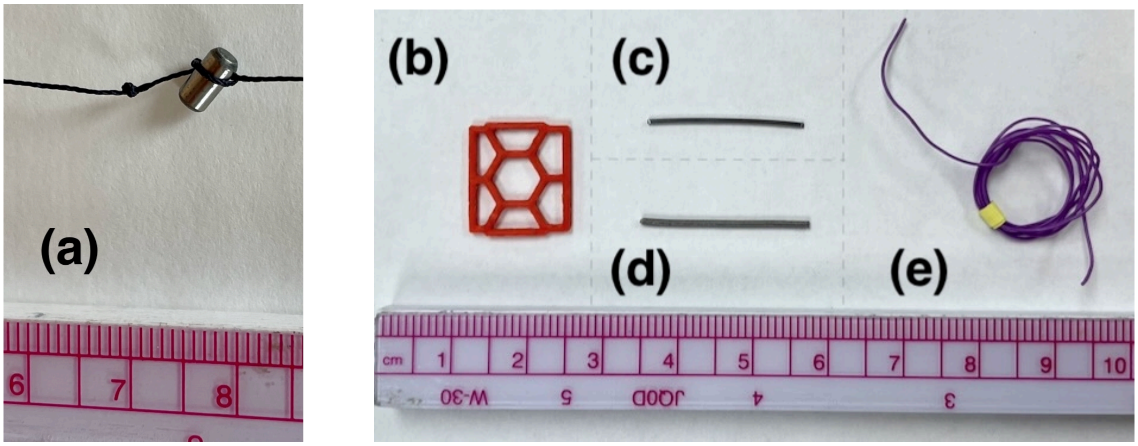


Figure 3.10: Test objects: (a) Ferromagnetic object and string attached. (b)3D Printed basket made of PLA. (c) 316L Stainless steel sample. (d) Titanium wire (Surgical Grade 1) sample. (e) Wrapped Medtronic STREAMLINE™ 6492 unipolar epicardial lead. On the bottom of each picture, a ruler in centimeters shows actual size.

3.2.2 Laboratory Methodology and Setup - Translational Force

The setup used the custom-made solenoid from Section 3.1.2 and the fixture described in Section 3.2.1. Figure 3.11 shows this experimental laboratory setup used to assess the methodology accuracy and reliability with all components. The test object, shown in Figure 3.10a, was attached to a common cotton sewing string suspended at 955 mm from the MRI safe fixture and the angle of deflection was measured visually with a ruler as the object displaced horizontally towards the coil. The gaussmeter probe was positioned to make magnetic field strength and angle deflection measurements at the same time.

The magnetic field characterization used here differs from Section 3.1.2. In that earlier mapping, the Gaussmeter probe was translated along the solenoid axis, while the current in the solenoid was held constant. For the translational force experiment, this approach was inverted. The pendulum was held stationary at a fixed position of 8 cm from the solenoid face, and the solenoid current was varied to increase through a range of magnetic field strengths B_0. This approach was chosen because it decouples the mechanical equilibrium of the pendulum from the field scanning motion, ensuring the fixture remains stationary while the magnetic force is varied. As a consequence, the z-coordinate of the resulting field maps represents the pendulum's displacement from its gravitational rest position (B = 0). Each data point therefore corresponds to a different applied current and field strength at the new equilibrium point. For example, the test object remains at rest (Figure 3.12a) once the current is flowing the magnetic field on the solenoid pulls the subject towards it (Figure 3.12b) the more current was applied to the solenoid the stronger the magnetic field and stronger the force.

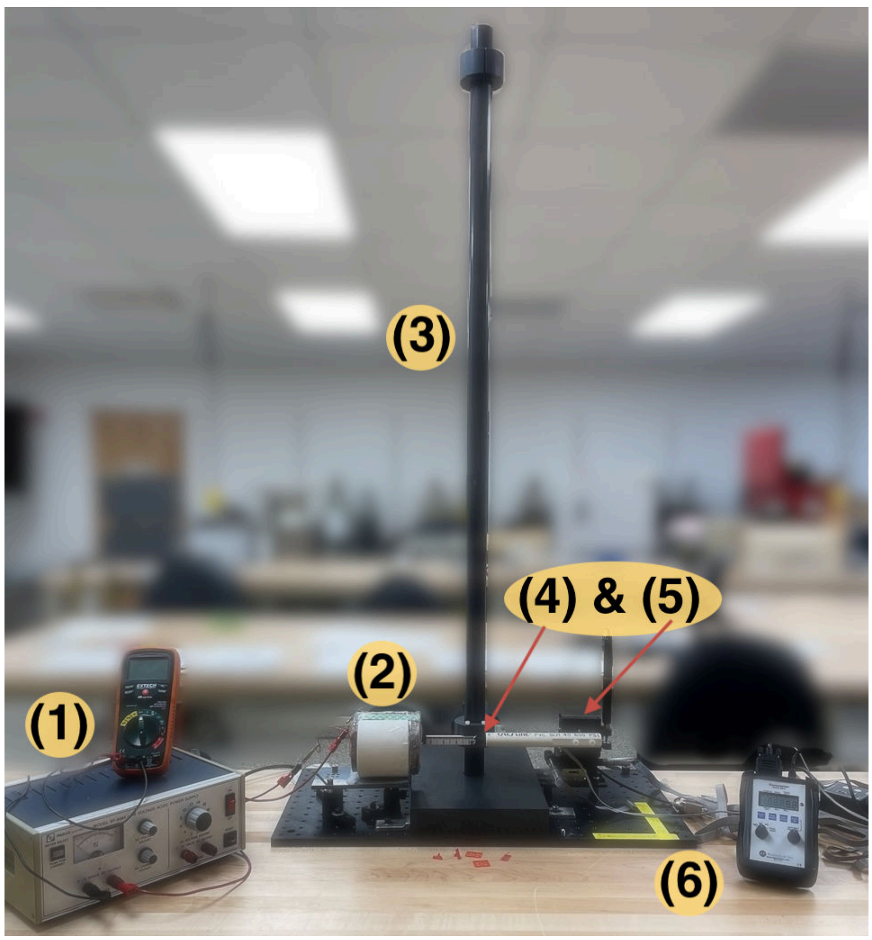


Figure 3.11: Laboratory setup for MRI translational force assessment. (1) Magnetic field generation instruments (left), (2) solenoid, (3) fixture with pole (center), (4) gaussmeter probe, (5) translation stage, and (6) gaussmeter read panel (right).

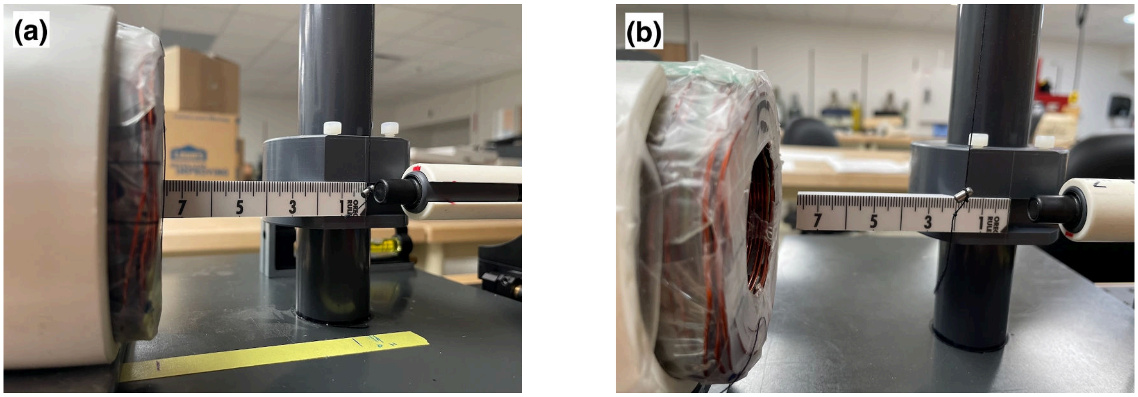


Figure 3.12: Comparison of the test object behavior with and without translational forces during magnetic field exposure. (a) Object hanging in pendulum at rest. (b) Magnetic force pulls the subject towards the magnetic field source.

Induced currents remained low because the ferromagnetic body barely moved, the translation stage's full range was used, the object never entered the solenoid unintentionally, and its's position was always known. For each current applied, the magnetic field was measured at the equilibrium point and recorded alongside the field flux. A magnetic-field map was acquired at each corresponding current level, ensuring the equilibrium position and its ± 0.5 mm neighbors were sampled at least nine times. With this method, the value of B_0 at equilibrium was known for every current level. The gradient was calculated for each point with the two neighbors as

$$\left.\frac{dB}{dz}\right|_{z=0} \approx \frac{B_{+0.5} - B_{-0.5}}{1 \text{ mm}}. \tag{3.3}$$

Using a 1 mm separation $(z_2 - z_1)$ keeps the approximation in the linear regime while remaining resolvable on the field map. The 1 mm increments in displacement was translated to angular displacement

$$\alpha = \tan^{-1}\left(\frac{x}{L}\right). \tag{3.4}$$

Furthermore, to determine the magnetic susceptibility χ_m of the test sample. We use Equations. (2.7) and (2.6),

$$\chi_m = \frac{\mu_0 m g}{V} \frac{\tan(\alpha)}{B_0 \frac{dB_0}{dz}}, \tag{3.5a}$$

$$\frac{\chi_m}{\rho} = \mu_0 g \frac{\tan(\alpha)}{B_0 \frac{dB_0}{dz}}. \tag{3.5b}$$

providing a direct estimation of magnetic susceptibility from angular deflection measurements, provided the field and its gradient are known. A plot of the deflection angle versus the product of field strength and field gradient was created. From it, a linear fit extracted χ_m from the slope following

$$\tan(\alpha) = \frac{\chi_m}{\rho \mu_0 g} \cdot B_0 \frac{dB_0}{dz}. \tag{3.6}$$

All measurements were logged in Microsoft Excel, and photos/videos were captured via smartphone to document behavior. A minimum of three trials were performed for each current level; however, only the final (most consistent) dataset was retained for analysis. Data were exported to a Jupyter Notebook for clean up and processing with `Pandas`, then plotted to evaluate $\tan\alpha$ vs. field gradient force. Propagated uncertainties in α and dB_0/dz were computed using standard error propagation formulas. Appendix C includes sample data from these trials and all analysis code and data are publicly available in [60].

3.2.3 In Clinic Measurement - Translational Force

To quantify magnetically induced translational forces on implantable samples within the MRI fringe field, the custom translational fixture was positioned on the patient couch, as shown in Figure 3.13. The fixture supported the lightweight baskets (Figure 3.9) suspended by a common sewing string of length 1010 mm, allowing samples to displace freely under magnetic force while measuring equilibrium position shifts relative to the zero-force state.

Figure 3.13: Measurements for induced translational force. Researcher visually measures displacement once equilibrium is reached.

The translational fixture and its central basket were aligned to scanner isocenter using the scanner lasers, with the zero position marked using tape on the couch surface. The distance from the couch edge to the fixture base was measured to enable reproducible placement of the equilibrium position. Due to the strong magnetic forces near the bore entrance, visible alignment of the basket with the ruler zero was only possible either outside the MR suite or with an empty basket. This meant repeated repositioning of the fixture and multiple attempts to establish a true zero reference, adding considerable time to the protocol.

Measurements were performed using a titanium (Ti) sample as a positive control, given its known paramagnetic response in clinical MRI contexts. Three replicate trials were conducted with the Ti sample, recording both the distance from isocenter and the observed displacement on the fixture scale at each position. The couch was advanced in 50 mm increments toward the bore entrance and 100 mm increments away from it to capture the spatial dependence of the force.

The titanium sample was configured as a loosely coiled spring-like shape, oriented horizontally so that its long axis aligned with the scanner's z-direction (longitudinal field) and its diameter was perpendicular to B_0. In contrast, a Medtronic STREAMLINE™ 6492 unipolar epicardial lead was coiled tightly and oriented vertically (perpendicular to B_0), a configuration dictated by its greater thickness and stiffness, which facilitated tighter packing. Recognizing that coil geometry and orientation strongly influence induced force and torque, additional trials were performed using straight segments of both the titanium material and 316L stainless steel. These straight configurations followed the identical incremental positioning protocol.

In total, eight raw data tables were acquired, three for the coiled titanium sample, three for the coiled epicardial lead, one for the straight titanium segment, and one for the straight 316L stainless steel segment. All displacement measurements were documented manually and later digitized for analysis (also publicly available at [60]). Deflection angles were used to extrapolate allowable spatial gradients at clinical field strengths following F2052-15 Appendix X3.

Once the translational force and material properties are validated through the combined field mapping and deflection measurements, we turned to the second key interaction, the torque induced by the magnetic field on the test object. The following section, experiment three, describes the fixture developed to measure this torque with high sensitivity. This includes sensor fabrication and methodology development. We record the induced rotation (or counter-deflection), convert it to torque magnitude, and compare the result against the expected torque from the known field strength and the material properties already confirmed in the translational experiment.

3.3 Torque Measurement via Counter-torque

The final part of this study involves the measurement of the magnetically induced torque on a horizontally-oriented object within the MRI scanner. ASTM 2213-17 [8] lists five different methodologies: calculation based on measured displacement force method, low friction surface method, suspension method, torsional spring method, and pulley method. In Section 1.1, the results of three of these methods (or a variation of them) were reviewed, including pulley, low friction, and a variety of the torsional spring method [24, 32, 33], which showed good results.

Similar to the suspension or string methods, a torque measurement device capable of quantifying the torque of a bendable test object was developed. This section describes the characteristics of the fixture, followed by an examination of its specialized components, including the sensor and associated circuit. Then, the sensor measurement limits are discussed, along with the calibration procedure employed. Finally, the experimental methodologies implemented using this device are presented.

3.3.1 Torque Measurement Apparatus

A non-magnetic fixture was fabricated with clinical considerations in mind. Figure 3.14 shows the main torque fixture, which consists of a pole, a base and a ring that supports the sensor. The main component is the pole made from Delrin™, 200 mm tall and 51 mm in diameter. Underneath it, a PVC base is attached for support with a nylon screw. On top, it features a space for a non-magnetic ceramic bearing, which is meant to reduce the friction for the rotation as much as possible. Finally, a ring with a clamp holds a custom made sensor marked in red (L) in Figure 3.14a, which will be explained in detail in section 3.3.1.1.

A torque transfer arm was added to provide support, because the physical model assumes a rigid dipole that spins uniformly with the magnetic field (B_0), while epicardial leads lack this rigidity. Figure 3.14b shows the top of the pole, where the ceramic bearing and a plastic notch holds this transfer arm. The transfer arm is 10 mm thick and is made from Kydex™, similar to the ruler described in Section 3.2.1.

This way, the magnetically susceptible test piece experiences a magnetic torque following equation (2.9) exerted around the attachment point. So, the torque trans-

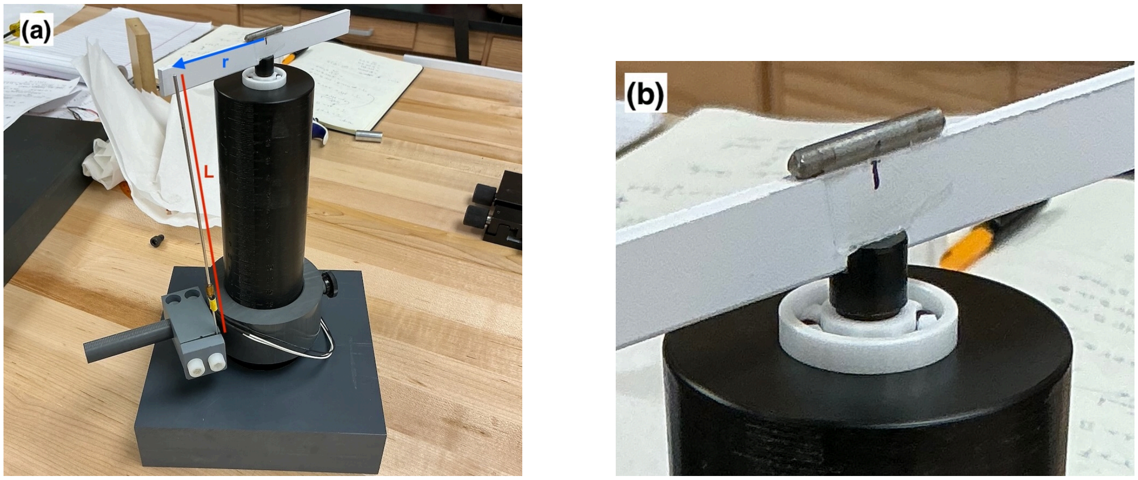


Figure 3.14: Non-magnetic torque fixture designed for horizontal test object orientation. (a) The torque fixture pole provides rigid support. The key adjustable parameters for maximizing measured torque are the sensor's tip position L (red) and the distance from center to sensor along the transfer arm r (blue). (b) Close up of the transfer arm on top of fixture (from bottom to top): magnetically susceptible test object, Kydex™ transfer arm, plastic alignment notch, and ceramic bearing for low-friction rotation.

mitted through the joint becomes a force, and is measured with a sensor located at a known distance (r_{sensor}) from the axis of rotation on the transfer arm, marked with blue in Figure 3.14a. The physics of this device follows that of a lever arm, $\tau_{max} = F_{sensor} r_{sensor}$, since the sensor provide a normal force that opposes that rotation. Because torque is conserved along the rigid body, it provides a practical means to extend the moment arm beyond the base geometry ensuring sufficient space for accurate force sensing while maintaining constant torque transmission.

In this case, the sensor is a thin rod bends following the cantilever beam deflection equations [61]. These equations indicate that the strain at the fixed end increases linearly with both the applied force and the beam length, suggesting a mechanical advantage for measuring small forces.

3.3.1.1 Sensor and Circuit Design for Torque Measurements

The sensor was paired with an amplifier circuit, made following the ideas described in Chiu *et al.*(2016)[62]. For our use case the sensor was fabricated from different non-magnetic materials to assess which would make the better sensor.

The sensor is formed by the beam and a pair of OMEGA® precision strain gauges (SGD-3/350-LY11) [63] with Gauge Factor $G_F = 2 \pm 5\%$ [64]. These strain gauges were epoxied on opposite sides of the rod around 3 to 5 cm from its end, depicted in Figure 3.15. The strain gauges are part of a Half-Wheatstone bridge configuration and are connected to the circuit with soldered jumper wires.

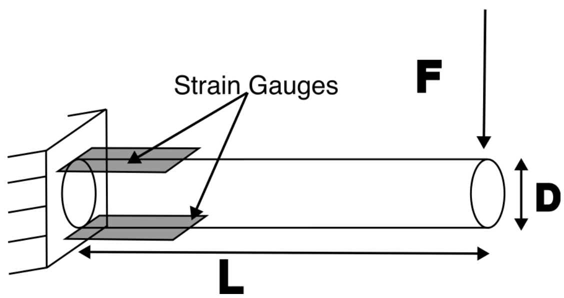


Figure 3.15: Cantilever setup showing the fixed cylinder and strain gauges "sandwiching" the rod at its base.

Figure 3.16 shows a sample set of three beam force sensors which were produced in the laboratory. The beam materials included glass, aluminum, titanium and plastic. Table 3.2 summarizes their characteristics. They were made of similar sizes to test repeatability.

Table 3.2: Characteristics of the sensor rods.

Rod ID	Material	Diameter (mm)	Length (mm)
(1)	Clear plastic	2.9	180
(2)	Aluminum	1.6	182
(3)	Glass	3.2	166.00

On the other hand, the circuit prototype of Chiu *et al.*(2016), shown in Figure 3.17 have four main sections: the Wheatstone bridge, the amplifier, a low pass filter

Figure 3.16: Rods fabricated during the torque experiment design: (1) Clear Plastic, (2) Aluminum, (3) Glass.

and the ground offset potentiometers. The potentiometers were an addition to the original design. A Kirchhoff analysis for the first two sections yielded the transmission function.

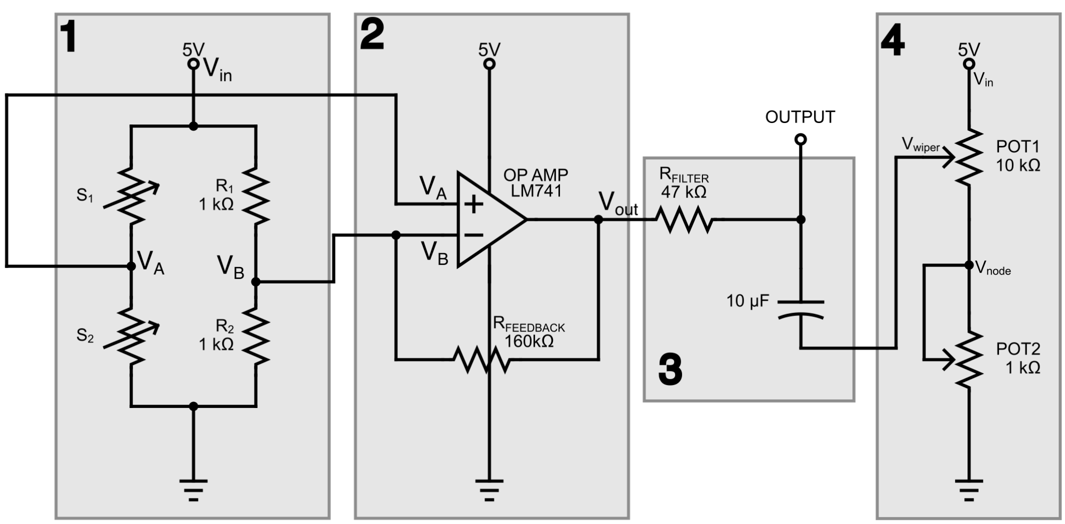


Figure 3.17: Circuit developed for the torque measurement setup, following the design principles of Chiu *et al.* (2016)[62]. Features include, 1. Half Wheastone Bridge Strain Gauge, 2. Amplifier, 3. Low-pass filter, 4. Coarse–fine adjustable reference network.

$$\frac{V_{out}}{V_{in}} = \frac{R_{FEEDBACK}}{R}\frac{S_2 - S_1}{S_2 + S_1} + \frac{S_2}{S_2 + S_1}, \tag{3.7}$$

where $R = R_1 = R_2$, S_1 and S_2 are the strain gauge variable resistance input (or signal coming from the sensor) and $R_{FEEDBACK}$ is the feedback loop needed for amplification. The output voltage of the circuit is driven by the relative difference between the two strain gauges compared to their total resistance. The operational amplifier multiplies this small imbalance by the gain factor $\frac{R_{FEEDBACK}}{R}$, in our circuit $\frac{160k\omega}{1k\omega} = 160$. The bridge sets a baseline through the voltage division $\frac{S_2}{S_2+S_1}$, whereas the amplified differential term $\frac{S_2-S_1}{S_2+S_1}$ determines the sensitivity of the circuit to strain. For the complete derivation see Appendix B.4.

After the instrumentation–amplifier stage, the output passes through a first–order RC low–pass filter composed of a 47 kΩ resistor in series with a 10 μF capacitor to ground, as in Figure 3.17 part 3. The time constant is then $\tau = 0.47$ s, yielding a 0.33 Hz cutoff frequency. Finally, a first–order system with this time constant has a 10 $\rightarrow$ 90% rise time of approximately 1.03 s. In other words, whatever the Wheatstone–op-amp transfer function generates, this RC stage only slows its dynamics rather than altering its steady–state gain. It suppresses high–frequency noise from the strain gauge bridge and smooths fast fluctuations that the rod could not physically produce.

In part 4 of Figure 3.17, the DC reference entering the filter is set by the two potentiometers (analyzed in Appendix B.4.1). Both potentiometers are connected in series, forming a programmable voltage divider from 0 to 5V. The capacitor of RC network simply charges toward whichever reference these pots define. From the voltage-divider derivation in Appendix B.4.1, the node voltage obeys

$$V_{\text{ref}} = V_{\text{in}}\frac{R_2}{R_1 + R_2}, \tag{3.8}$$

so sweeping the fine pot moves R_2 over a narrow range ($\sim$ 0–1 kΩ), which shifts the reference only within a small, controllable window. The coarse pot changes the large R_1 segment ($\sim$ 10 kΩ), expanding or collapsing the overall span. Connections for this circuit included power supply to a Data Acquisition (DAQ) card, 5V output from the DAQ card to the circuit, differential signal lines (plus and minus) for torque readout, DAQ-to-USB interface for real-time logging, and a 4-pin connection for sensor input.

Parts 1–3 of the circuit follow directly from Chiu *et al.* (2016)[62]. Part 4, the coarse–fine potentiometer network, was added to the original design prior to this work. The description presented here represents our best interpretation of its function based on empirical observation and circuit tracing. In practice, the network was found to provide DC offset control sufficient for the purposes of this work.

Understanding both the strain at the sensor and the electrical physics of the circuit enables identification of the measurement device's capabilities. We now proceed to validate our sensor construction against the reference provided by Chiu *et al.* (2016)[62].

3.3.1.2 Smallest Measurable Torque

For torque measurements, the sensitivity of cantilever-based torque sensors is determined primarily by the relationship between the applied tip load and the strain induced at the fixed end of the beam. Figure 3.18 shows a diagram for a thin cylindrical beam of length L fixed at one end and subjected to a transverse point load F at the free tip.

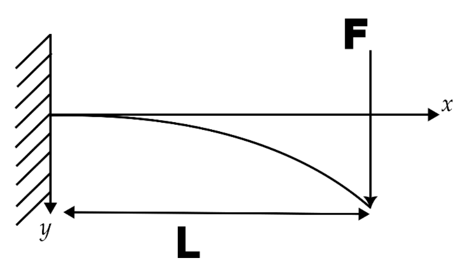


Figure 3.18: Cantilever Beam of length L with concentrated load F at the free end.

Classical beam theory [61, 62] gives the maximum surface strain as

$$\epsilon = \frac{32FL}{\pi E D^3}, \tag{3.9}$$

where D is the diameter, and E is the Young's modulus. This relationship indicates that the strain is inversely related to the Young's modulus and the cube of the diameter and directly proportional to beam length. As a result, longer and thinner rods increase the observable signal by producing greater strain for the same applied force.

Following the geometry used by Chiu *et al.* (2016)[62], a wire of $L = 69.85$ mm and $D = 0.64$ mm fabricated from Nitinol ($E = 66.2\,GPa$) achieved an empirical detection limit of 0.49 mN. Using Equation 3.9, this translates to a geometric sensitivity factor on the order of $L/D^3 \approx 2.66 \times 10^8\,m^{-2}$, isolating the geometry-dependent amplification of strain and therefore serving as a design metric for mechanical sensitivity.

A similar configuration can, in principle, achieve micro-Newton sensitivity when paired with a low-noise amplification circuit, provided the strain gauge and readout maintain linear response and the system noise floor remains below the equivalent strain corresponding to the smallest measurable force.

In the actual implementation, a metallic foil strain gauge bonded to the beam surface, which its position is shown in Figure 3.15 and 3.16, converts the mechanical strain into a measurable change in electrical resistance. For small elastic deformations, the fractional change in resistance is linearly related to strain through the gauge factor G_F [65, 66, 67],

$$\frac{\Delta R}{R} = G_F\,\epsilon, \tag{3.10}$$

where G_F is typically in the range of 2.0–2.2 for constant foil gauges. Combining Equations 3.9 and 3.10 gives the total resistance change due to an applied tip force.

$$\Delta R = R_0\,G_F\,\frac{32FL}{\pi E D^3}. \tag{3.11}$$

This resistance change is converted to a differential voltage when the gauge is part of a Wheatstone bridge circuit. For a half-bridge configuration, where two active strain gauges form adjacent arms of the bridge, the small-signal output voltage under excitation V_{exc} is approximately [65, 68]

$$V_{out} \approx \frac{V_{exc}}{2} G_F \, \epsilon, \tag{3.12}$$

assuming perfect linearity, little temperature drift, and perfect bridge balancing. The direct proportionality between applied force and bridge voltage may be obtained by combining Equation 3.9 into 3.12,

$$V_{out} = 16 \, G_F \, V_{exc} \frac{L}{\pi E D^3} \, F. \tag{3.13}$$

In practice, the smallest detectable force is governed by the combined noise of the gauge, bridge, and amplifier system. If the noise-equivalent output voltage V_{noise} corresponds to a minimum detectable strain $\epsilon_{min} = \frac{2V_{noise}}{(V_{exc} G_F G)}$, then the force detection limit follows from Equation 3.9 as

$$F_{min} = \frac{\pi E D^3}{32 L} \, \epsilon_{min}. \tag{3.14}$$

Theoretically, the sensor can resolve forces in the milli-Newton range by optimizing the geometric ratio L/D^3, minimizing circuit noise, and selecting materials with moderate elastic modulus (e.g., aluminum or Nitinol). Using the parameters reported by Chiu *et al.* (2016), such as bridge noise of $\sigma_V = 3$ mV and gain of 100, Equation 3.14 yields a minimum detectable force of $F_{\mathrm{min}} = 0.161$ mN (see Appendix B.3 for sample calculations). This theoretical floor assumes ideal amplification and no parasitic noise; in practice, real circuit noise raises the empirical detection limit, as Chiu et al. report 0.49 mN approximately three times above the theoretical floor of 0.161 mN, consistent with real-world imperfect conditions.

For the present design, the limit of detection is expressed in terms of torque. This is obtained by multiplying the minimum detectable force by the transfer arm length

r, as described in Sections 3.3.1 and 3.3.1.1. Using the parameters summarized in Table 3.3, the minimum detectable force is $\approx$ 0.094 mN, corresponding to a torque resolution of $\sigma_\tau = 4.70$ μN·m under ideal conditions. The experimental test results for glass and plastic are reported in Appendix E with aluminum results on Section 4.3.1, both following the calibration methodology of Section 3.3.1.3.

Table 3.3: Summary of the sensor and circuit parameters used for force and torque resolution estimation. Values are set for an aluminum-based sensor, paired with the circuit explained in Section 3.3.1.1

Parameter	Symbol	Value
Young's modulus	E	68 GPa
Wire diameter	d	1.6 mm
Beam length	L	182 mm
Bridge noise	σ_V	0.0005 V
Excitation voltage	V_{exc}	5 V
Gauge factor	G_F	2
Gain	G	160
Min. Force	F_min	0.094 mN
Min. Torque	τ_min	4.70 μ N·m

These restrictions can be decreased by averaging (σ scales as $1/\sqrt{N}$), by increasing mechanical leverage (slope grows almost linearly with cantilever length), or by enhancing the bridge/amplifier noise performance. For quasi-static measurements, the limits of detection may be moved from tens of mg into the single-mg or sub-mg range using a combination technique (longer/thinner cantilever + averaging + low-noise amplifier); dynamic measurements will be constrained by bandwidth and 1/f drift.

3.3.1.3 Sensor Calibration

Prior to any torque measurement, each sensor or rod from Table 3.2 was tested or calibrated to accomplish two things: ensure the use of the most sensitive sensor and to acquire the conversion factor or slope of voltage to weight. Additionally, the orientation and position of the rods was tested to ensure optimal values on these parameters for torque measurements.

For calibration in the laboratory, the sensor was set in a cantilever position as shown in Figure 3.19, so the bar faces horizontally and a small cone-shaped basket made out of tape was attached to hold increments of water. Water was added in μL increments (equivalent to milligrams) simulating torque loads from 0 to 800 mg. Initial increments of 10 to 200 μL covered the expected force range of epicardial leads weight, followed by additional 50 μL increments to demonstrate linearity and sensor stability.The calibration took about 10 minutes to weight all 800 μl. Then it was cleaned up and tried again in different days for consistency.

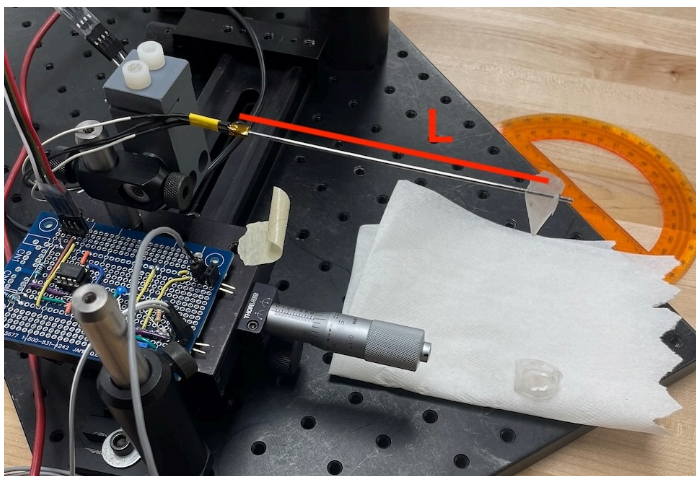


Figure 3.19: Circuit and sensor. The sensor was clamped with a custom made PVC holder to act as a cantilever beam. A weigh boat at length L was added, then a pipette was used to add water and calibrate voltage changes with weight.

The sensor voltage vs. water mass was plotted and fit to a line to determine the calibration constant. This was done using python `Pandas` library in a Jupyter notebook and is publicly available [60].

To find the best position for the sensor, the sensor was reoriented from its cantilever configuration to an upright (vertical) position on the main fixture. A pair of neodymium magnets was positioned above the ceramic bearing to provide a localized magnetic field, and a platform angled at 45° was used to position the sensor relative to the field source. The radial distance from the field source (r) and the effective contact position along the rod length (L) were varied independently, as illustrated in Figure 3.20. Radial measurements were performed by translating the sensor inward from approximately 90 mm to 50 mm relative to the magnetic source, while axial measurements were conducted independently by varying the contact point along the rod length.

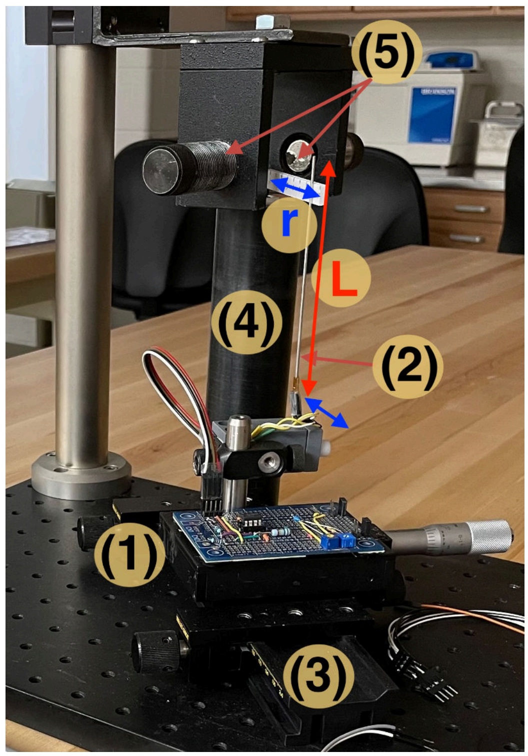


Figure 3.20: (1) Circuit and (2) sensor mounted on a (3) translation stage angled 45° relative to the (4) fixture pole. In this configuration, the sensor is held upright to vary the contact point with the transfer arm in two ways: axially along the sensor length L (red) and radially along the transfer arm length r (blue). A pair of (5) neodymium magnets is positioned above to generate a constant magnetic field at the bearing.

Results from this calibration and optimization are detailed in Appendix E. The aluminum-based sensor was selected for all subsequent measurements due to its reliability. Based on the calibration results, the sensor response depended on the point of force application relative to its geometry: loading closer to the strain gauges reduced the output signal, while loading closer to the rotation axis increased it. Accordingly, for all subsequent experiments the sensor was positioned 5 cm from the rotation axis, with its tip used as the contact point with the transfer arm.

3.3.2 Laboratory Methodology and Setup - Torque

With the transfer arm length and the sensor position optimized and confirmed, the method for the experiment goes as follows: an increasingly stronger magnetic field was applied to the test object, initially positioned such that it makes a approx 45° angle with the magnetic field, following Equation 2.9a. The magnetic field is measured and recorded using the AlphaLab Inc. Gaussmeter. The test object, experiences a torque given by Equation 2.9b, transferring it to the arm. There, it is stopped by the sensor described in section 3.3.1.1, which measures voltage change proportional to the force it receives from the rotating arm. This voltage reading collection was done using a DAQ card and a computer.

Figure 3.21 shows the experimental apparatus. In it each relevant piece is labeled and numbered: (1) and (2) are the power supply and Helmholtz coil generating the magnetic field. (3) the gaussmeter probe, which is connected to a computer. (4) Our circuit. (5) Is the DAQ card, also connected to the same computer. And (6) the main torque fixture with sensor, transfer arm and sample.

To assess different torque strengths, two test objects of the same material and different lengths were considered. First a long steel solid cylinder of about 20 cm length and 5 mm diameter was used; this way a strong dipole moment could be induced in it, serving as a visual confirmation that the system is working. Second, a smaller cylinder of about 2 cm long and 5 mm diameter was used; with it we attempted to measure the smallest magnetically induced torque possible. Following the same logic as in the translational experiment this torque scales proportionally.

Figure 3.22 shows two different setups for magnetic field generation. This was done to ensure a wide range of magnetic field strengths. Initially, a Helmholtz Coil

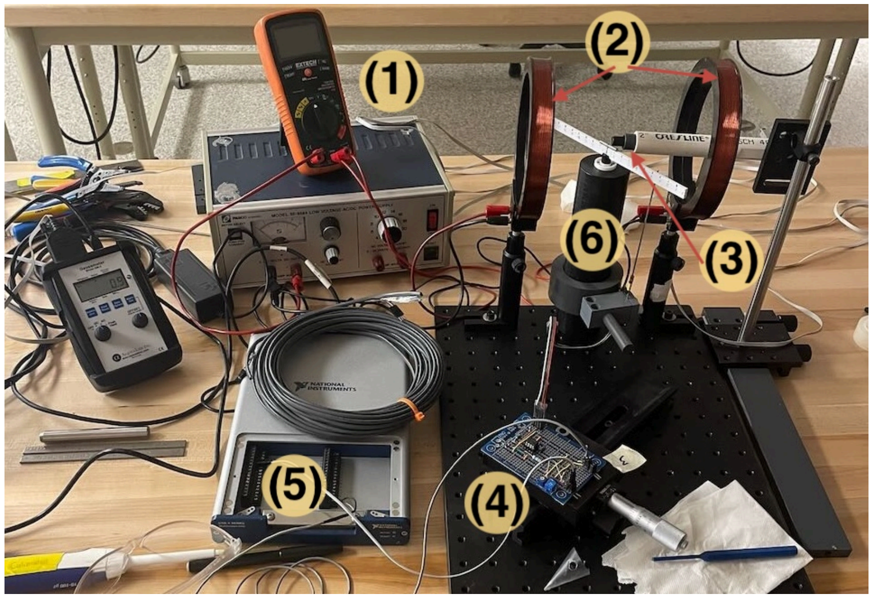


Figure 3.21: Experimental setup for torque measurements. Includes magnetic field generation and measurement devices, (1) through (3). (4) is the sensor circuit, (5) is the DAQ system connected to a laptop. (6) is the main device where the test object sits. The object is positioned at 45° to the static field, corresponding to the angle of maximum torque for a paramagnetic material.

EM-6722 from PASCO Scientific (Figure 3.22a) created a controlled, nearly uniform static magnetic field that could be varied in strength and observed in real time. This Helmholtz coil consist of two identical coils, each wound with 200 turns, they were wired so current flowed in the same direction. They had an inner radius of about 10 cm and outer radius of 10.6 cm, and they sit on a optical base that lets us adjust the center-to-center spacing close to where the field is most uniform (about one diameter apart). The coils used a maximum DC current of 2 A, which we supply with the PASCO Scientific SF-9584 DC power supply to increase the magnetic field incrementally; as current increases, so too does the magnetic flux density between the coils, allowing us to observe corresponding torque changes.

Figure 3.22b shows a second setup, a set of neodymium magnets. The magnets were mounted symmetrically on a rigid mechanical frame, with their magnetization axes oriented to produce a static magnetic field across the region occupied by the torque arm. Threaded lateral adjustment screws and a central support allow con-

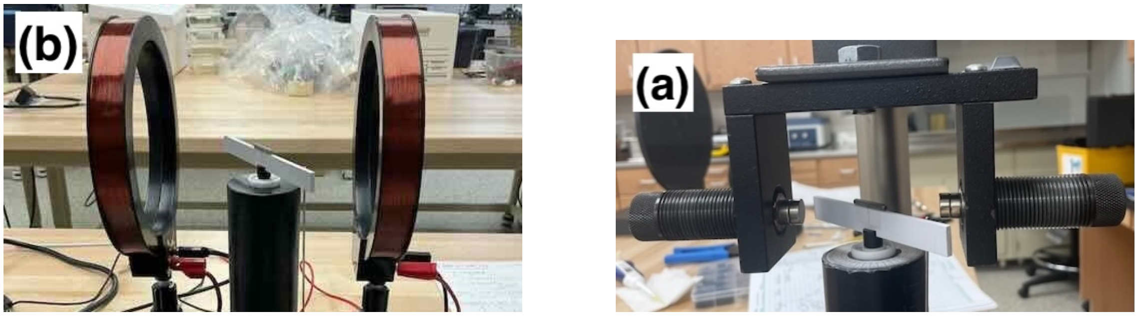


Figure 3.22: Comparison of the two magnetic field generation setups used for torque measurements: (a) Helmholtz coil generates a controllable and uniform static magnetic field at the coil center. The coils are driven with equal currents in the same direction to superpose their magnetic fields at the center. (b) Four neodymium magnets symmetrically mounted on a rigid frame to generate a static magnetic field across the torque arm. Adding more magnets increases the field at the center.

trolled positioning and alignment of the magnets relative to the rotating element. Each magnet came magnetized through their thickness, with nominal dimensions of 1/4 in thickness and 5/16 in outer diameter.

3.3.3 In Clinic Measurement - Torque

The custom torque fixture was assembled, integrating the torque sensor with lightweight balanced lever arms on the patient couch, as shown in Figure 3.23. The circuit and data acquisition (DAQ) card were set up and connected adjacent to the acquisition computer in the MRI control room. Similar to the gaussmeter probe, a 10,m extension cable was routed from the sensor in the MR suite through the waveguide/port in the wall into the control room.

Once fully connected, the sensor input was functionally tested by applying gentle manual pressure against the lever arm, confirming that deflection produced a clear signal response before entering the scanner room. The assembled fixture was aligned to scanner isocenter using the built-in room lasers, with the zero-torque reference position permanently marked using tape on the couch surface for positional reproducibility across trials.

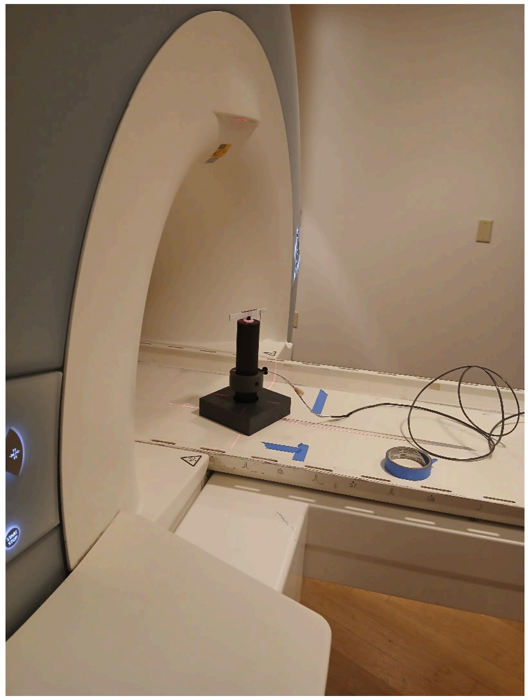


Figure 3.23: Torque devices at the bore entrance ready to enter the MRI scanner. A 10m extension cable allows for communication with circuit at control room.

Trials commenced with cylinder-shaped samples of non-magnetic stainless steel (316L) and titanium mounted on the lever arms. The rods were oriented to maximize potential rotational response relative to the fringe field gradient, and the couch was incrementally advanced toward the bore entrance to expose the samples to increasing field strength and gradient.

3.4 Experimental Setup Summary

This chapter outlined the development and validation of three custom non-magnetic test fixtures designed to quantify MRI-induced translational forces and torques on post-surgical epicardial pacing leads, in accordance with ASTM standards [7, 8, 9, 10]. Each fixture was tested in a low-field laboratory environment (∼30 G) using scaled ferromagnetic samples, before testing non-magnetic sample. Initial clinical measurements on a Siemens Magnetom Trio™ 3T MRI scanner at CUMC–Bergan Mercy were performed following a general experimental plan.

The first fixture focused on characterizing the static magnetic field strength and gradient. In the lab fields were generated via a custom air-core solenoid and probe was positioned using a translational stage with servomotors for sampling. In the clinic, the platform and probe were mounted on the patient couch, aligned to isocenter and translated axially providing volumetric field maps essential for force/torque predictions.

The second fixture employed a pendulum-based deflection apparatus (ASTM F2052-15 [7]) to assess forces through angular displacement balanced against gravity. In the lab, we used the low field solenoid and deflection angles were measured and converted to forces. Additionally, we established the lower limit of resolved forces down to millinewtons, which were ultimately constrained by the shape and accuracy of visual inspection. In the clinic, the fixture was positioned on the patient table and incrementally advanced into the bore entrance. Deflections on cylinder-shaped samples (316L stainless steel and titanium) and leads were recorded manually.

Finally, the third fixture used a pole and low-friction bearing with transfer arms, algonside a strain-sensor to quantify torque. In the lab, sensors were calibrated by applying known weights, establishing linear force-voltage relationships. Then, the smallest measurable torque was analyzed based on rod stiffness and signal resolution. The setup included solenoid/neodymium fields, a DAQ system, with LabVIEW for data acquisition. In the clinic, the fixture was aligned on the couch, the test objects were oriented to maximize response and torque on non-magnetic samples was documented.

With these setups the transition from laboratory to clinical is completed. Workflow layout, instrument operation validation, and fundamental physics has been established. Furthermore, a preliminary clinical data on epicardial leads was captured.

Next, Chapter 4 shows all findings related to each experiment in laboratory and clinical settings, while Chapter 5 discusses the validity of our findings and post-clinical notes on future experiment improvement.

Chapter 4

Results

Usually temporary epicardial leads are labeled MRI-conditional at best, leaving uncertainties on whether patients with these leads can undergo a MRI scan [28, 30]. The physical effects that these leads can potentially experience include, magnetically induced forces and torques, and electrical currents produced by gradient encoders [18]. Chapter 2 presented the fundamental concepts and safety considerations of MRI, while Chapter 3 detailed the experimental design, instrumentation, and methodologies used in this study.

In this chapter, the results obtained from each experiment are presented and analyzed. Each experiment incorporates both laboratory and clinical measurements. In the laboratory setting, we conducted measurements under controlled conditions to characterize behavior and inform expected clinical outcomes. Clinic measurements were used to validate the methods described on Chapter 3. This distinction divides the chapter in three experiment each divided in two with their proper result presentation and interpretation.

4.1 Magnetic Field Mapping

The first experiment consists of mapping the magnetic field in both laboratory and clinical environments, as described in section 3.1. Equation 2.7 predicts that the magnetically induced force experienced depends on both the field strength and its spacial gradient. As such, we present these quantities for the solenoid described in Section 3.1.2 and for the 3 T field from the clinical MRI scanner.

4.1.1 Mapping for a Solenoid in the Laboratory Setting

The main objective of the first experiment was to find the spatial gradient of the static magnetic field produced by a solenoid using a gaussmeter. A constant current of 4 A was applied to the air-core solenoid, and measurements were taken at three positions across its circular face to check for homogeneity. The gaussmeter probe was translated along the solenoid axis, from 20 cm outside the aperture toward and away from the bore.

Figure 4.1 shows three plots from the field mapped. First, Figure 4.1a shows the magnetic field flux B in gauss as a function of time as the probe moved toward and away from the solenoid front face. Figure 4.1b shows the magnetic field B as a function of z-position for the forward (towards) and backward (away) passes in dashed and solid lines respectively, where z was estimated by linear interpolation from video-synchronized timestamps recorded every centimeter. The three colors in the plot correspond to the probe locations for each trial: center, east, and southwest. The on-axis magnetic field (center) B reached a maximum strength of 182.2 G at 0.25 cm from the solenoid face. Measurements at both off-axis positions (east and southwest, 10 cm away from center at 0° and 225°), showed maximum values of 165.1G at 0.35 cm, with the expected radial fall-off from the axis. The close agreement between the east and southwest edge measurements confirms azimuthal field homogeneity at r=10 cm.

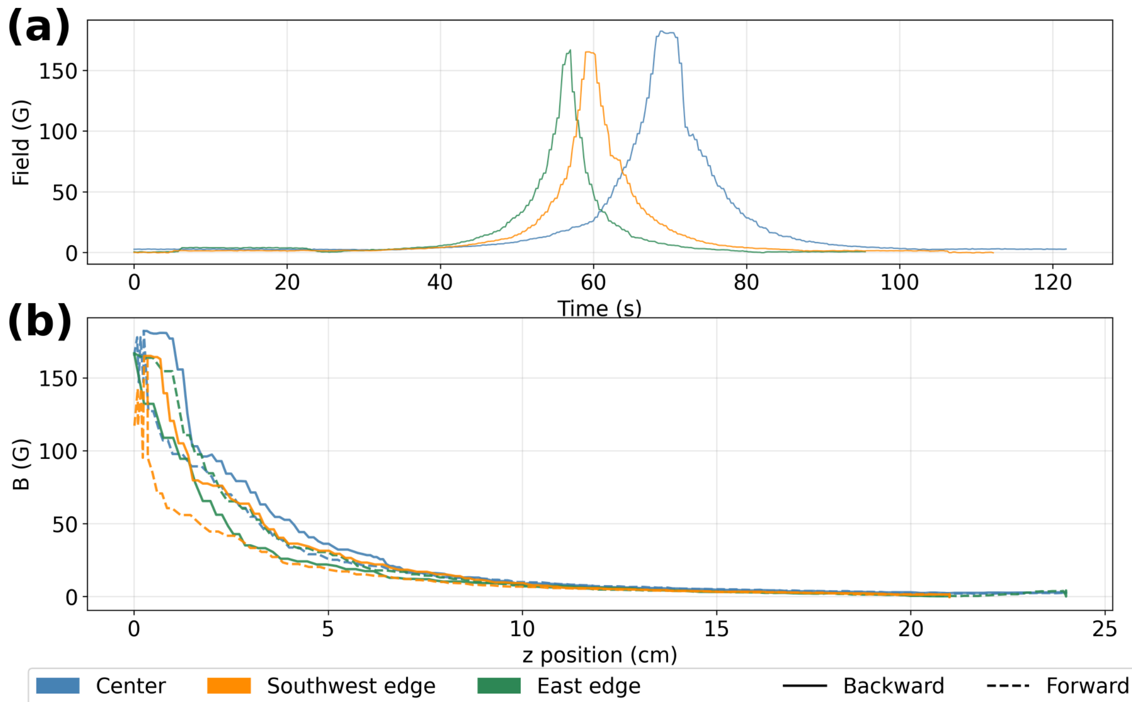


Figure 4.1: Magnetic field measurements plotted against time and axial displacement. (a) Shows the complete measurement in time as the probe travels. (b) The probe approaches the solenoid. (c) The probe retreats from the solenoid.

A small discrepancy between the forward and backward passes is visible in the 0–10 cm region across all three positions. This offset does not necessarily reflect hysteresis but rather a systematic difference in probe speed between the inward and outward trajectory. In the 0–10 cm zone, the average probe speed differed between passes by 9–35% depending on position (most pronounced at the east edge: 0.72 cm/s inward vs. 0.97 cm/s outward), which introduces a small z-assignment error through the linear interpolation of video-synchronized timestamps. The resulting position uncertainty is estimated at sub-centimeter scale and does not affect the qualitative field characterization presented here.

Figure 4.2 shows the spatial gradient computed with central difference (Equation 3.1), and the product $B\frac{dB}{dz}$ for both forward and backward passes. Both quantities increased as the probe approaches the entrance of the magnet (z=0), consistent with the behavior shown in Figure 2.2. The traces that show a drop toward zero near z = 0 is an artifact of the probe pausing at that position during data collection. The east edge (green) trace do not exhibit this artifact. The probe did not enter the solenoid bore in any run, and the interior field profile is not captured here, as the mapping objective was to characterize the fringe field decay outside the solenoid.

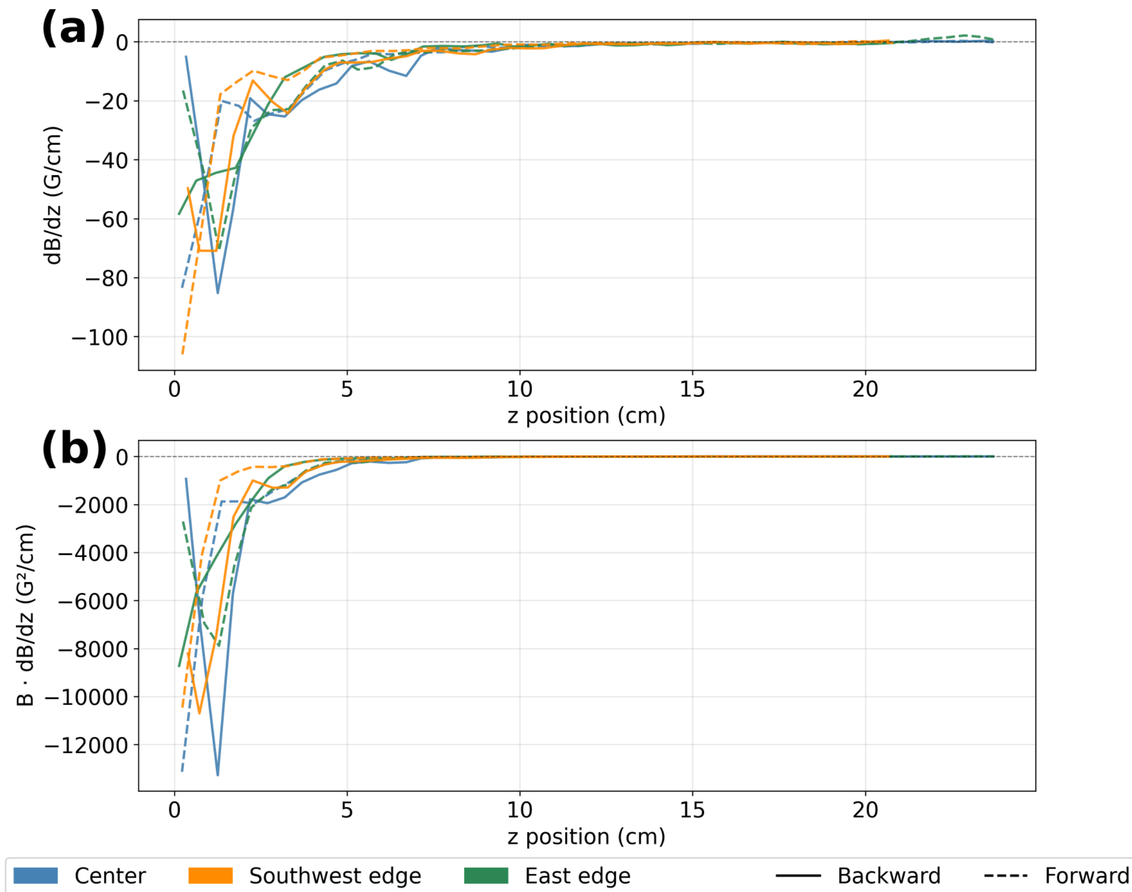


Figure 4.2: Computed magnetic field gradients and the $B\nabla B$ product used for translational force characterization.

The decision not to map the interior of the solenoid bore was intentional, as this experiment was primarily done to validate the measurement approach and establish physical intuition before moving to the clinical setting. The following section presents the field characteristics of the 3 T MRI scanner at CUMC–Bergan Mercy, mapped using the same methodology.

4.1.2 Mapping for 3T Siemens Magnetom™ MRI Scanner

In the clinic, we used the mapping platform described in Section 3.1.1. Its coordinates, shown in Figure 4.3, marked positions within the axial plane (X-Y) as the probe moved along the z-axis, as it would during a clinical scan. Each number represent a radial distance along the mapping platform while each letter corresponds to a position along the angular direction. Color-coded dots represent the positions selected for this experiment, from top to bottom: A6, A3, 00, D3, G3, G6. As in the solenoid experiment, the probe was translated along the patient couch from a starting position near the bore to approximately 1 m away and back as it measured the magnetic field strength.

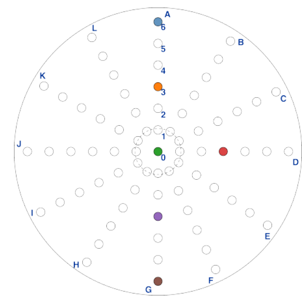


Figure 4.3: Coordinated grid used for mapping MRI scanner B.

Figure 4.4 shows the three relevant quantities (B, dB/dz, $B \cdot dB/dz$) for the coordinate G6 (lowest along the central axis, brown in Figure 4.3) as a function of axial position (z). Each plot contains four lines. Line style (solid vs. dashed) denotes travel direction, away from the bore (backward pass) or toward the bore (forward pass). Color shade (full vs. shaded) distinguishes sub-runs A and B, a division introduced in Section 3.1.3 to address sampling resolution limitations. Figure 4.4a shows the field strength (B) in Tesla. The gradient (Figure 4.4b) from sub-runs A (solid color) shows a more pronounced saw-tooth artifact, reflecting larger inter-point differences, compared to its sub-runs B (shaded) counterparts. Finally, $B \cdot dB/dz$ is presented in Figure 4.4c.

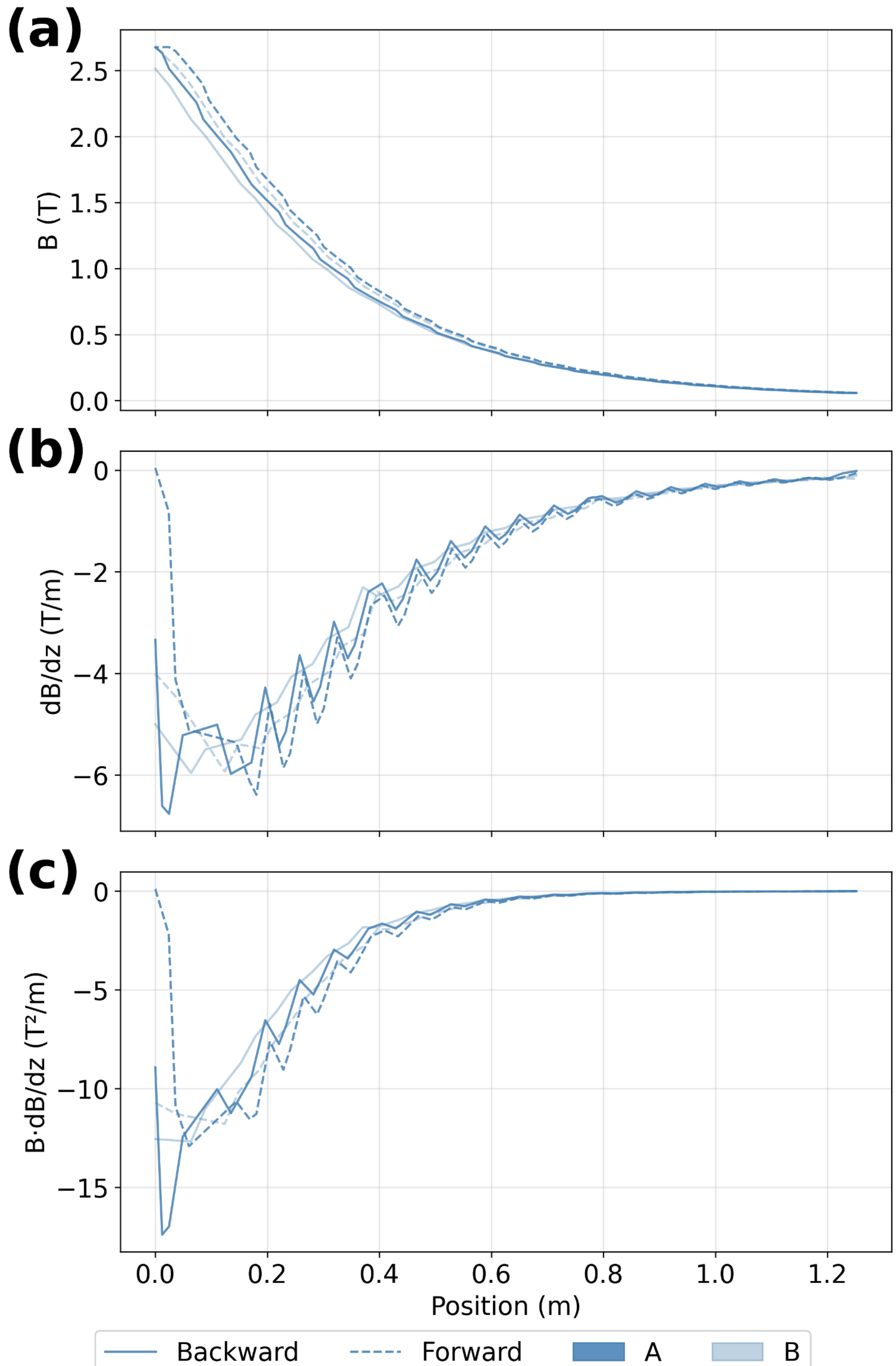


Figure 4.4: Magnetic field with corresponding gradient and product along the measurement path after separating repeated measurements into two independent sequences.

Plots like this one, for each of the points highlighted on Figure 4.3 are presented on Appendix D. All six points on the axial plane exhibited the expected monotonic increase in field strength as the probe approached the bore entrance, with B decreasing by approximately 2.5–2.9 T over the full 1252 mm couch travel. At the bore entrance, the peak field strengths ranged from 2.51 T at G6 to 2.95 T at A6 (highest along the central axis, blue in Figure 4.3), consistent with the scanner's 3 T isocenter field attenuating into the fringe region sampled by the fixture.

The spatial gradient $|\nabla B|$ ranged from 5.96 T/m at G6 to 10.39 T/m at A6, and the force-relevant product $|B\nabla B|$ ranged from 12.68 T^2/m at G6 to 22.95 T^2/m at A6. The near-axis coordinates (00, G3, G6) agreed within approximately 2% in peak B, confirming lateral homogeneity in that region and consistent with the homogeneity observed in the solenoid mapping of Section 4.1.1.

G6 was selected as the primary reference coordinate based on the practical constraints of the clinical session. While it yields the lowest $|B\nabla B|$ product among all sampled coordinates (12.68 T^2/m), it serves as a representative baseline for demonstrating the measurement method in the clinical environment. The field conditions $|B_0|$= 2.67 T and $|\nabla B|$= 5.93 T/m are used as the test location input for translational force analysis in Section 4.2, where results from all six coordinates are compared. This conditions were picked from the data of G6 (same position as our pendulum), Forward pass (matches translational test geometry), sub-B (cleaner gradient sample).

Together, these field and gradient maps provide the spatial landscape of the magnetic field to determine the translational force measurements meaning.

4.2 Translational Force

Translational force measurements follow directly from the field maps of Section 4.1. Using the pendulum fixture described in Section 3.2, the deflection angle of a test object under magnetic loading is recorded as the patient couch approaches the bore, yielding the force ratio FR = $\tan(\alpha)$ between the magnetically induced force and the object's weight.

4.2.1 Translational Force Measurements in the Laboratory Setting

In the laboratory, the pendulum fixture was held at a fixed position while the solenoid current was varied to sweep the field product $|B\nabla B|$ across a range of values, to replicate the spatial variation of field strength encountered along the bore axis in the clinic. The objective was to validate the force model of Equation 3.6 by calculating the magnetic susceptibility χ_m of a ferromagnetic reference sample from the measured deflection angles, following the method described in Section 3.2.2.

Figure 4.5 shows a plot of the observed FR, $[\tan(\alpha)]$ against the magnetic field properties (B∇B) in T^2/m, which were determined from mapping the magnetic field. Figure 4.5a is a representation of Equation 3.6, it shows a linear fit with a slope of 160 ± 8 and intercept 0.0160 ± 0.003. Figure 4.5b, shows the residuals for the linear fit suggesting this fit is appropriate.

Following Equation 3.6, it can be quickly inferred that the magnetic susceptibility of the our sample is given by

$$\chi_m = m(\mu_0 \rho g), \tag{4.1}$$

where m is the slope found with Figure's 4.5a fit, μ_0 the vacuum permittivity, ρ the material's density and g the acceleration due to gravity. The test sample was an unidentified ferromagnetic cylinder of measured dimensions and mass in Table 4.1. We obtained a density of 8600 ± 60 kg/m^{3}and a magnetic susceptibility $\chi_m = 17.3 \pm 0.1$. For reference, carbon steels such as AISI 1018 have a density of approximately 7870

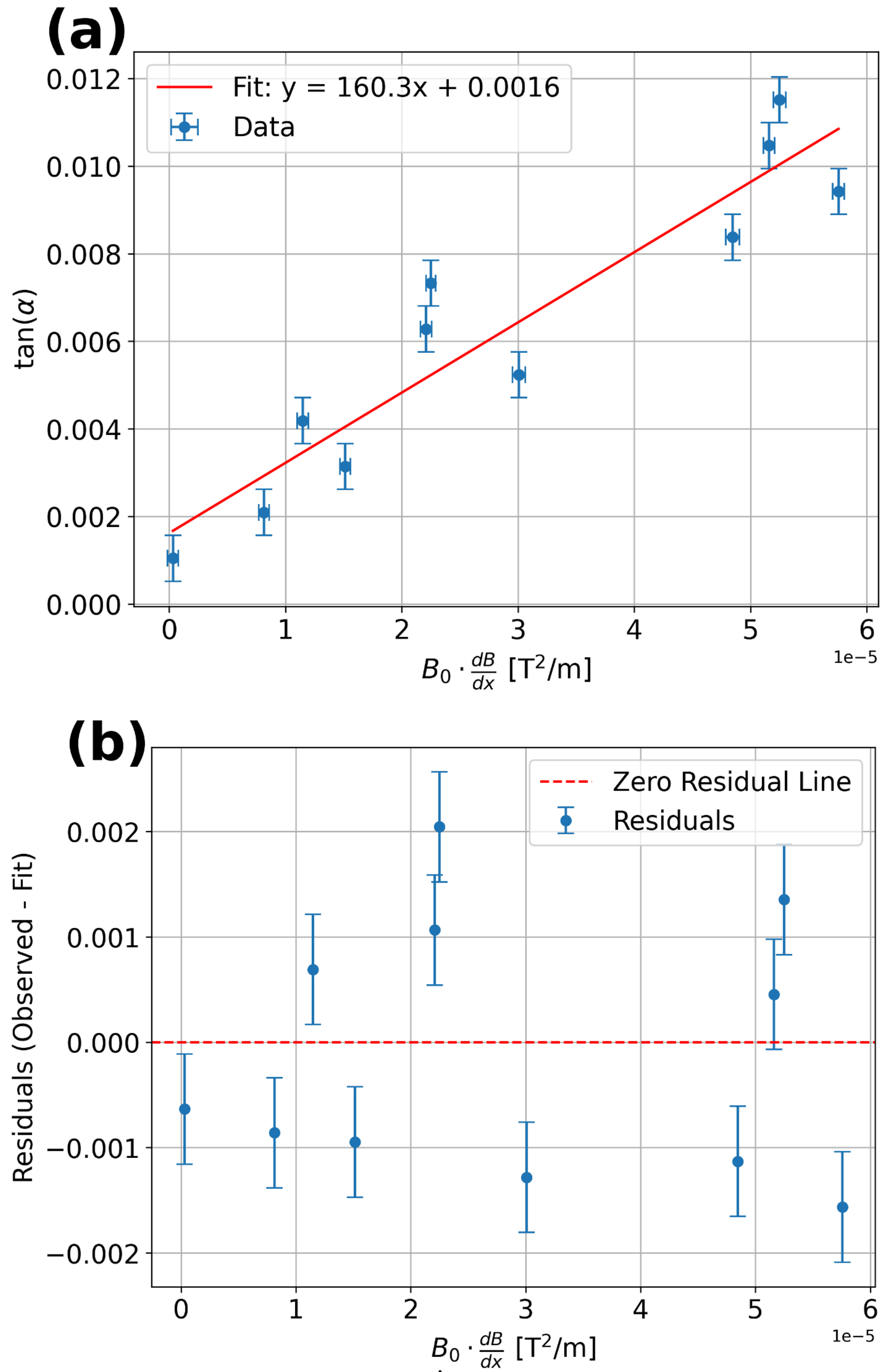


Figure 4.5: Curve fit and residual analysis for the ferromagnetic reference sample (Figure 3.10a) under the laboratory solenoid. (a) Measured $\tan(\alpha)$ versus applied field product $B_0 \cdot \frac{dB_0}{dz}$, with linear fit (Equation 3.6). (b) Residuals from the linear fit. Error bars were calculated from the uncertainty of the ruler for the distance displaced and the reported uncertainty of the gaussmeter then propagated accordingly.

kg/m^3 [69], while pure nickel is approximately 8857 kg/m^3 (0.32lb/in^3)[70], suggesting the sample is a nickel-rich alloy or a high-density ferromagnetic compound of unknown grade. Additionally, when brought near to a neodymium magnet, the sample was attracted and showed no remnant magnetization upon separation.

Table 4.1: Measured and calculated characteristics of our ferromagnetic object.

Property	Value	Unit
Length	6.00 ± 0.01	mm
Diameter	3.00 ± 0.01	mm
Mass	0.3648 ± 0.0001	g
Volume	$(4.24 \pm 0.03) \times 10^{-8}$	m^3
Density	8600 ± 60	kg/m^3
Mass magnetic susceptibility $[\frac{\chi}{\rho}]$	$(2.01 \pm 0.05) \times 10^{-3}$	m^3/kg
Magnetic susceptibility	17.3 ± 0.1	unitless

Importantly, in Figure 4.5a, the FR $\tan(\alpha)$ is in the range of 0.002 to 0.012. According to ASTM 2052 [7], the FR threshold for MRI safety indicates 1 or, equivalently, a deflection angle of 45° putting our laboratory measurements far from that threshold. This confirms that the pendulum fixture resolves deflection angles at the sub-degree scale and that the ferromagnetic reference sample behaves as expected under the force model of Equation 3.6. Therefore, the method is validated for application in the clinical setting, where the force-relevant field product $|B\nabla B|$ at the bore entrance (12.68–22.95 T^2/m, Section 4.1.2) exceeds the laboratory conditions by several orders of magnitude. In that same line, the test objects of clinical interest are the temporary epicardial leads rather than a ferromagnetic sample, and their considerably lower magnetic susceptibility keeps the force response within the linear regime assumed by Equation 3.6.

4.2.2 Translational Forces Observed in a 3T Siemens Magnetom™ MRI Scanner

In the clinic, the four test objects or Materials Under Test (MUTs) from section 3.2.1.2 were tested following the ASTM F2052-15 [7] pendulum protocol at the bore entrance of the 3T Siemens Magnetom™ scanner. A 1010 mm pendulum was used, with displacement uncertainty $\delta z = 0.005$ mm. The ASTM F2052-15 acceptance criterion is a force ratio $\mathrm{FR} = \tan(\alpha) < 1.0$ (equivalently, $\alpha < 45°$), which corresponds to the magnetically induced displacement force being less than the gravitational weight of the device [7].

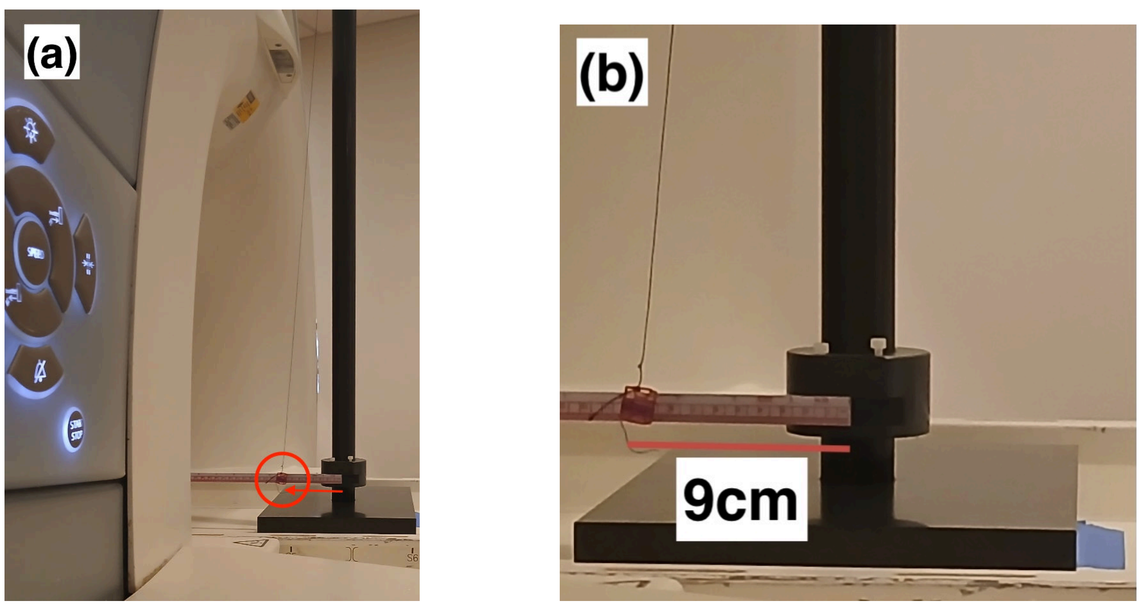


Figure 4.6: Temporary epicardial lead at the bore entrance showed a displacement of 9 cm from the induced translational force. (a) Shows the position of our fixture with respect the entrance of the scanner. (b) A zoomed-in view of our measurement.

Figure 4.6 shows the epicardial lead mounted on the pendulum fixture at the bore entrance. A visible displacement of up to 9 cm was recorded for the Medtronic STREAMLINE™ 6492 unipolar epicardial lead, making it the most strongly deflected device among those tested. Figure 4.7 shows the FR as a function of z-axis position along the bore axis for our MUTs. The force ratio increases monotonically as the

MUT approaches the bore entrance (smaller z), consistent with the increasing $B|\nabla B|$ product expected from the magnetic field map. Critically, all MUTs remained well below the FR = 1.0 threshold across the entire measured range. The Medtronic STREAMLINE™ 6492 unipolar temporary epicardial lead (a) and the 316L stainless steel reference sample (b) reached the highest force ratios, with maximum mean values of $\tan(\alpha) = 0.1666$ ($\alpha = 9.5°$) and $\tan(\alpha) = 0.1490$ ($\alpha = 8.5°$), respectively. The titanium straight wire (c) and the titanium coil (d) showed substantially lower deflections, with maximum mean force ratios of $\tan(\alpha) = 0.0068$ ($\alpha = 0.4°$) and $\tan(\alpha) = 0.0098$ ($\alpha = 0.6°$). The titanium straight wire showed minimal deflection. However, trials on the straight wires (b) and (c) were measured only once and should be repeated.

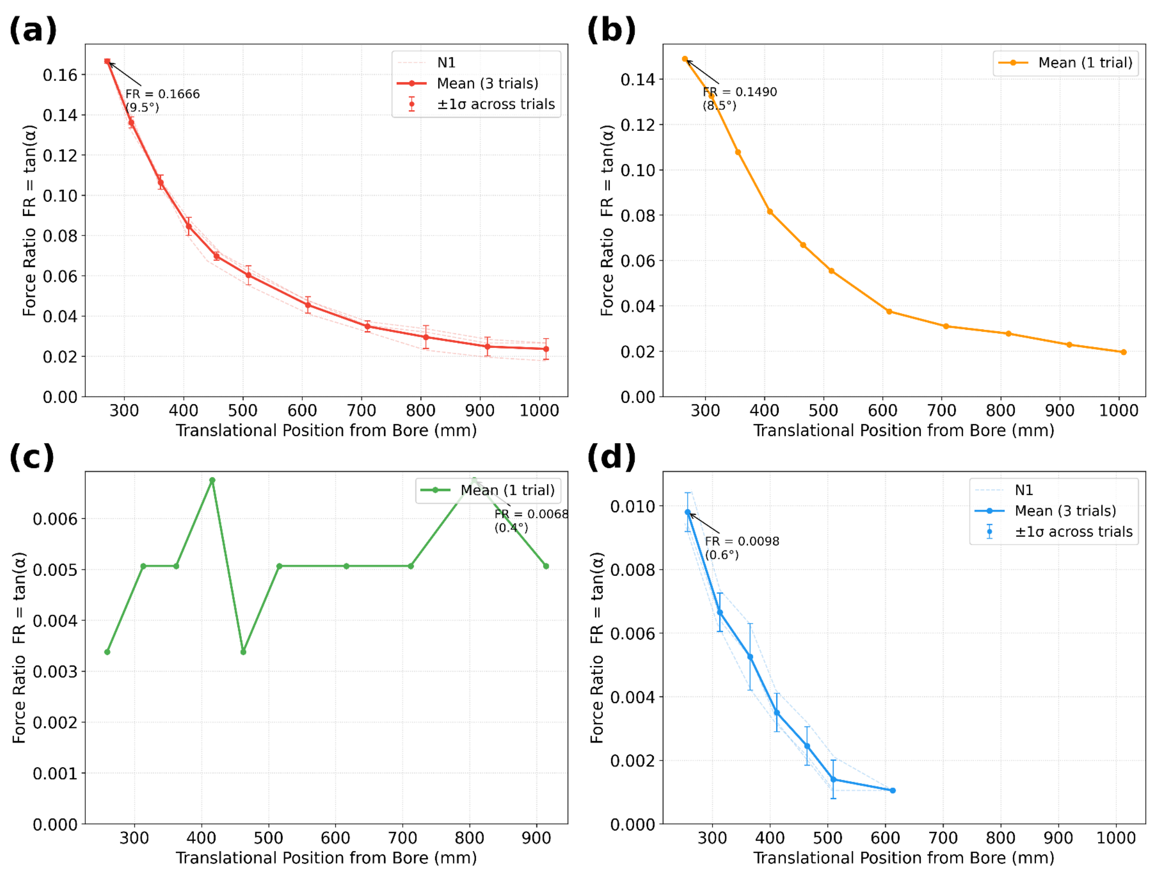


Figure 4.7: Magnetically Induced Displacement Force – ASTM F2052-15[7]. Force Ratio FR = $\tan(\alpha)$ vs. z-axis Position. (a) Unipolar Epicardial Lead. (b) 316 Stainless Steel straight wire. (c) Titanium straight wire. (d) Titanium coiled wire.

Figure 4.8 summarizes the peak FR per MUT. All four MUTs passed the F2052-15 criterion; however, it is notable that the unipolar epicardial lead, the primary device of clinical interest, produced a force ratio roughly ten times larger than the titanium reference samples. The 316L stainless steel, included as a higher-susceptibility control material, yielded a peak force ratio comparable to the unipolar lead, as expected given its known magnetic properties. Notably, the coiled geometry of the titanium wire (d) produced larger deflections than their straight counterpart (c), suggesting that conductor geometry contributes meaningfully to the force response independently of material susceptibility. The epicardial lead was also coiled around the basket perpendicular to the field, which could have increased its interaction, therefore additional measurements are needed.

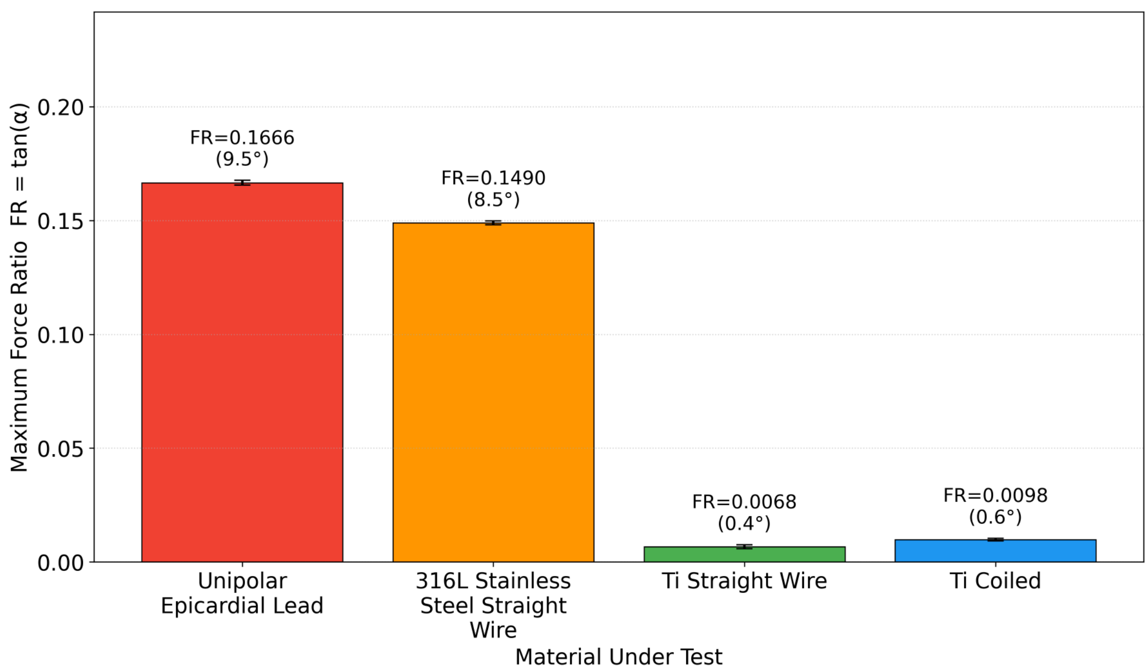


Figure 4.8: Summary of the measured maximum magnetically induced FR per MUT – ASTM F2052-15[7]

Figure 4.9 presents two complementary views of the spatial distribution of translational loading across the mapping fixture. Panel (a) shows the measured $|B\nabla B|$ product at each coordinate from the forward pass field map (Section 4.1.2). Panel

(b) shows the predicted FR at each coordinate for all four MUTs, obtained by scaling the measured FR at the G6 test location linearly with $|B\nabla B|$, following the proportionality of Equation 3.6.

Coordinate A6, nearest to the bore top, yields the highest $|B\nabla B|$ product (21.22 T^2/m) and consequently the highest predicted FR for all MUTs, approximately twice the values at G6. This positional dependence is consistent with the expected peak of $B|\nabla B|$ near the bore entrance reported in literature [39]. All predicted FR values remain well below the FR = 1.0 threshold even at the worst-case coordinate.

Additionally, the allowable maximum spatial gradient was extrapolated using the ASTM F2052-15 Appendix X3 extrapolation, which projects measured deflection angles at a test location onto target field conditions (19 T/m for 1.5 T systems and 17 T/m for 3.0 T) [7]. Given our case, we input the mean values $B_{0L} = 2.67T$ and $|\nabla B|_L = 5.93T/m$ as test field location field conditions, α_L for each MUT and project it to find the gradient necessary to produce a $\alpha_C = 45°$ deflection on either 1.5 T or 3.0 T compared to their typical values.

This extrapolation is shown in Figure 4.10 and answers the question: under what field conditions would our MUTs reach the FR = 1 safety threshold? That is, what spatial gradient a clinical scanner would need to produce in order to deflect each device to 45°.

Of most interest, the unipolar lead yielded an allowable gradient of 63.5 T/m at 1.5 T and 31.8 T/m at 3.0 T, both far beyond the scanner's physical maximum gradient, indicating that the translational force alone does not pose a safety concern under standard clinical conditions for this particular sample. Titanium samples showed a deflection angle below 2°, Ti Coil ($\alpha = 0.56°$) and Ti Rod ($\alpha = 0.39°$), which were conservatively floored to 2°, as F2052-15 recommends. With it, they both yield an allowable gradient of T/m at 1.5 T and 151 T/m at 3.0 T.

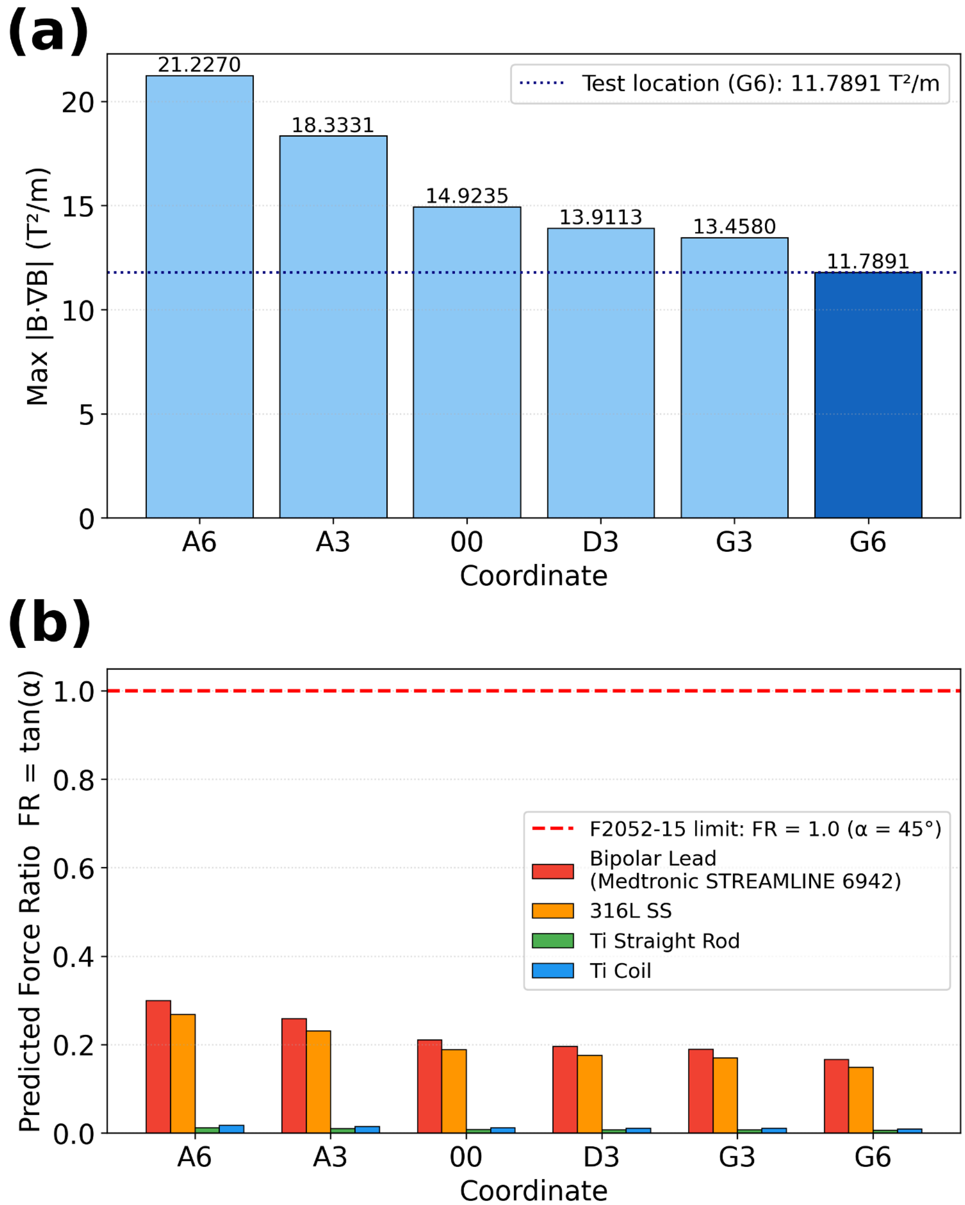


Figure 4.9: Field map and translational cross analysis. (a) Measured $|B \cdot \nabla B|$ per fixture coordinate on the forward pass, sub run B. (b) Predicted Force Ratio (FR) per device at each fixture coordinate at bore entrance.

It should be noted, that per F2052-15 [7], a passing result for translational force is not sufficient alone to establish MR safety. RF-induced heating (ASTM F2182 [9]) and torque (ASTM F2213 [8]) must also be assessed. The Ti Rod and 316L stainless steel samples were tested in a single trial only, falling short of the three-repetition minimum required by the standard. These results should therefore be

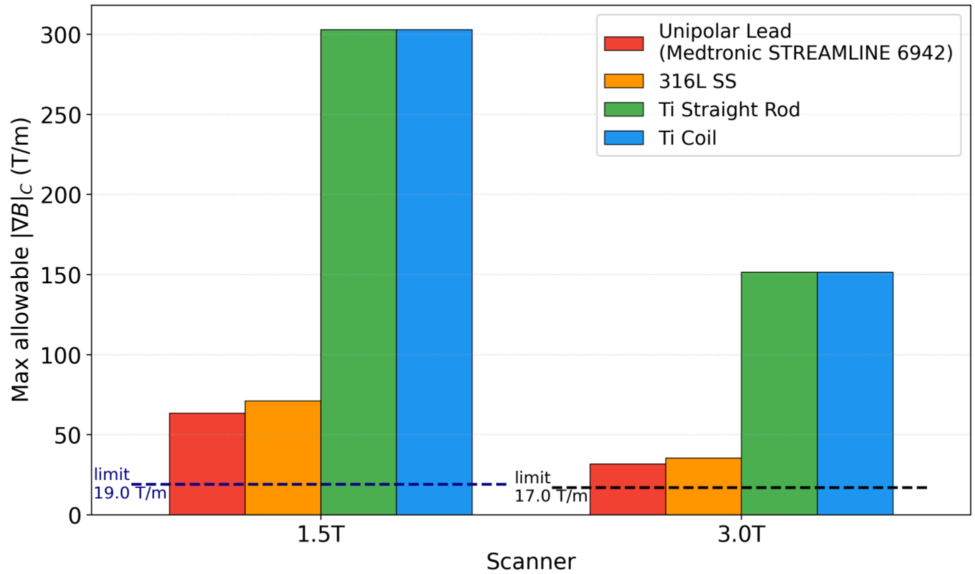


Figure 4.10: Maximum allowable spatial gradient. The measured test location was $B_{0L} = 2.67$ T, $|\nabla B|_L = 5.93$ T/m, with actual maximum gradient marked in red.

treated as indicative rather than fully compliant. Additional follow-up experiments will be performed.

Taken together, these results demonstrate that the custom pendulum fixture successfully resolved magnetically induced deflection at clinically relevant scales, and that all four MUTs satisfy the ASTM F2052-15 translational force criterion under the field conditions of the 3T scanner. The temporary epicardial lead, remains well within the safety threshold despite producing the largest deflection among the devices tested.

The last piece of the study was the torque measurement. Laboratory work included sensor calibration, with extended details found in Appendix E. In laboratory measurements, torque was successfully detected, exhibiting the expected quadratic dependence on magnetic field strength ($\tau \propto B^2$). At the clinic, no measurable torque were obtained from positive control samples.

4.3 Torque

In parallel with Section 3.3, this section is divided into laboratory and clinical work. This experiment has its bases on applying a magnetic field to a sample that is free to rotate on a low friction surface and align with the magnetic field B_0. The sample experiences a torque given by Equation 2.9b and the counter-force applied by the level arm is measured in our sensor as voltage change.

4.3.1 Torque Measurements in the Laboratory Setting

The aluminum-based sensor was calibrated prior to any experiment following methodology described in Section 3.3.1.3. Figure 4.11 shows the resulting calibration curve, from which the slope was obtained in units of Volts per Newton.

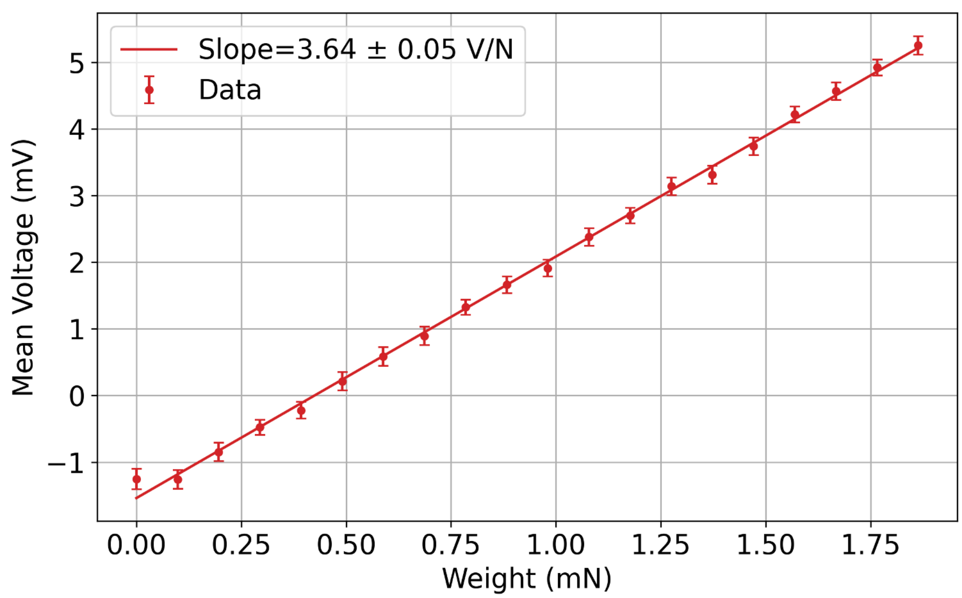


Figure 4.11: Calibration of an aluminum sensor for applied forces from 0 to 2 mN, using known water volumes (0–200 μl) as mass references.

The calibration slope gives a force sensitivity of $k = 1/m = 1/3.64 = 0.275$ N/V. Given a minimum measurable voltage was $\Delta V = 0.000885$, the smallest measurable weight was $W_{min} = 0.243 \pm 0.003$ mN,corresponding to minimum measurable torque $\tau_{min} = 0.012 \pm 0.002$ mN·m for the 5 cm transfer arm.

As described in Section 3.3.2, torque measurements were performed under two different experimental configurations. First, an Helmholtz coil configuration that increases magnetic field strength at its center with varying current. Second, a set of neodymium magnets with field gradient controlled by distance and number of magnets. Both were measured with a gaussmeter.

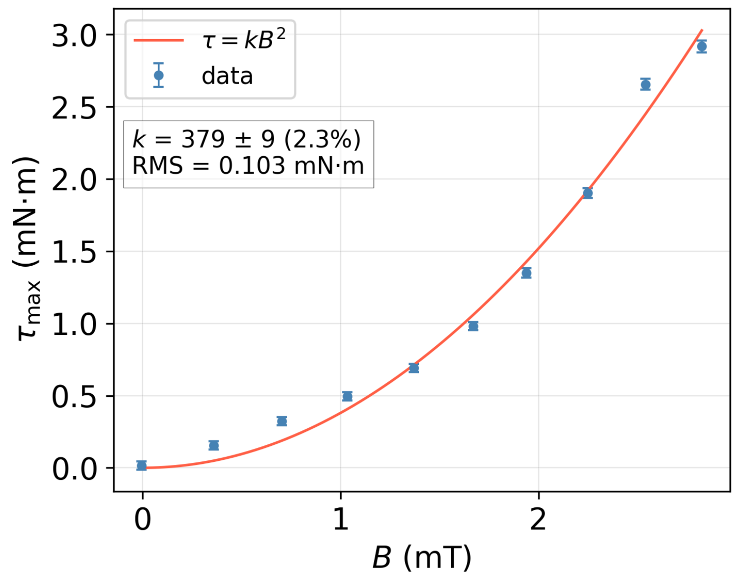


Figure 4.12: Maximum measured torque in mN·m as a function of applied magnetic flux density in mB for the longer cylinder under a Helmholtz coil configuration.

When a very small uniform magnetic field ranging from 0.335 to 3.85 mT was applied using Helmholtz coils, the torque produced by the short (2 cm) ferromagnetic test cylinder remained at or below the detection limit of the measurement system. Only the longer rod yielded reliable results under a similar magnetic field range (0.36 to 2.825 mT). Figure 4.12 shows the maximum torque measured against the magnetic flux, exhibiting a quadratic dependence on B. The data was also fit to a second order polynomial, however the extra fitting parameters did not significantly improve the fit.

The dominant quadratic term (kB^2) is in agreement with the theoretical expectation for the torque on a magnetic dipole in a uniform field, where $\tau \propto B^2$.

Because of the difficulty quantifying torque for the short cylinder (2 cm) under the low magnetic field from the Helmholtz coil, a second set of measurements was performed using significantly stronger magnetic fields generated by four neodymium magnets. Figure 4.13 shows a polynomial relationship. In that case, notice that approximately an order of magnitude stronger field was required to produce torque values only 1.5 to 2 times higher than those in Figure 4.12.

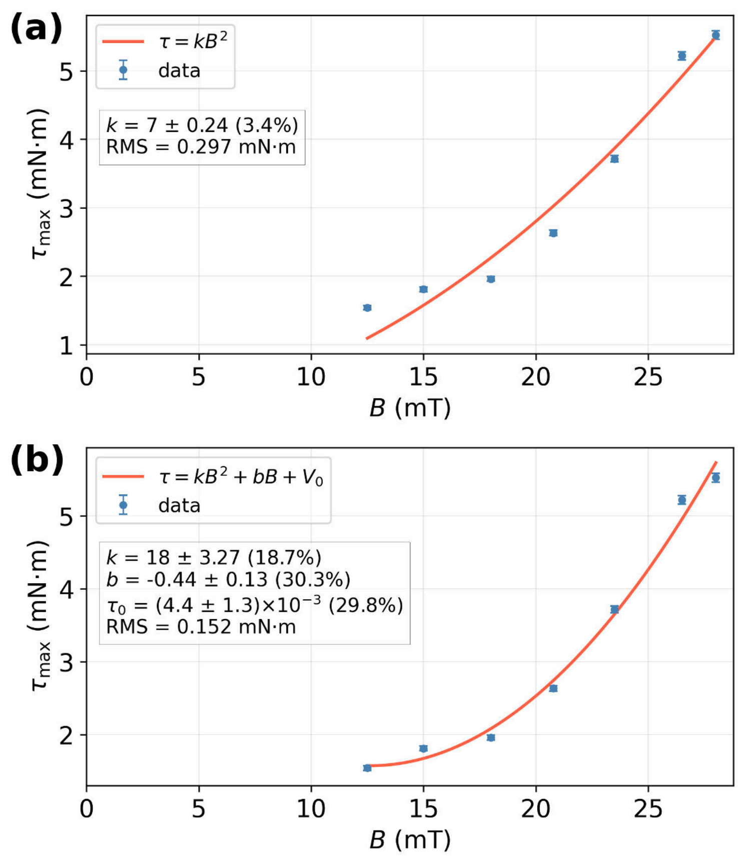


Figure 4.13: Maximum torque as a function of applied magnetic flux density for a 2 cm long ferromagnetic cylinder under stronger fields (Neodymium magnets). Blue points are the experimental measurements while the orange curve represents a polynomial fit.

The zero-intercept fit yielded $k = 7 \pm 0.2\,(3.4\%)\mathrm{N} \cdot \mathrm{m}/\mathrm{T}^2$, consistent with the quadratic dependence observed for the long rod. The three-parameter fit returned a comparable $k = 18 \pm 3.27(18.7\%)\mathrm{N} \cdot \mathrm{m}/\mathrm{T}^2$ but with a linear coefficient b and offset τ_0 both carrying uncertainties above 29%, indicating these terms are not statistically significant. The zero-intercept model is therefore sufficient to describe both datasets.

From the zero-intercept fit, the extracted k values imply effective susceptibilities of χ_m=45±1.5 for the short rod and χ_m=221±5 for the long rod. Both values fall within the broad range associated with ferromagnetic steels, which we used as samples. Though the significant spread in that range suggests there is better material identification methods. Nonetheless, both configurations confirm that the sensor resolves the expected $\tau \propto B^2$ dependence, validating the measurement system prior to the clinical session, whose results are described in the next section.

4.3.2 Torque observed in a 3T Siemens Magnetom™ MRI scanner

In the clinic, the torque fixture was positioned on the scanner couch and samples were advanced toward the bore entrance following the same couch travel path described in Section 3.3.3. The fixture pole was aligned to maximize the expected torque response, with samples free to rotate against the sensor arm. Both the circuit and the MRI-safe torque measuring fixture were verified to be functioning properly and circuit calibration was appropriate for clinical measurements. Based on the magnetic field map acquired in the same session (Section 4.1.2), the maximum field experienced by the samples during the closest positioning was approximately $B \approx 1.5$ T at ∼0.2 m from the bore entrance.

Prior to mounting samples on the fixture, a qualitative assessment was performed by manually bringing cylindrical samples of titanium and stainless steel near the bore entrance. A slight but perceptible attraction was felt on the stainless steel sample, consistent with its ferromagnetic response observed before the session. The titanium sample produced no such response.

Neither sample produced a sensor-measurable torque deflection when mounted on the fixture at the maximum measured field strength position. The signal remained within the noise baseline throughout. The stainless steel sample used in the fixture was the same small piece from the translational experiment. Absence of a signal might be related to the sample dimensions falling below the effective detection threshold at the available field strength given that torque is related to volume by Equation 2.9b. Possible explanations and implications for sample selection are addressed in Chapter 5.

4.4 From the Bench to the Scanner

The successful collaboration between Creighton University and CUMC–Bergan Mercy made possible in-clinic data acquisition on a working 3T Siemens Magnetom™ scanner. All custom fixtures proved MRI-safe, non-interfering, and functional in the clinical environment. Acquiring real-world translational force and field-mapping data constitutes a significant achievement in itself and validates the entire experimental framework developed in this thesis.

Mapping the magnetic field for the solenoid in the laboratory confirmed the expected spatial gradient behavior and established that the platform could reliably produce the $B\nabla B$ product needed as input for the force model. The ferromagnetic reference in the pendulum yielded force ratios in range of 0.002–0.012 and a $\chi_m = 17.3$, validating the linear force model and confirming that materials with low susceptibility, such as an epicardial lead, would remain within its linear regime at clinical field strengths. Sensor calibration produced a consistent linear voltage-to-force slope and a minimum detectable torque of 0.12 mN·m (5 cm arm). This detection floor was the reference threshold against which all subsequent measurements were evaluated. Trials for induced torque confirmed the $\tau \propto B^2$ dependence for two ferromagnetic rods of different sizes under both Helmholtz coil and neodymium magnet configurations, validating the system and establishing a nonzero torque response as the expected baseline for clinical measurements.

In the clinic, mapping the 3T scanner yielded $B\nabla B$ between 12.68 and 22.95 T^2/m at the bore entrance along the vertical points in our fixture, with A6 (nearest the bore top) measurement yielding the highest product. All four MUTs (Medtronic STREAMLINE™ 6492 unipolar lead, 316L stainless steel, titanium coil, titanium wire) tested for magnetically induced translational force passed the ASTM F2052-15 criterion (FR $<$ 1.0). The temporary unipolar epicardial lead reached the highest deflection $\alpha = 5.32°$ and force ratio FR $= 0.093$, a larger-than-anticipated displacement of 9 cm, though still well within the safe regime. The coiled geometry likely contributed to this response, as the titanium coil also deflected more than the straight titanium rod despite identical material, suggesting configuration plays a role beyond bulk susceptibility. The The 316L and titanium references behaved as anticipated from laboratory trials, lending confidence to the measurement chain. That both references were measured in a single trial is a limitation, but it does not undermine their role in that session.

The field strength experienced at the test position (G6 $B_{0L} = 2.67$ T) served as the test location input for the extrapolation, which showed allowable gradients of 107 T/m at 1.5 T and 54 T/m at 3.0 T for the unipolar lead, both beyond the scanner's physical maximum of 19 T/m. The torque fixture and its strain-gauge sensor operated without interference in the MRI suite, completing the set of three functional custom instruments in the clinical environment. In the clinical torque run, no measurable torque was obtained from either the titanium or stainless steel samples at B $\approx$ 1.5 T. Given that torque scales with sample volume (Equation 2.9b), the small stainless steel piece used is likely to have produced a torque below the characterized detection floor at the available field strength.

Chapter 5

Discussion and Conclusion

Patients with retained epicardial pacing leads are frequently denied MRI access because of theoretical safety concerns, yet the quantitative evidence supporting that restriction is limited. This thesis developed, deployed, and validated three custom non-magnetic fixtures for use in the clinic to measure the effects of force and torque produced by the static field as the primary mechanical hazard of concern under ASTM F2052-15 [7] and ASTM F2213-17 [8].

5.1 Summary of Findings

The three fixtures included a field-mapping platform, a pendulum-based deflection apparatus, and a strain-gauge torque sensor/fixture. Each was validated in a controlled low-field laboratory environment before deployment on the Siemens Magnetom™ 3 T scanner at CUMC–Bergan Mercy. All three operated without measurable interference in the MR suite, which is itself a meaningful result. This opens the door for systematic device-by-device characterization of forces that does not currently exist in the literature.

For the translational force experiment, the Medtronic STREAMLINE™ 6492 unipolar temporary epicardial lead produced the largest deflection among the four MUTs, reaching a force ratio of $\mathrm{FR} = \tan\alpha = 0.166$ ($\alpha = 9.5°$) at the bore entrance under a field product of $|B\nabla B| = 12.68\,\mathrm{T}^2/\mathrm{m}$. The measured magnetic force on the lead at that location corresponds to $F_m \approx 0.22\,\mathrm{mN}$, four orders of magnitude below the force required for a pleural drainage needle to perforate cardiac tissue under clinical conditions [71]. The ASTM F2052-15 extrapolation further showed that an allowable spatial gradient of 63.5 T/m at 1.5 T and 31.8 T/m at 3.0 T would be required to bring this lead to the safety threshold, both values far from the physical maximum of any current clinical scanner ($\leq$ 19 T/m [7]).

No measurable torque was observed for either the titanium or 316L stainless steel samples at $B \approx 1.5\,\mathrm{T}$. This outcome is most likely attributable to sample volume. Torque scales with material volume per Equation 2.9b, and the stainless steel piece used in the clinical session was trimmed to match the mass of the epicardial lead for the translational experiment. A secondary contributor may be the circuit architecture, placing the offset trim resistors at the amplifier output, rather than directly in the Wheatstone bridge arm, partially suppresses the measurable signal, a limitation noted during calibration and independently identified by circuit reviewers. The null result therefore reflects a measurement constraint rather than an absence of torque and should not be interpreted as a safety finding.

These results do not establish that the lead is broadly MRI safe. Translational force and torque is one component of a multi-parameter safety assessment. RF-induced heating, and gradient-induced electrical stimulation remain independent concerns that must be addressed before clinical recommendations can be revised.

5.2 Scope and Limitations

First, this study tested a single lead type in a single session, in a coiled configuration that is not representative of how a retained epicardial lead typically appears *in* *situ.* The coiled geometry likely increased the effective volume of material interacting with the field relative to a straight or gently curved wire fragment. The observation that the coiled titanium sample deflected approximately 2.3 times more than its straight counterpart, despite identical material, suggests that geometry contributes meaningfully to the force response independently. Whether this reflects increased effective volume, altered orientation relative to the field gradient, or both cannot be resolved from the current datasets. A straight or loosely curved lead configuration is expected to yield a lower force ratio.

Second, the translational measurements were performed at coordinate G6, chosen for practical reasons during the clinical session, rather than at A6, which the field maps identified as the location of peak $|B\nabla B|$ ($22.95\,\mathrm{T}^2/\mathrm{m}$ versus $12.68\,\mathrm{T}^2/\mathrm{m}$ at G6). The cross-coordinate analysis in Figure 4.9 projects the expected force ratios at all sampled positions, but direct measurement at A6 or at scanner isocenter would provide a more clinically representative dataset.

Third, the tested lead is a modern post-2000 device. Even within this category, vendor guidance remains conservative. MEDTRONIC's MRI SureScan™ leads are indicated for use under defined scanning conditions, and explicitly excludes abandoned or fractured leads from its indications [72]. Vuorinen et al. (2021)[21] noted that one of the two adverse events in their cohort occurred in patients with leads implanted before 2000, while no adverse events were detected in patients with modern leads. Whether differences in material composition, insulation, or lead geometry between older and newer devices contribute to that pattern remains an open question. Testing leads from different manufacturing eras would be a straightforward way to address it.

5.3 Future Directions

The immediate next step is a dedicated translational force session with a broader sample set. This should include multiple epicardial lead types from different vendors, tested in at least two orientations (straight along the z-axis and perpendicular to it), alongside matched titanium and 316L stainless steel controls in identical configurations. Measurements should be performed at the bore-entrance position of peak $|B\nabla B|$, identified in the current field map as coordinate A6. Such a session would directly address the generalization gap and could, in principle, produce a vendor-agnostic reference table that a clinical physicist or radiologist could consult when evaluating a specific retained lead.

For the torque measurement, a dedicated bench session should precede any further clinical deployment. The circuit should be modified to place the balancing network within the Wheatstone bridge rather than at the amplifier output. A dedicated clinical torque session, separate from the translational measurement to avoid time pressure, would allow proper characterization of the torque response at clinically relevant field strengths. Positive control samples of sufficient volume should be confirmed to produce a measurable sensor response under the scanner field.

Longer term, the methodology described in this thesis is one component of a broader safety assessment framework. A clinical physicist testing the MRI scanner for field homogeneity and safety prior to a scan session for a patient with a retained epicardial lead would need to integrate translational force data of the kind produced here with RF-heating characterization following ASTM F2182 [9], for which the groundwork was laid by Aboyewa et al. [34] using the same scanner, and with gradient-induced current considerations. The field maps acquired with methods from this work provide the spatial input required for force predictions at any bore position, accompanied by vendor documentation of the scanner, could serve as a shared reference dataset for multi-parameter analyses.

The pendulum fixture is simple enough for a trained resident to operate and provides direct visual feedback. A lead deflecting under a known field makes the physics of static-field interaction tangible in a way that a table of susceptibility values does not. Incorporating a demonstration of this kind into MRI safety training could build the intuitive understanding of implant behavior that informs clinical decision-making at the point of care.

Finally, all three measurements could be acquired in a single couch pass if the fixtures are staged in order along the patient table axis, with the torque device nearest the bore, the pendulum at mid-position, and the mapping platform trailing. Dedicated computers for signal acquisition, or implementing a multiplexed logging protocol, would enable this. This represents a workflow optimization rather than a redesign, and is a reasonable target for a follow-on study.

5.4 Concluding Remarks

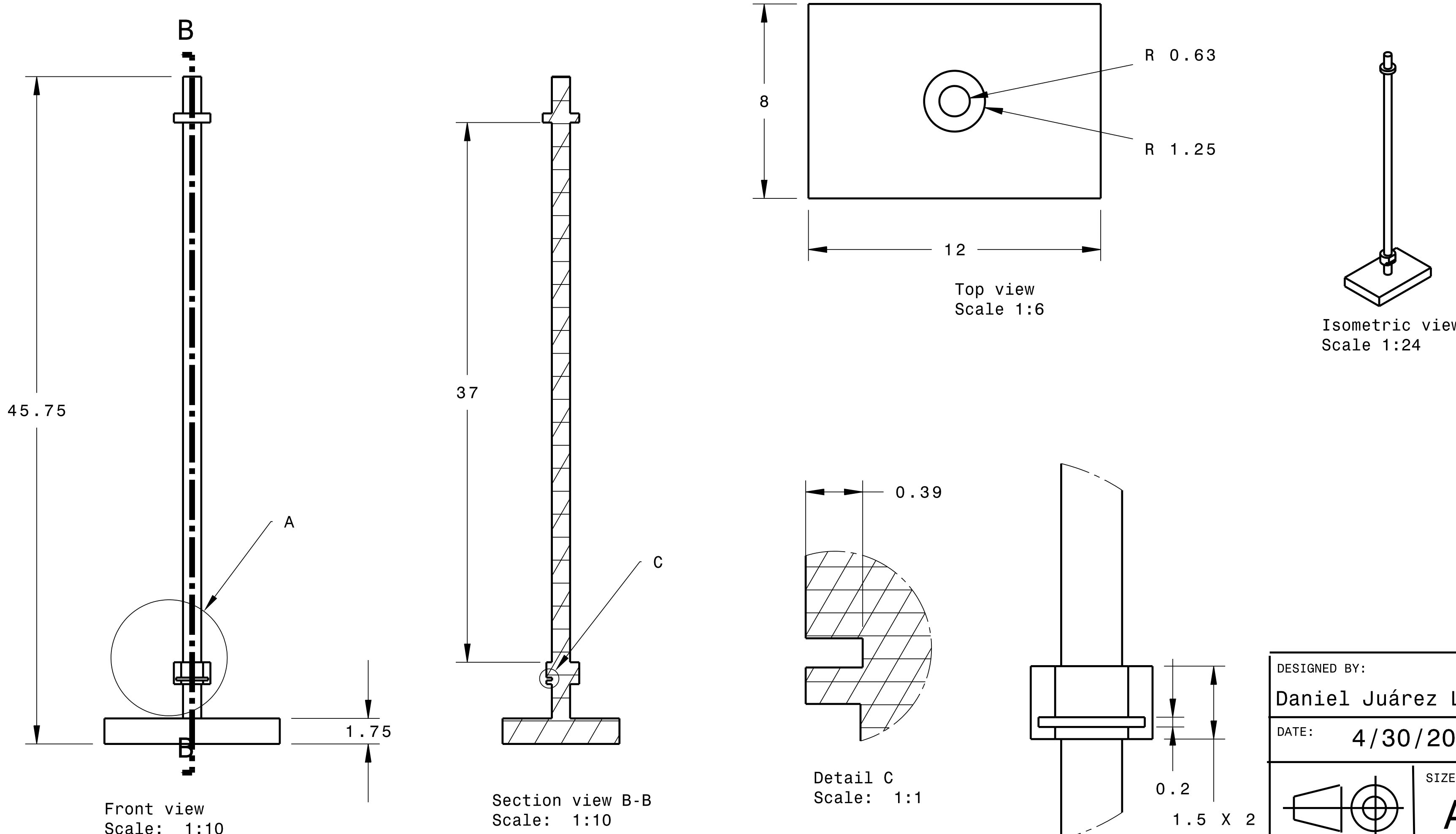
With the number of CIED implants continuing to grow each year [6], the population of patients who carry retained epicardial lead fragments and require MRI will grow with it. This study contributes to the systematic, quantitative, device-specific characterization required to inform evidence-based MRI access for this population.

The static magnetic field of a 3 T MRI scanner exerts a measurable translational force on the Medtronic STREAMLINE™ 6492 unipolar temporary epicardial lead. At 9.3% of the ASTM safety threshold and four orders of magnitude below clinically relevant tissue-perforation forces, it does not provide physical justification for the broad MRI exclusion applied to patient population with this type of lead. The fixtures and protocols developed here make it possible to answer, on a device-specific basis, whether a given epicardial lead can be safely exposed to the static magnetic field of a 3T MRI scanner.

Appendix A

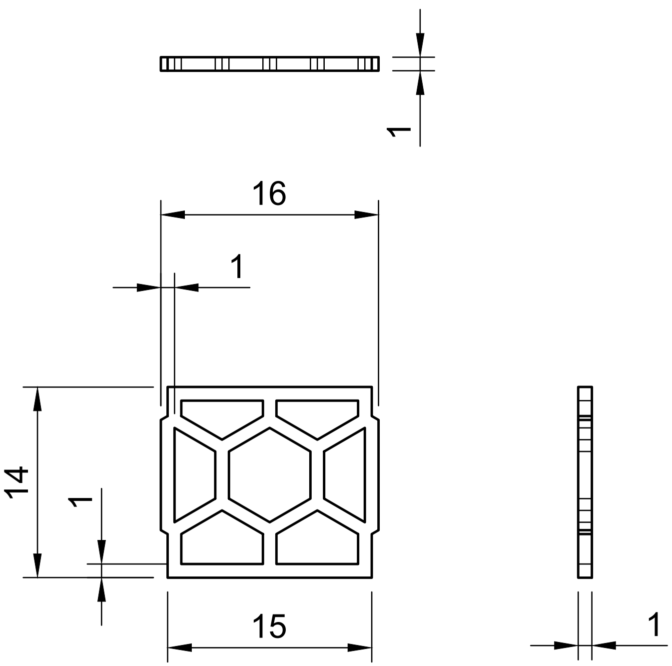
Engineering Drawings

This appendix contains the engineering drawings for the two custom fixtures fabricated for this work: the lead holder basket used in the translational force experiment and the translational force fixture pole. The fixture was designed in CATIA, while the basket was designed in Autodesk Fusion 360. Both are referenced in Chapter 3. The lead holder drawing corresponds to the hexagonal mesh basket shown in Figures 3.9 and 3.10b, used to suspend test objects from the pendulum string. The pole drawing corresponds to the main structural component of the translational force fixture shown in Figure 3.7. Dimensions are given in millimeters. All components were fabricated from non-magnetic materials as required for use in the MRI environment.



DESIGNED BY:	Daniel Juárez Lun
DATE:	4/30/2025
SIZE	A4
MATERIAL:	PVC
WEIGHT:	11.6 lb

Properties	
Material:	PLA
Weight:	0.124 g



Dept.	Technical reference	Created by	Approved by
		Daniel Juarez 8/5/25	
		Document type	Document status
		Title: Lead Holder Bee Comb Mesh MRI Translational Force	DWG No.
			Rev. / Date of issue / Sheet 1/1

Appendix B

Sample calculations and Methodology Notes

This appendix collects sample calculations and methodology notes that support the experimental chapters. Each section corresponds to a specific step in the analysis and is intended to provide enough detail for reproducibility without interrupting the flow of the main text. Section B.1 derives the minimum magnetic susceptibility required to produce a measurable deflection under reduced laboratory field conditions, supporting the scaling argument presented in Chapter 2 and Table 2.2. Section B.2 works through the smallest measurable translational force for the pendulum fixture introduced in Chapter 3. Section B.3 provides the equivalent calculation for the torque sensor. Sections B.4 through B.4.2 contain the Kirchhoff analysis of the strain gauge circuit and an explanation of the coarse-fine trim resistor behavior. Section B.5 documents practical lessons learned during the laboratory and clinical measurement sessions that are relevant to future iterations of the protocol.

B.1 Sample Calculation for Scaling at 30 G

This appendix supports Section 2.3.3.1 and Table 2.2. The goal is to determine the minimum magnetic susceptibility a laboratory test object must possess to produce a measurable deflection under reduced field conditions, and to verify that the same force-to-weight ratio encountered clinically can be reproduced at bench scale.

From the ASTM F2052 force balance (Appendix X3 therein [7]), the equilibrium deflection angle satisfies

$$\tan(\alpha) = \frac{\chi_m}{\rho\,\mu_0\,g} \cdot B_0 \frac{dB_0}{dz}, \tag{B.1}$$

where $\mu_0 = 4\pi \times 10^{-7}\,\mathrm{H/m}$, $g = 9.807\,\mathrm{m/s^2}$, ρ is the material density, and $B_0\,dB_0/dz$ is the magnetic force product at the test location. The product χ_m/ρ and the field term appear symmetrically: to reproduce the same $\tan(\alpha)$ at a lower field, the ratio χ_m/ρ must increase in inverse proportion,

$$\left.\frac{\chi_m}{\rho}\right|_{\mathrm{lab}} = \frac{\mu_0\,g\,\tan\alpha}{\left(B_0\,\dfrac{dB_0}{dz}\right)_{\mathrm{lab}}}. \tag{B.2}$$

The on-axis field of a finite solenoid of radius R and length L, normalized to a central field B_0, is

$$B_z(z) = \frac{B_0}{2\,B_{ctr}} \left[\frac{z + L/2}{\sqrt{R^2 + (z + L/2)^2}} - \frac{z - L/2}{\sqrt{R^2 + (z - L/2)^2}} \right], \tag{B.3}$$

where $B_{ctr} = (L/2)/\sqrt{R^2 + (L/2)^2}$ is the analytic center value. The force product $B_z\,dB_z/dz$ peaks near the coil exit, which is the maximum-sensitivity test location prescribed by ASTM F2052 [7].

A bench solenoid with $R = 5\,\mathrm{cm}$ and $L = 20\,\mathrm{cm}$ was modeled at $B_0 = 30\,\mathrm{G}$ $= 3 \times 10^{-3}\,\mathrm{T}$. The maximum force product occurs at $z \approx 8.7\,\mathrm{cm}$ from the solenoid center (near the coil exit), where

$$\left(B_0 \frac{dB_0}{dz}\right)_{\text{lab}} = 6.09 \times 10^{-5}\ \text{T}^2/\text{m}. \tag{B.4}$$

Assuming a target deflection of $\tan\alpha = 0.01$ (1 cm over a 1 m pendulum string) and substituting into Equation (B.2),

$$\frac{\chi_m}{\rho} = \frac{(4\pi \times 10^{-7})(9.807)(0.01)}{6.09 \times 10^{-5}} = 2.02 \times 10^{-3}\ \text{m}^3/\text{kg}. \tag{B.5}$$

For a laboratory test material with $\rho = 7900\,\text{kg/m}^3$ that of common grades of steel,

$$\chi_m = 2.02 \times 10^{-3} \times 7900 \approx 15.98. \tag{B.6}$$

Comparison with clinical conditions

ASTM F2052-15 reports an upper bound of $|B_0\,\nabla B_0| < 48\,\text{T}^2/\text{m}$ for current 3 T clinical systems [7]. The ratio of the clinical to laboratory force product is

$$\frac{(B_0\,\nabla B_0)_{3\,\text{T}}}{(B_0\,\nabla B_0)_{\text{lab}}} = \frac{48}{6.09 \times 10^{-5}} \approx 7.9 \times 10^5, \tag{B.7}$$

and the required χ_m/ρ scales inversely by the same factor. The full results across field levels are summarised in Table 2.2.

For reference, Woods et al. [32] measured deflection angles for 11 common implant alloys on a Siemens Trio 3 T system under $|B_0\,\nabla B_0| \approx 3.5$–$4.6\,\text{T}^2/\text{m}$. Back-calculating from their measured force ratios, the highest product of strength and gradient yielded a χ_m/ρ of $4.2 \times 10^{-7}\,\text{m}^3/\text{kg}$ for 316L stainless steel. This is $\sim$4800 times smaller than the laboratory threshold in Equation (B.2), confirming that none of the tested clinical alloys would produce a detectable deflection under the assumed bench field conditions.

B.2 Smallest Measurable Force of Translational Fixture

Following section 3.2.1.1 For a specimen of $m = 0.1\,\mathrm{g} = 1 \times 10^{-4}\,\mathrm{kg}$, the smallest measurable magnetic force is given by equation 3.2

$$F_{m,\min} = mg\frac{z}{L}, \quad \text{(Eq. 3.2)}$$

where $g = 9.81\,\mathrm{m/s^2}$, $z = 1 \times 10^{-3}\,\mathrm{m}$, and $L = 1\,\mathrm{m}$. Substituting these values yields

$$\begin{aligned} F_{m,\min} &= (1 \times 10^{-4})(9.81)\left(\frac{1 \times 10^{-3}}{1}\right) \\ &= 9.81 \times 10^{-7}\,\mathrm{N} \approx 1.0 \times 10^{-6}\,\mathrm{N}. \end{aligned} \tag{B.8}$$

Thus, for a 0.1 g mass, the smallest detectable magnetic force is approximately $1\,\mu\mathrm{N}$. The corresponding force ratio remains

$$\frac{F_m}{F_g} = \frac{z}{L} = 10^{-3}, \tag{B.9}$$

indicating that the setup can detect magnetic forces as small as one-thousandth of the object's weight.

B.3 Smallest Measurable Torque from Sensor

Following Section 3.3.1.2, we evaluate the strain, voltage response, and force detection limit using the geometry reported by Chiu, *et al.* (2016) [62] For the Nitinol wire,

$$L = 69.85\,\text{mm} = 6.985 \times 10^{-2}\,\text{m},$$

$$D = 0.64\,\text{mm} = 6.604 \times 10^{-4}\,\text{m},$$

$$E = 66.2\,\text{GPa} = 6.62 \times 10^{10}\,\text{Pa}.$$

The geometric sensitivity term appearing in Equation 3.9 is

$$\frac{L}{D^3}. \tag{B.10}$$

Substituting numerical values,

$$\begin{aligned} D^3 &= (6.604 \times 10^{-4})^3 = 2.88 \times 10^{-10}\,\text{m}^3, \\ \frac{L}{D^3} &= \frac{6.985 \times 10^{-2}}{2.88 \times 10^{-10}} = 2.43 \times 10^{8}\,\text{m}^{-2}. \end{aligned} \tag{B.11}$$

This agrees with the order-of-magnitude sensitivity reported in Section 3.3.1.2.

Using Equation 3.9, the surface strain produced by the minimum detectable force $F = 0.49\,\text{mN} = 4.9 \times 10^{-4}\,\text{N}$, as reported by Chiu, *et al.* (2016) [62] is

$$\epsilon = \frac{32FL}{\pi E D^3}. \tag{B.12}$$

Substituting,

$$\begin{aligned} \epsilon &= \frac{32(4.9 \times 10^{-4})(6.985 \times 10^{-2})}{\pi(6.62 \times 10^{10})(2.88 \times 10^{-10})} \\ &= 1.82 \times 10^{-5}. \end{aligned} \tag{B.13}$$

Thus, the minimum detectable force corresponds to a strain of approximately

$$\epsilon \approx 20\,\mu\varepsilon. \tag{B.14}$$

Assuming a half-bridge configuration (Equation 3.12) with

$$G_F = 2,$$
$$V_{exc} = 5\,\mathrm{V},$$

the predicted output voltage is

$$\begin{aligned} V_{out} &\approx \frac{V_{exc}}{2} G_F \epsilon \\ &= \frac{5}{2}(2)(2.0 \times 10^{-5}) \\ &= 5(2.0 \times 10^{-5}) \\ &= 1.0 \times 10^{-4}\,\mathrm{V}. \end{aligned} \tag{B.15}$$

Therefore,

$$V_{out} \approx 0.1\,\mathrm{mV}. \tag{B.16}$$

Additional amplification is therefore required to reach the experimentally observed $\sim 3\,\mathrm{mV}$ signal level.

However, using Chiu, *et al.* (2016) reported gain and assuming noise at

$$V_{noise} = 3mV,$$
$$G = 100,$$

We got an ϵ_{min} of

$$\epsilon_{min} = \frac{2V_{noise}}{V_{exc}G_F \quad \text{gain}} \tag{B.17}$$
$$= \frac{2(0.003)}{5(2)(100)} \tag{B.18}$$
$$= 6\mu\varepsilon. \tag{B.19}$$

Using Equation 3.14, the minimum detectable force for a given minimum strain ϵ_{min} is

$$F_{min} = \frac{\pi E D^3}{32L}\,\epsilon_{min}. \tag{B.20}$$

If the system resolves strains on the order of $1\,\mu\varepsilon = 6 \times 10^{-6}$, then

$$F_{min} = \frac{\pi(6.62 \times 10^{10})(2.88 \times 10^{-10})}{32(6.985 \times 10^{-2})}(6 \times 10^{-6})$$
$$= 0.161\text{mN}. \tag{B.21}$$

This confirms micro-Newton sensitivity becomes achievable with reduced noise, increased L/D^3, and signal averaging.

B.4 Kirchhoff analysis of the Circuit

Using Kirchhoff's Current Law (KCL), in Figure B.1 for node V_A which corresponds to the positive leg of the op amp, or non inverting voltage, we have

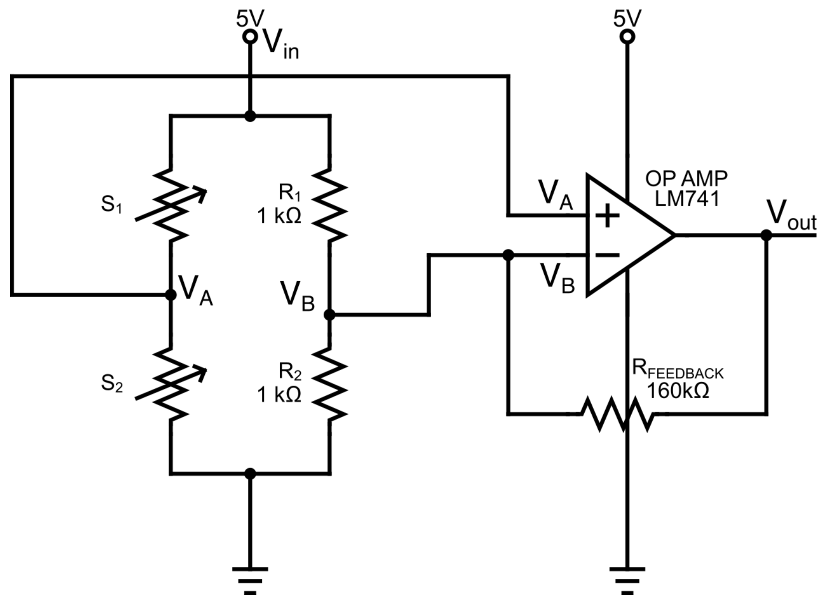


Figure B.1: Schematic of the Wheatstone bridge with strain gauges $S1$ and $S2$, fixed resistors $R = R1 = R2$, and operational amplifier with feedback resistor R_f. This configuration corresponds to the transfer function in Equation (3.7).

$$
\begin{aligned}
\frac{V_A - V_{in}}{S1} + \frac{V_A}{S2} &= 0 \\
\frac{S2(V_A - V_{in}) + S1V_A}{\cancel{S1S2}} &= 0 \\
V_A(S2 + S1) - S2V_{in} &= 0 \\
V_A &= \frac{S2V_{in}}{S2 + S1}.
\end{aligned}
$$

Similarly, for the node V_B corresponding to the inverting input of the op-amp

$$
\begin{aligned}
\frac{V_B - V_{in}}{R} + \frac{V_V}{R} + \frac{V_B - V_{out}}{R_f} &= 0 \\
\frac{R_f(2V_B - V_{in}) + R(V_B - Vout)}{\cancel{RR_f}} &= 0 \\
V_B(2R_f + R) - R_fV_{in} - RV_{out} &= 0 \\
V_B &= \frac{R_fV_{in} + RV_{out}}{2R_f + R}.
\end{aligned}
$$

At this point, we make a standard ideal op-amp assumption, the op-amp has infinite open-loop gain, meaning any finite output voltage corresponds to a vanishingly small input voltage difference. That is, for the difference amplifier

$$V_{out} = A_{\mathrm{ol}}(V_A - V_B), \quad A_{\mathrm{ol}} \to \infty, \tag{B.22}$$

which implies $V_A \approx V_B$. Substituting $V_A = V_B$ into the previous equations allows us to solve for the closed-loop transfer function

$$\begin{aligned}
\frac{S2V_{in}}{S2 + S1} &= \frac{R_f V_{in} + RV_{out}}{2R_f + R} \\
\frac{V_{out}}{V_{in}} &= \frac{R_f(S2 - S1) + S2R}{R(S2 + S1)} \\
\frac{V_{out}}{V_{in}} &= \frac{R_f}{R}\frac{S2 - S1}{S2 + S1} + \frac{S2}{S2 + S1}
\end{aligned} \tag{3.7}$$

In summary, the output voltage is composed of two contributions: a differential term amplified by the ratio $R_f/R = 160$, which captures the imbalance between the strain gauges $S1$ and $S2$, and a baseline voltage division term $S2/(S2+S1)$ that represents the inherent bridge offset. This expression clearly shows how the Wheatstone bridge imbalance is amplified by the op-amp to produce the measurable output.

B.4.1 Explaining the behavior of Coarse-Fine Tuning with trim resistors in Circuit

Starting with KVL around the loop

$$V_{\mathrm{in}} - IR_1 - IR_2 = 0, \tag{B.23}$$

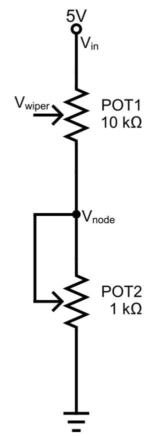


Figure B.2: Schematic of the coarse–fine trim resistors (POTs) configuration used for the voltage-divider derivations in Figure 3.17.

Factor out the current

$$V_{\text{in}} = I(R_1 + R_2), \tag{B.24}$$

Solve for the series current

$$I = \frac{V_{\text{in}}}{R_1 + R_2}, \tag{B.25}$$

The node voltage is the drop across the lower resistor R_2

$$V_{\text{node}} = I\,R_2 = \left(\frac{V_{\text{in}}}{R_1 + R_2}\right) R_2. \tag{B.26}$$

Final voltage-divider form

$$\boxed{V_{\text{node}} = V_{\text{in}}\,\frac{R_2}{R_1 + R_2}}. \tag{3.8}$$

For these evaluations

$$V_{\text{in}} = 5.0\ \text{V}, \qquad R_1 = 10\ \text{k}\Omega, \qquad R_2 \in [0, 1000]\ \Omega. \tag{B.27}$$

Halfway Fine Adjustment $R_2 = 500\ \Omega$

$$V_{\text{node}} = 5.0\,\frac{500}{10000 + 500} = 5.0\left(\frac{500}{10500}\right) = 5.0(0.047619) = 0.2381\ \text{V}. \tag{B.28}$$

$$V_{\text{node}} = 5\,\frac{R_2}{10000 + R_2}. \tag{B.29}$$

Fine Pot Setting	R_2 (Ohms)	V_{node} (V)
Max	1000	0.4545
Mid	500	0.2381
Min	0	0

Now fix the fine pot at

$$R_2 = 500\ \Omega. \tag{B.30}$$

The 10 k coarse pot wiper divides R_1 into two portions. Let the coarse wiper position be $\alpha \in [0, 1]$, where

$$\alpha = 0 : \text{wiper at top}, \quad \alpha = 1 : \text{wiper at bottom}. \tag{B.31}$$

The wiper voltage is a weighted average between the supply (5 V) and the node voltage computed with the fixed fine setting

$$V_{\text{node,fine=500}} = 0.2381\ \text{V}. \tag{B.32}$$

Thus the coarse-wiper voltage is

$$V_{\text{wiper}} = (1 - \alpha)(5.0) + \alpha(0.2381). \tag{B.33}$$

Coarse-Wiper Table (Fine Fixed at Half)

Coarse Setting	α	V_{wiper} (V)
Top (Min Coarse)	0	5.000
Midpoint	0.5	2.619
Bottom (Max Coarse)	1	0.238

At $\alpha = 0$ (coarse at minimum), the wiper sits at the full input voltage

$$V_{\text{wiper}} = 5 \text{ V}, \tag{B.34}$$

which is independent of the fine adjustment. At $\alpha = 1$ (coarse at maximum), the wiper voltage collapses into the range controlled solely by the fine pot

$$V_{\text{wiper}} = V_{\text{node,fine}}. \tag{B.35}$$

Thus the coarse pot sets the overall span, and the fine pot determines the precise lower-region tuning.

B.4.2 Circuit Design Caveat

From the torque methodology developed in Section 3.3.2, a set of neodymium magnets was introduced to increase the magnetic field strength to 300 G, compared to the 30 G provided by the Helmholtz coil, because the short rod in the smaller fields produced signals that were too small to yield reliable results. The calibration established a detection limit of $\tau_{\text{min}} = 0.012, \text{mN} \cdot \text{m}$. For the short rod in the Helmholtz configuration the maximum field strength was $B_{\text{max}} = 3.85, \text{mT}$), and using the χ_m obtained from the neodymium-based fit, the expected torque was $\tau_{\text{expected}} = 0.26, \text{mN} \cdot \text{m}$. This corresponds to a ratio of $\tau_{\text{expected}}/\tau_{\text{min}} \approx 21.7$, which is well above the detection threshold in principle.

A likely explanation is the influence of noise in the measurement circuit, particularly from the trim resistors when the signal approaches the detection limit. During calibration, a noise level of approximately 0.8 mV was observed when the trim resistors were used to null the signal prior to applying load. However, this was variable across sensors, as noted by Dr. Rutel (University of Oklahoma), placing the balancing trim resistor at the output of the amplifier, Figure B.2 right after Figure B.1, effectively suppresses part of the measurable signal. Incorporating the trim resistors directly into the Wheatstone bridge arm would instead allow null balancing at the bridge level, eliminating residual offset at the source and improving sensitivity for low-torque measurements.

B.5 Methodology Notes

The following paragraphs describe the methods troubleshooting and practical improvement on the methodology. For a complete description of the final version of the methodologies applied on this work see 3

For the translational force experiment in the lab (Section 3.2.2), initial trials attempted to map the magnetic field from the furthest position of the pendulum, 8 cm from the solenoid face, at a constant solenoid current in a single run. The idea was to obtain (B_0) at the pendulum's resting position and then compute the field gradient at any point from that map. Afterward, the pendulum was allowed to be slowly pulled toward the solenoid to its equilibrium position so its displacement (in cm) could be recorded. This was repeated while increasing the solenoid current in 0.5 A steps, aiming to fit the resulting equilibrium positions using Eq. 3.6.

These attempts produced inconsistent results that pointed to errors inherent to the setup. Because the current, rather than the entire apparatus position, was changed, the value of B_0 at the pendulum's resting position did not correspond to the required by Eq. 3.6. Even using the mapped field, estimating the correct value for the magnetic field for each equilibrium position was unreliable. Moreover, predicting equilibrium positions before running the experiment proved difficult. Additional complications came from solenoid imperfections, current leakage, and induced currents, all of which became significant once displacements exceeded 1 cm. At higher fields the pendulum was sometimes pulled fully into the solenoid.

Another factor was the gaussmeter restriction on accurate motion tracking to roughly 11–12 mm on the translation stage. While this approach resembles the proposed clinical protocol, the scale of the apparatus and the nonuniform fields make it impractical. Once the approach was inverted, as explained in Section 3.2.2 the methodology, deflection angle measurments resulted in χ_m values with 0.5% precision, validating the methodology.

In the clinic, a single mapping session gave both B and ∇B at any point, and left the angle measurement as the critical step (see Section 3.2.3). However, the initial attempt revealed practical challenges with string attachment to the baskets. Knotting proved time-consuming and inconsistent across different basket designs, leading to variability in suspension length and tension. To streamline future sessions, pre-prepared assemblies are recommended: additional 3D-printed baskets should be fabricated, each paired with a fixed-length string segment of approximately 1.3 m, with MUTs attached pre-weighted, and labeled for rapid deployment. Similarly on the torque pipeline (see Section 3.3.3), multiple lever arms of identical size should be fabricated in advance, each with MUT samples pre-attached to ensure consistent mounting and to facilitate rapid swapping during trials.

Appendix C

Code and Data Samples

This appendix provides representative samples of the raw data collected during the field mapping and translational force experiments, included here to illustrate the data structure and support reproducibility. The complete datasets, along with all analysis code, are available in the GitHub repository documented in Appendix C.3. Section C.1 shows a subset of the stray magnetic field measurements recorded along the patient couch during the backward pass at the central grid coordinate, corresponding to the data processed in Figure 3.5. Section C.2 shows a representative subset of the displacement and field measurements from the laboratory translational force experiment, corresponding to the data used to produce the susceptibility fit in Figure 4.5.

C.1 Sample Data for Stray Magnetic Field Mapping Along the Patient Couch (Backward Pass at (0,0))

As mentioned in Section 3.1.3, Table C.1 presents a representative subset of the raw dataset from the first backward (retracting) couch translation pass along the longitudinal (z) axis at the central grid coordinate (0,0). Data were logged continuously using the Hall-effect gaussmeter probe. Flux values are raw instrument readings in gauss. The err_byte column is the gaussmeter's reported error code.

Uncertainties: position ±1 mm (couch display and alignment tolerance), time ±0.01 s, flux ±0.2% of reading or ≈ ±50 G near full scale.

time [s]	flux [G]	err_byte
0.25	26823	64.039
0.50	26823	64.289
1.00	26822	64.788
2.00	26824	65.788
3.50	24245	67.290
5.00	20287	68.791
7.00	14331	70.788
12.00	5886	75.803
15.00	3563	78.807
20.00	1607	83.805
25.00	794	88.798
30.00	570	93.801

Table C.1: Subset of measurements from the first retracting translation along the z-axis at isocenter (grid 00). Flux decreases as the probe moves away from the bore entrance. Full datasets (120 rows) and all 15 CSV files were imported into Python with pandas for cleaning, alignment of forward/backward pairs, and numerical computation of the longitudinal gradient $\partial B/\partial z$.

C.2 Sample Data for Translational Force Experiment

Table C.2 presents a representative subset of the displacement–current datasets used for field–gradient and angle calculations for experiment at Section 3.2.2 shown in Figure 4.5. Uncertainties: position ±0.05 mm, current ±0.001 A.

N	x [mm]	I [A]	$B_{-0.5}$	B_0	$B_{+0.5}$	dB/dx	θ	$B_0\,dB/dx$
1	0	0.000	–	–	–	–	–	–
2	1	0.554	2.86	2.81	2.84	0.1111	0.001047	0.3123
3	2	1.221	5.56	5.66	5.70	1.4444	0.002094	8.1691
4	3	1.636	7.03	7.22	7.24	2.1111	0.003141	15.2469
5	4	2.217	9.32	9.44	9.44	1.2222	0.004188	11.5432
6	5	2.471	10.27	10.40	10.56	2.8889	0.005235	30.0444
7	6	2.752	11.61	11.66	11.80	1.8889	0.006282	22.0160
8	7	2.966	12.58	12.63	12.76	1.7778	0.007329	22.4593
9	8	3.196	13.92	14.02	14.27	3.4444	0.008377	48.2988
10	9	3.507	15.52	15.69	15.89	3.6667	0.009424	57.5259
11	10	3.775	17.01	17.28	17.31	3.0000	0.010471	51.8333
12	11	3.921	18.03	18.17	18.32	2.8889	0.011518	52.4815

Table C.2: Representative dataset of displacement, current, magnetic field samples at ± 0.5 mm and equilibrium, computed gradients, and corresponding angular measurements.

C.3 GitHub repository

This thesis is accompanied by a public GitHub repository containing all analysis notebooks, raw experimental data, and publication figures. The repository is available at the following URL and is cited throughout this work as [60]:

`https://github.com/Zslw/mri-translational-force-n-torque-analysis`

Because GitHub repositories are public and editable after submission, this appendix serves as the authoritative printed record of what existed at the time of publication. Any content added after commit `8435c9b` is supplementary and not part of the reported work.

The following folder inventory describes the contents of the repository as they existed at commit `8435c9b`. All paths are relative to the `JupyterLab/` subfolder, which is the root of the analysis code within the repository.

- `BfieldSIM/` – Solenoid field simulation and gaussmeter mapping notebooks. Produces Figures 2.2 on simulation, and 4.1 and 4.2 for mapping.

Field	Value
Repository	https://github.com/Zslw/mri-translational-force-n-torque-analysis
Branch	main
Thesis sync commit	8435c9b
Commit message	Final thesis sync: updated notebooks, figures, and appendix calculations
Date of sync	April 23, 2026
Scanner	Siemens Magnetom 3T, CUMC-Bergan Mercy, Omaha NE
Device tested	Medtronic STREAMLINE 6492 unipolar temporary epicardial lead
Standards followed	ASTM F2052-15, ASTM F2213-17
Primary result	FR = 0.093 (alpha = 5.32 deg) – passes F2052-15 (FR less than 1.0)

Table C.3: Repository archive record at time of thesis submission.

- `Translational/` – Laboratory translational force experiment. Susceptibility fitting and tan(alpha) vs B times grad B analysis. Produces Figure 4.5. Includes the STL file for the 3D-printed test object basket (Figure 3.9).
- `Torque/Calibration/` – Torque sensor fabrication and calibration data. Results appear in Figure 4.11 and Appendix E.
- `Torque/MRIForces/` – Laboratory torque measurements under Helmholtz coil and neodymium magnet fields. Produces Figures 4.12 and 4.13.
- `ClinicalData/MappingBField/` – Clinical magnetic field mapping pipeline for all fixture coordinates. Produces Figures 4.4 and Appendix D.
- `ClinicalData/TranslationalExperiment/` – ASTM F2052-15 pendulum deflection protocol applied to the epicardial lead and reference samples. Produces Figures 4.7 and 4.8.
- `ClinicalData/` root – Cross-analysis notebook linking field mapping and translational results. Produces Figures 4.10 and 4.9.
- `SmallestTorquecalcs.xlsx` – Spreadsheet for the sample calculations in Appendix B.3.

Appendix D

Field Mapping - Additional Results

This appendix presents the complete set of magnetic field mapping figures for all fixture coordinates measured during the clinical session at CUMC–Bergan Mercy. The main text discusses the results at coordinate G6 in detail, as it served as the primary reference coordinate for the translational force analysis (Section 4.2.2). The figures here extend the field presentation to all six sampled coordinates: 00, A3, A6, D3, G3, and G6 (Figure 4.3). For each coordinate, three quantities are shown as a function of axial position: magnetic flux density B, spatial gradient dB/dz, and the force-relevant product B times grad B. A summary figure comparing all coordinates is included at the end. These results also support the cross-coordinate force ratio predictions reported in Figure 4.9.

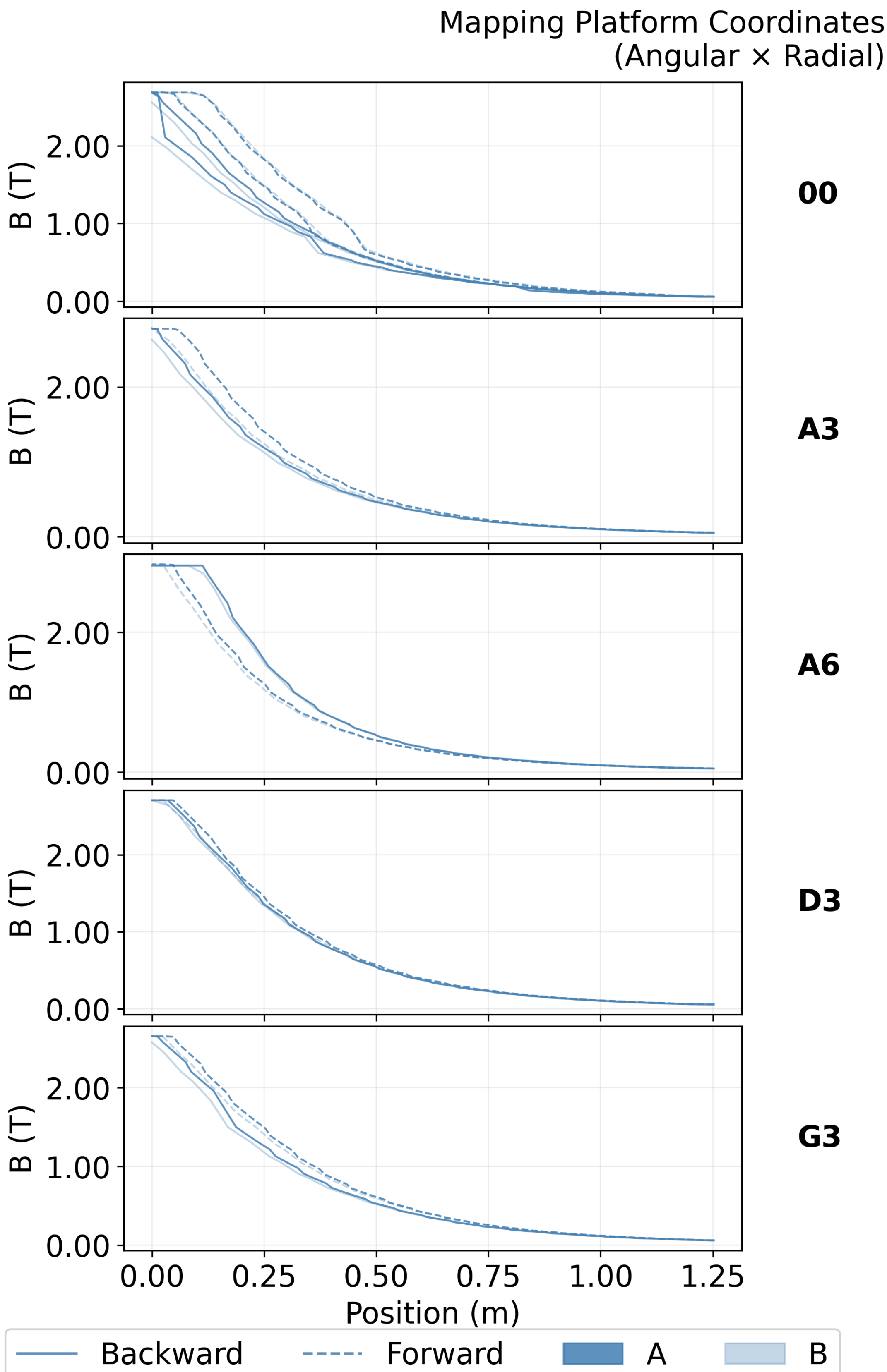


Figure D.1: Magnetic field along the measurement path.

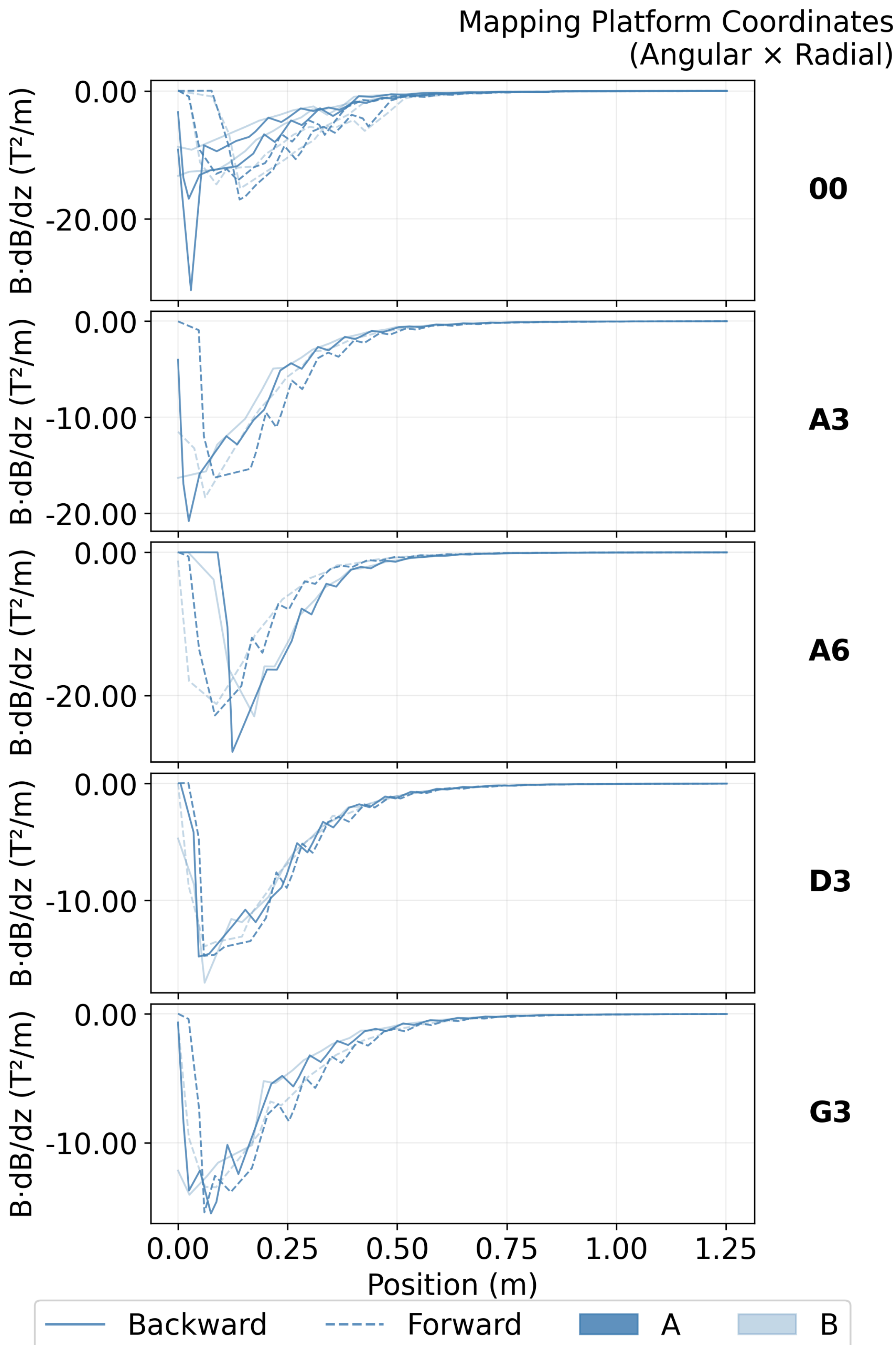


Figure D.2: Magnetic field spatial gradient along the measurement path

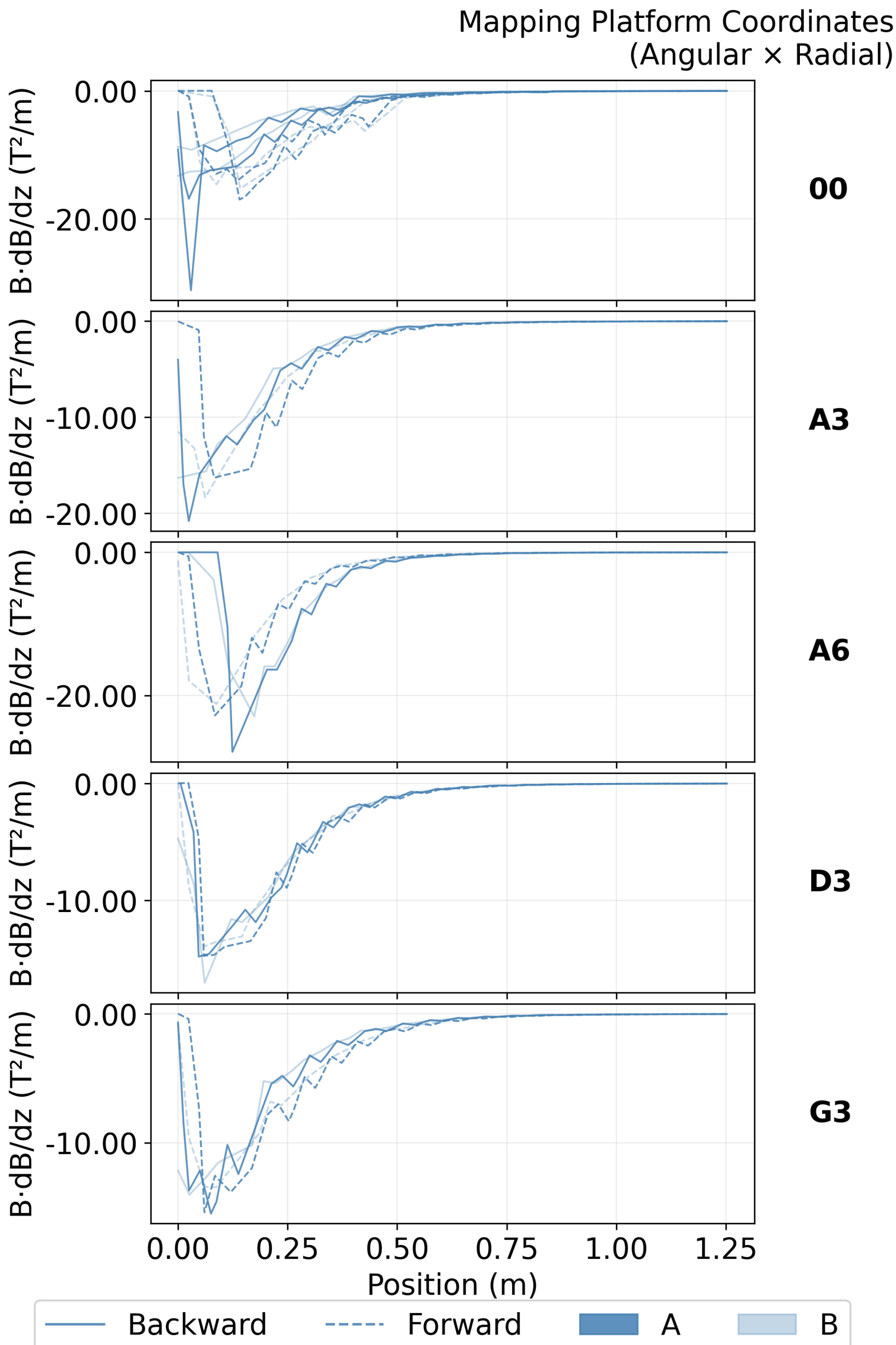


Figure D.3: Magnetic field and gradient product along the measurement path

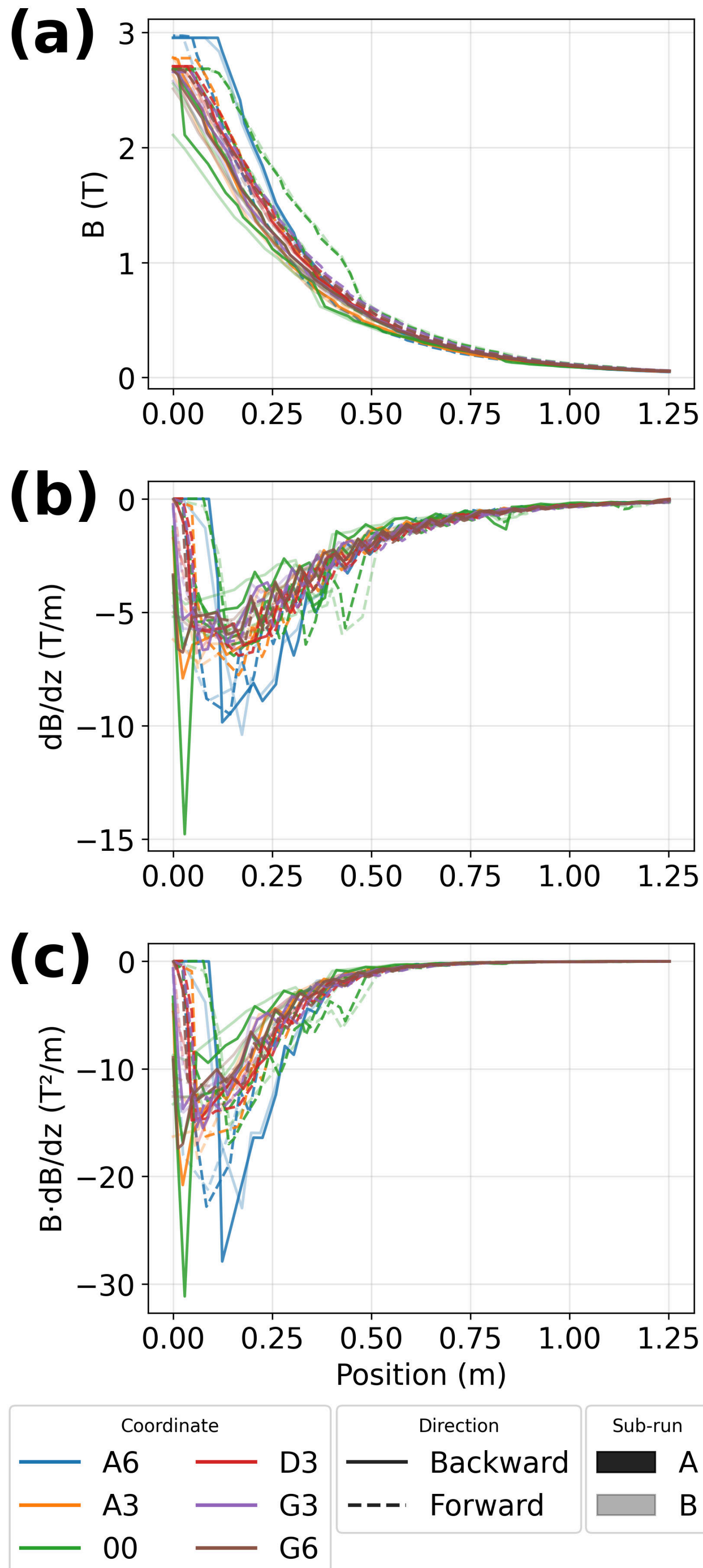


Figure D.4: Magnetic field with corresponding gradient and product along the measurement path. Color represents a coordinate along the fixture A6 is the top, 00 the center and G6 the bottom. Each has two directions, represented with solid/dashed lines, while each direction was split into two sub-runs represented by shade.

Appendix E

Torque Sensor Calibration - Additional Results

This appendix presents supplementary results from the torque sensor calibration process described in Section 3.3.1.3. The aluminum-based sensor was selected for all reported measurements based on its consistency and sensitivity relative to the other candidate materials. The figures here document the calibration trials for the glass and clear plastic sensors, which showed insufficient linearity and voltage range for reliable use. Together these results justify the sensor selection and positioning choices reported in Chapter 3 and use for results presented in Chapter 4.

The sensor calibration consisted in incrementally adding weight to our sensor as described in section 3.3.1.3. The strain sensors studied included clear plastic, aluminum, and glass. Each sensor was tested for linearity voltage vs. weight up to 800 mg. Figure E.1 shows calibration plots for two of the trials done. (a) The glass sensor showed some linearity, but small mean voltage range of operation even at max weight. The slope found for each fit is too different and non reproducible. (b) The clear plastic sensor showed stiffer, showing better slope trend consistency but still being too uncertain.

These results demonstrated that these sensors were unreliable, and characterized by unpredictable changes in voltage when weight was applied. Additionally, both sensor had a sub-millivolt range, ineffective for measurements. These two rods were the thickest of all available rods in Table 3.2. The observed behavior is expected as explained by Equation 3.13.

In contrast the aluminum rods, 2 and 3, showed a clear linear trend along all trials and measurements (Figure E.2). This plot features three different sensor rotation configurations (Fig E.2b). Either 0° or 180° gave maximum deformation of the strain gauges, while 90° was ineffective.

From these results, we used rod 3 oriented in either 0 or 180 degrees, since it showed remarkable consistency across calibration trials. Focusing on the weight range(0 to 200 mg), we found similar slopes in repeated calibrations over a period of one week.

Figure E.3 shows two lines, along the lever arm 90 and 50 mm (r) away from the rotation axis position. On the x-axis position along the sensor length measurements demonstrated greater sensitivity for greater distances from the strain gauges. And we can see that the 50 mm line deviates from the 90 mm line showing that the closest to the rotation axis, the higher the signal reading for the same force.

Figure E.4 demonstrates the expected inverse relation of force and position along the transfer arm from the rotation axis ($1/r$) in two ways. First Figure E.4a shows a log-log plot voltage vs position with a fit n value very close to -1. Additionally, a A/r fit (Figure E.4b) provides the constant A, also very close to -1. This signifies, it is a good method to measure torque via the counter force.

Once the sensor and circuit were calibrated and the best position for measurement in the torque fixture are affixed, we moved on to the in laboratory measurements of torque in Section 3.3.1.3.

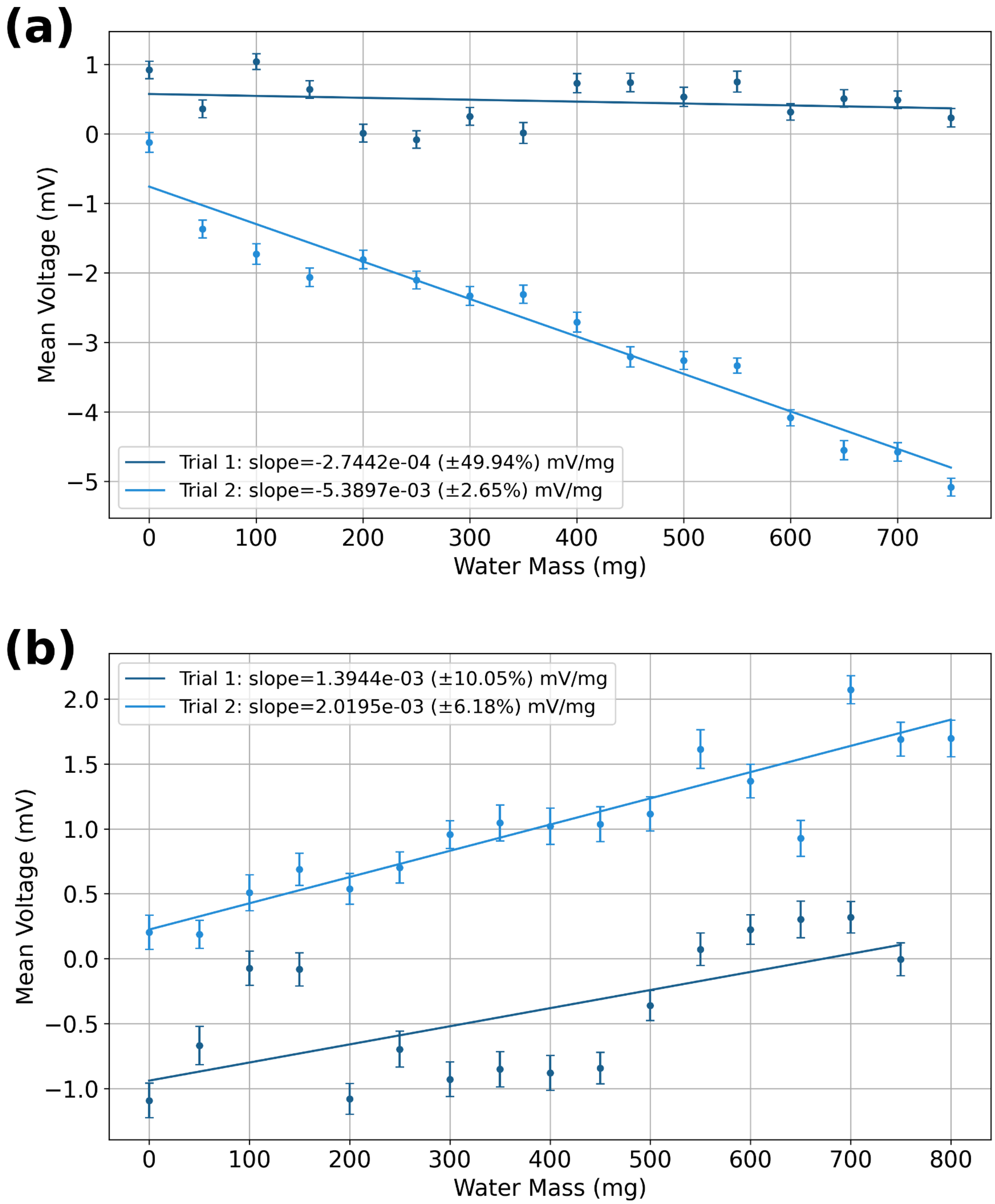


Figure E.1: Trial examples for Rods 5 and 7. Both of them showed inconsistency across trials showing minimal to no change in voltage with weight added. (a) Voltage readings for Rod5 (Glass) with up to 750 mg of water in increments. (b) Voltage readings for Rod7 (Clear Plastic) with up to 800 mg of water in increments.

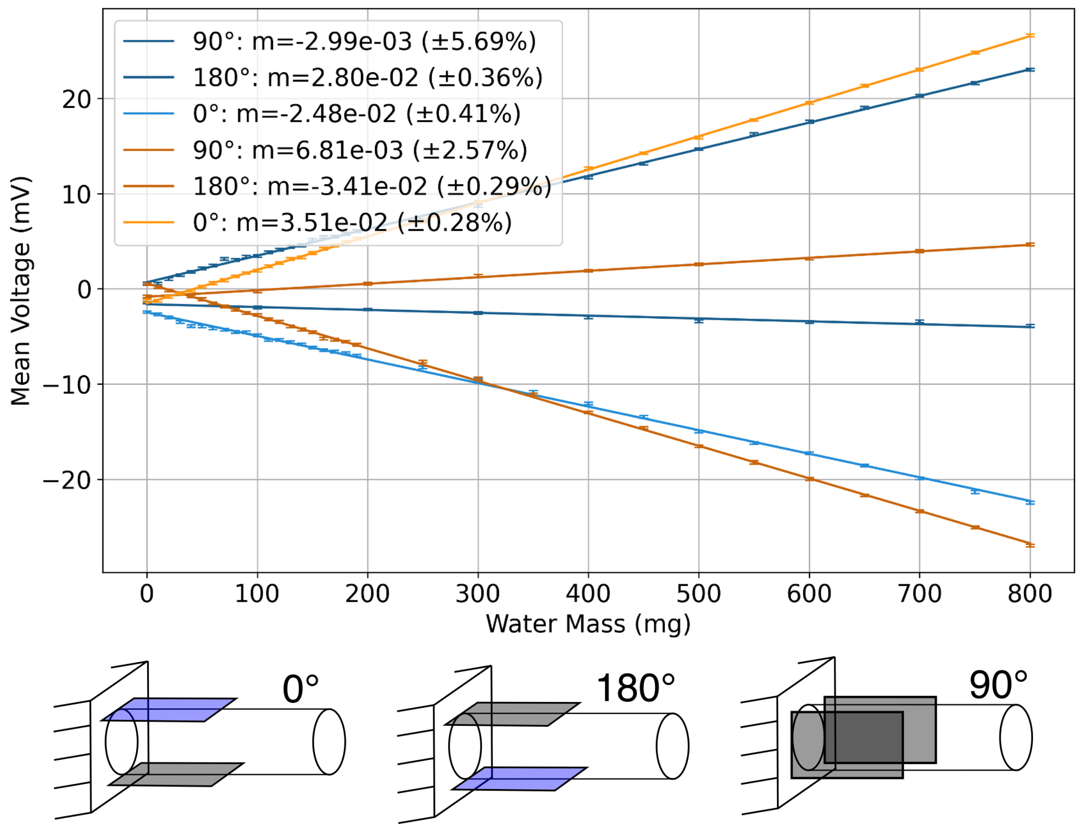


Figure E.2: Linear fits show consistent response across orientations, confirming the reliability of the lab-built sensor. (a)Calibration curves for two aluminum (AL) rods in three different orientations. (b)Three configurations tested, 0° and 180° are equivalent but flip voltage sign with respect of each other. 90° gives less sensibility as strain gauges does not bend in the proper axis.

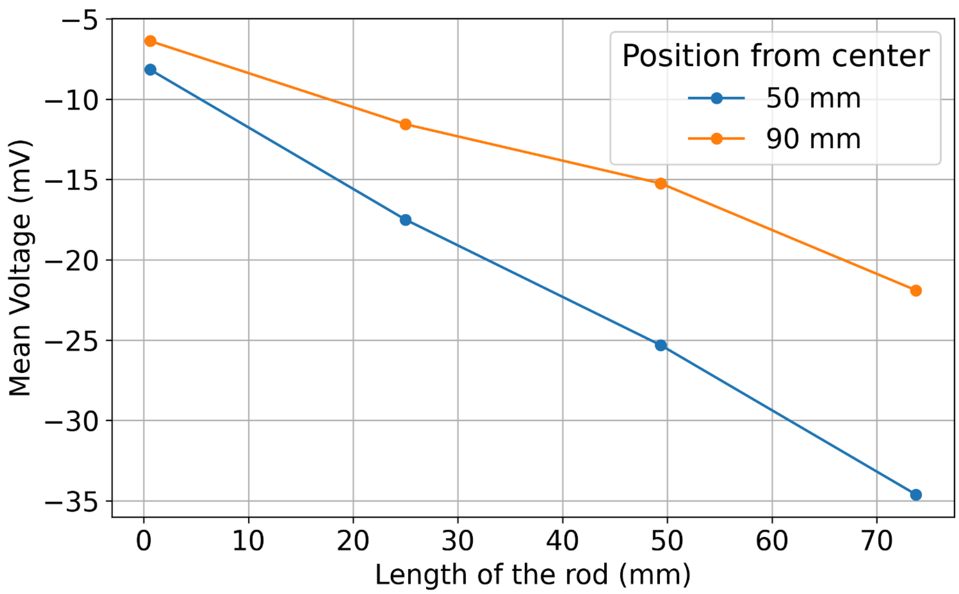


Figure E.3: Mean voltage response of the sensor as a function of length (L) for different distances (r) from the center (50–90 mm). At larger lengths (farther from the strain gauges) an increase in signal reading. Further from the rotation axis the measurements diverge, showing higher sensitivity near the center (50 mm).

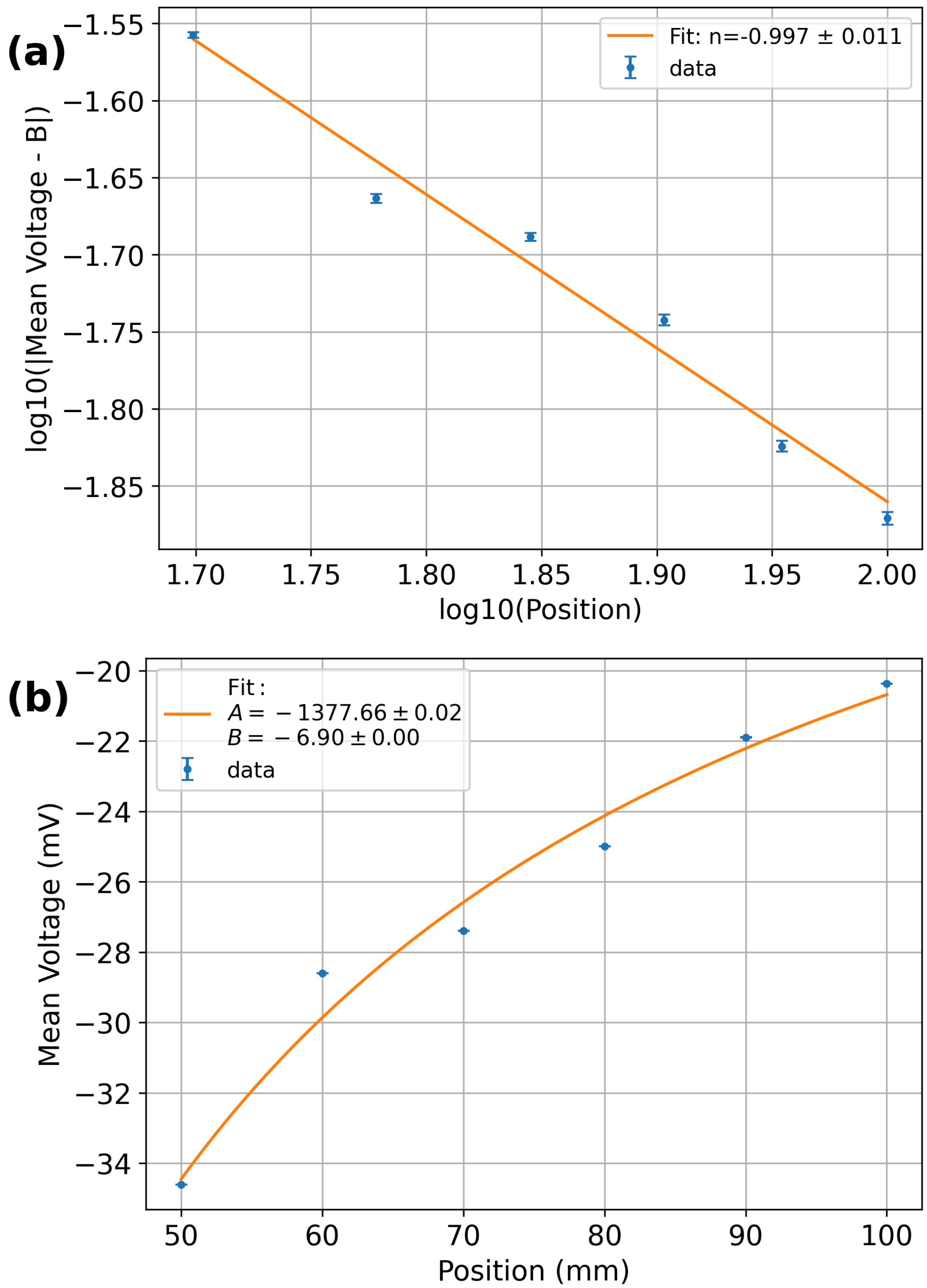


Figure E.4: Demonstration of the expected inverse-distance dependence of the force (and thus torque) with respect to the distance r from the rotation axis. Both analyses confirm the $1/r$ dependence characteristic of the torque measurement via counter-force method. (a) Log-log plot of effective measured voltage (proportional to force) versus position r, showing a power-law fit with exponent $n = -0.997 \pm 0.011$, very close to the theoretically expected value of -1. (b) Direct fit of the form $F(r) = A/r + B$, yielding $A = -1.378 \pm 0.015$ (very close to -1 in normalized units) and a negligible offset $B = -0.0069 \pm 0.0002$.

Appendix F

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Glossary

AIMD Active Implantable Medical Device. 8, 9, 10, 11, 15

CIED Cardiac Implantable Electronic Device. 1, 2, 3, 4, 5, 24, 29, 38, 103

CMS Centers for Medicare & Medicaid Services. 29, 38

CRT Cardiac Resynchronization Therapy. 24

CSV Comma Separated Values. 47

CUMC–Bergan Mercy CHI Health Creighton University Medical Center–Bergan Mercy. 21, 40, 41, 75, 81, 97, 99, 125, 136

DAQ Data Acquisition. 65, 71, 72, 73, 75

ECG Electrocardigram. 2

ESC European Society of Cardiology. 29, 38

FEG Frequency Encoding Gradient. 19

FR Force Ratio. 34, 84, 86, 87, 89, 90, 91

HRS Heart Rhythm Society. 1, 29, 38

ICD Implantable Cardioverter-Defibrillator. 9, 24

KCL Kirchhoff's Current Law. 113

MRI Magnetic Resonance Imaging. 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 20, 21, 22, 23, 24, 28, 29, 30, 31, 32, 33, 38, 39, 40, 41, 42, 43, 50, 52, 53, 54, 55, 57, 58, 59, 73, 74, 75, 77, 78, 81, 86, 96, 97, 98, 99, 100, 102, 103, 104

MUT Material Under Test. 87, 88, 89, 90, 92, 98, 99, 120

PEG Phase Encoding Gradient. 19

PLA Polylactic Acid. 53, 54

PNS Peripheral Nerve Stimulation. 31

PVC Polyvinyl Chloride. 12, 41, 42, 50, 51, 60, 69

RF Radiofrequency. 1, 2, 3, 10, 11, 16, 17, 18, 19, 28, 30, 31, 32, 100, 102

SAR Specific Absorption Rate. 32

SSG Slice Select Gradient. 18

VI LabVIEW Virtual Instrument. 42, 43

References

[1] Jerrold T. Bushberg, J. Anthony Seibert, Edwin M. Leidholdt, and John M. Boone. *The Essential Physics of Medical Imaging*. Lippincott Williams & Wilkins, 3rd edition, 2011.

[2] Haris M. Haggani, Laurence M. Epstein, and Joshua M. Cooper. Engineering and construction of pacemaker and icd leads. In Kenneth A. Ellenbogen, Bruce L. Wilkoff, G. Neal Kay, and Chu-Pak Lau, editors, *Clinical Cardiac Pacing, Defibrillation, and Resynchronization Therapy*, chapter 4, pages 127–143. Saunders / Elsevier, Philadelphia, PA, 2011.

[3] Fathima S Mubarak, Yevinka Ellepola, Kadijah Nyosha Chamba, Sanjay Agrawal, and Maged Makhoul. Complications of epicardial pacing wire removal following adult cardiac surgery: A systematic review. *Cureus*, 15(11):e49076, Nov 2023.

[4] Pennsylvania Patient Safety Authority. Pennsylvania patient safety advisory—volume 3, issue 2 (june 2006): "mitigating look-alike/sound-alike medication errors". https://patientsafety.pa.gov/ADVISORIES/Pages/200603_08.aspx, June 2006. Accessed: 2025-07-09.

[5] M. C. Reade. Temporary epicardial pacing after cardiac surgery: a practical review. *Anaesthesia*, 62(4):364–373, 2007.

[6] Heart Rhythm Society. Heart rhythm 2024 spotlights new developments in cardiac implantable electronic devices and other treatments. https://www.hrsonline.org/news/hr24-spotlights-new-developments-cieds/, May 2024. Accessed: 2025-07-07.

[7] Astm f2052-21 standard test method for measurement of magnetically induced displacement force on medical devices in the magnetic resonance environment, 2021. ASTM International.

[8] Astm f2213-17 standard test method for measurement of magnetically induced torque on medical devices in the magnetic resonance environment, 2022. ASTM International.

[9] Astm f2182-19 standard test method for measurement of radio frequency induced heating on or near passive implants during magnetic resonance, 2019. ASTM International.

[10] ASTM International. Standard practice for marking medical devices and other items for safety in the magnetic resonance environment. ASTM F2503-23e01. https://www.astm.org/f2503-23e01.html, 2023. DOI: 10.1520/F2503-23E01.

[11] G G Hartnell, L Spence, L A Hughes, M C Cohen, R Saouaf, and B Buff. Safety of mr imaging in patients who have retained metallic materials after cardiac surgery. *American Journal of Roentgenology*, 168(5):1157–1159, 1997. PMID: 9129404.

[12] Frank G. Shellock. Ecg (ekg) electrodes. Safety Topic/Article, MRIsafety.com, n.d. Accessed: 2026-01-25. Shellock R & D Services, Inc.

[13] Frank G. Shellock. Monitoring patients in the mr environment. Safety Topic/Article, MRIsafety.com, n.d. Accessed: 2026-01-25. Copyright 2004– by Shellock R & D Services, Inc. and Frank G. Shellock, Ph.D.

[14] P. G. Poh, C. Liew, C. Yeo, L. R. Chong, A. Tan, and A. Poh. Cardiovascular implantable electronic devices: a review of the dangers and difficulties in mr scanning and attempts to improve safety. *Insights into Imaging*, 8(4):405–418, 2017.

[15] Rahul G. Muthalaly, Nitesh Nerlekar, Yin Ge, Raymond Y. Kwong, and Arthur Nasis. Mri in patients with cardiac implantable electronic devices. *Radiology*, 289(2):281–292, 2018.

[16] American College of Cardiology. Current State of MRI With Cardiac Devices. https://www.acc.org/Latest-in-Cardiology/Articles/2024/05/21/10/30/Current-State-of-MRI-With-Cardiac-Devices, May 2024. Expert Analysis, published May 21, 2024.

[17] John F. Schenck. Safety of strong, static magnetic fields. *Journal of Magnetic Resonance Imaging*, 12(1):2–19, jul 2000.

[18] Lawrence P. Panych and Bruno Madore. The physics of mri safety. *Journal of Magnetic Resonance Imaging*, 47(1):28–43, 2018.

[19] Claudia Meier, Carsten Israel, Michel Eisenblätter, Annika Hoyer, Ferdinand Valentin Stoye, Ali Yilmaz, and Stephan Gielen. Safety of magnetic resonance imaging in patients with cardiac implantable electronic devices and abandoned or epicardial leads: A systematic review and meta-analysis. *Europace*, 26(6):euae165, 2024.

[20] Frank G. Shellock. Mri considerations for patients with temporary epicardial pacing leads, temporary intracardiac pacing leads, permanent intracardiac pacing leads, and permanent epicardial pacing leads. Safety Topic [286], MRIsafety.com, 2021. Accessed: 2026-01-25.

[21] AM Vuorinen, R Paakkanen, J Karvonen, J Sinisalo, M Holmstrom, S Kivisto, J Peltonen, and T Kaasalainen. Mri safety with abandoned or functional epicardial pacing leads. *European Heart Journal - Cardiovascular Imaging*, 22(Supplement 2), 2021.

[22] D. Grainger. Safety guidelines for magnetic resonance imaging equipment in clinical use. Technical report, Medicines and Healthcare Products Regulatory Agency (MHRA), London, UK, 2014. MHRA Safety Guidelines.

[23] Jana G. Delnoa, Daniel M. Krainak, Stephanie A. Flesher, and Donald L. Miller. Mri-related fda adverse event reports: A 10-yr review. *Medical Physics*, 46(10):e1107–e1116, 2019. U.S. Government work, public domain in the USA.

[24] Dan Stoianovici, Karun Sharma, and Kevin Cleary. Measurement of magnetically induced torque on lightweight medical devices in the magnetic resonance environment for astm f2213. *IEEE Transactions on Biomedical Engineering*, 71(1):36–44, 2024.

[25] Jennifer R. Sawyer, Lance M. Relland, Molly S. Hagele, and Joseph D. Tobias. Ingested magnets found inadvertently during elective magnetic resonance imaging. *Cureus*, 16(7):e65321, 2024.

[26] Madeline Holcombe. Keith McAllister dies after being pulled into an MRI machine by a metal chain used for weight training. CNN Health, July 2025. Accessed: March 31, 2026.

[27] Mojtaba Ghadimi and Alex Thomas. *Magnetic Resonance Imaging Contraindications*, page Online. StatPearls Publishing, Treasure Island (FL), 2025. [Updated 2025 May 4].

[28] U.S. Food & Drug Administration. Testing and labeling medical devices for safety in the magnetic resonance (mr) environment: Guidance for industry and

food and drug administration staff. Guidance Document, October 2023. Docket No. FDA-2019-D-2837; "Content current as of 10/10/2023".

[29] American College of Radiology. *ACR Manual on MR Safety*, 2024. Accessed: 2025-10-28.

[30] Robert D. Schaller, Tamara Brunker, Michael P. Riley, Francis E. Marchlinski, Saman Nazarian, and Harold Litt. Magnetic resonance imaging in patients with cardiac implantable electronic devices with abandoned leads. *JAMA Cardiology*, 6(5):549–556, 2021.

[31] A. Ferreira. Predicting angular displacement of medical devices in a mri scanner bore using comsol multiphysics. Master's thesis, Drexel University, 2017.

[32] Terry Woods, Jana Delfino, and Sunder Rajan. Assessment of magnetically induced displacement force and torque on metal alloys used in medical devices. *Journal of Testing and Evaluation*, 49:20190096, 03 2021.

[33] Hyo-Chang Seo, Yongkwan Lee, and Segyeong Joo. A simple apparatus for safety assessment of magnetically induced torque on active implantable medical devices (aimds) under 1.5 t and 3.0 t mri. *Magnetic Resonance Materials in Physics, Biology and Medicine*, 34(5):767 – 774, 2021. Cited by: 3.

[34] O. Aboyewa. *Measurement of Radiofrequency Induced Heating in Patients with Retained Post-Surgical Epicardial Leads During MRI.* PhD thesis, Creighton University, 2021.

[35] M.G. Nichols. Safety assessment of translational forces and torques on post-surgical epicardial leads during mri, 2021. Creighton MRI Safety Proposal (unpublished internal document).

[36] David J. Griffiths. *Introduction to Electrodynamics*. Pearson, 4th edition, 2013.

[37] Jerry L. Prince and Jonathan Links. *Medical Imaging Signals and Systems*. Pearson, 2nd edition, 2015.

[38] William Lowrie. *Fundamentals of Geophysics*. Cambridge University Press, Cambridge, 1997. Illustrated reprint edition; discusses Earth's magnetic field and related geophysics topics.

[39] Frank G. Shellock. The most dangerous place in the mri suite. https://mriquestions.com/most-dangerous-place.html, n.d. MRIQuestions.com. Educational MRI safety resource. Accessed March 13, 2026.

[40] Frank G. Shellock. Gradient safety overview. https://mriquestions.com/gradient-safety-overview.html, n.d. MRIQuestions.com. Educational MRI safety resource. Accessed April 4, 2026.

[41] Frank G. Shellock. Rf safety overview. https://mriquestions.com/rf-safety-overview.html, n.d. MRIQuestions.com. Educational MRI safety resource. Accessed April 4, 2026.

[42] Aren Shah and Shima Aran. A review of magnetic resonance (mr) safety: The essentials to patient safety. *Cureus*, 15(10):e47345, Oct 2023. Review article. Open access via PMC.

[43] R. Jason Stafford. The physics of magnetic resonance imaging safety. *Magnetic Resonance Imaging Clinics of North America*, 28(4):517–536, 11 2020. Available at: https://mriquestions.com/uploads/3/4/5/7/34572113/stafford_safety_1-s2.0-s1064968920300519.pdf.

[44] Pascal Defaye, Mauro Biffi, Mikhael El-Chami, Serge Boveda, Michael Glikson, Jonathan Piccini, and Marco Vitolo. Cardiac pacing and lead devices management: 25 years of research at ep europace journal. *Europace*, 25(8):euad202, 2023.

[45] Andrea M. Russo, Milind Y. Desai, Monika M. Do, Javed Butler, Mina K. Chung, Andrew E. Epstein, Maya E. Guglin, Wayne C. Levy, Jonathan P. Piccini, Nicole M. Bhave, Andrea M. Russo, Milind Y. Desai, Monika M. Do, Amrut V. Ambardekar, Nancy C. Berg, Kenneth C. Bilchick, G. William Dec, Rakesh Gopinathannair, Janet K. Han, Liviu Klein, Rachel J. Lampert, Gurusher S. Panjrath, Ryan R. Reeves, Danita Marie Yoerger Sanborn, Lynne W. Stevenson, Quynh A. Truong, Paul D. Varosy, Todd C. Villines, Annabelle S. Volgman, and Karolina M. Zareba. Acc/aha/ase/hfsa/hrs/scai/scct/scmr 2025 appropriate use criteria for implantable cardioverter-defibrillators, cardiac resynchronization therapy, and pacing. *JACC*, 85(11):1213–1285, 2025.

[46] Hospital for Sick Children / AboutKidsHealth. Pacemaker: Caring for your child at home, 2009. Last updated December 17, 2009.

[47] Hiroomi Murayama, Masanobu Maeda, Hajime Sakurai, Akihiko Usui, and Yuichi Ueda. Predictors affecting durability of epicardial pacemaker leads in pediatric patients. *The Journal of Thoracic and Cardiovascular Surgery*, 135(2):361–366, 2025/09/30 2008.

[48] HowToPace Editors. Basics of pacing leads. https://www.howtopace.com/basics-of-pacing-leads/, 2023. HowToPace. Accessed October 12, 2025.

[49] Frank G. Shellock. Epicardial pacemakers and leads. https://mriquestions.com/epicardial-pacersleads.html, n.d. MRIQuestions.com. Accessed April 8, 2026.

[50] William D. Callister and David G. Rethwisch. *Materials Science and Engineering: An Introduction*. Wiley, 10th edition, 2018.

[51] Carpenter Technology Corporation. Mp35n alloy datasheet. https://www.cartech.com/alloys/mp35n.

[52] Mathias Davids, Bastien Guérin, Matthias Malzacher, Lothar R. Schad, and Lawrence L. Wald. Predicting magnetostimulation thresholds in the peripheral nervous system using realistic body models. *Scientific Reports*, 7(1):5316, 2017.

[53] Deborah A. Langman, Ira B. Goldberg, J. Paul Finn, and Daniel B. Ennis. Pacemaker lead tip heating in abandoned and pacemaker-attached leads at 1.5 tesla MRI. *Journal of Magnetic Resonance Imaging*, 33(2):426–431, February 2011.

[54] AlphaLab, Inc. Gaussmeter model GM2. https://www.alphalabinc.com/products/gm2/, 2023. Laboratory gaussmeter for measuring AC/DC magnetic fields. Accessed December 26, 2025.

[55] AlphaLab Inc. Alphaapp. https://www.alphalabinc.com/alphaapp/, 2023. Software for data logging, streaming, and plotting; lacks adjustable sampling frequency control.

[56] AlphaLab Inc. Alphalab data acquisition communication protocol. https://www.alphalabinc.com/wp-content/uploads/2018/02/alphaapp_comm_protocol.pdf, 2018. Used to implement custom sampling control via LabVIEW VI.

[57] Thorlabs, Inc. Mt1-z8 – 12 mm (0.47") one-axis motorized translation stage, 1/4"-20 taps. Product page and manual, 2009–2025. Thorlabs motorized translation stage used for precise positioning in the pendulum fixture setup. Travel: 12 mm (approx. 0.5"). Accessed February 2026.

[58] Thorlabs, Inc. Kinesis motion control software. Software package and API reference (Thorlabs.MotionControl.DotNet_API.chm), ongoing. Used for controlling motorized stages in the experimental setup. Includes .NET API, GUI panels,

and programming interfaces for custom applications. Installed version accessed February 2026.

[59] TEMCo Industrial. Surgical grade 1 titanium wire (non-resistance). https://temcoindustrial.com/wire-electrical/wire-cable/non-resistance-wire, 2025. Accessed: February 2026. Commercially pure titanium wire, spooled in USA; used for straight wire positive control (measured diameter 1.02 mm).

[60] Daniel Juarez Luna. Magnetically induced forces on epicardial leads: Analysis code and data. https://github.com/Zslw/mri-translational-force-n-torque-analysis, 2025. Accessed: April 23, 2026.

[61] R.C. Hibbeler. *Mechanics of Materials*. Pearson Education, 9th edition, 2016.

[62] Johannes Chiu. Characterization of a low-cost millinewton force sensor for ionic polymer metal composite actuators. https://api.semanticscholar.org/CorpusID:139053109, 2016.

[63] Omega Engineering. *Precision Strain Gage: precision LINEAR PATTERN for static and dynamic applications*, n.d. SGD_LINEAR1-AXIS series strain gages datasheet.

[64] OMEGA Engineering. Strain gage installation guide. https://studylib.net/doc/18162390/strain-gage-installation. Accessed: 2026-04-14, Gauge factor reported as nominal 2.0 for constantan foil strain gages (see page 3).

[65] Jacob Fraden. *Handbook of Modern Sensors: Physics, Designs, and Applications*. Springer, Cham, Switzerland, 5th edition, 2016.

[66] Micro-Measurements. What is a strain gage? https://www.micro-measurements.com/what-is-a-strain-gage, 2023. Accessed: 2025-11-06.

[67] OMEGA Engineering. Strain gage measurement—a technical reference. https://www.omega.co.uk/techref/pdf/straingage_measurement.pdf, 2023. Accessed: 2025-11-06.

[68] National Instruments Corporation. Strain gauge measurement — a tutorial (application note 078). Technical Report 341023C-01, National Instruments Corporation, August 1998. Application Note AN078; available online.

[69] MatWeb, LLC. AISI 1018 Steel, Cold Drawn. https://www.matweb.com/search/datasheet.aspx?MatGUID=3a9cc570fbb24d119f08db22a53e2421, 2025. Accessed: April 2025.

[70] McMaster-Carr. Corrosion-Resistant 200 Nickel Rod, 1/2" Diameter, Part No. 9136K23. https://www.mcmaster.com/9136K23/, 2025. Material: 200 Nickel, 99.5% Ni max, density 0.32 lb/in^3 (8,857 kg/m^3). Accessed: April 2025.

[71] Dominic L. C. Guebelin, Matthias P. Hilty, Michael J. Thali, and Stephan A. Bolliger. Left ventricular perforation resistance during pleural drainage puncture. *International Journal of Forensic Science & Research*, 2(1), 2025.

[72] Medtronic. Indications, safety, and warnings: MRI SureScan pacing leads. https://www.medtronic.com/en-us/l/isw/surescan-mri-pacing-leads-indications-safety-warnings.html, 2022. Document MAJ_69581. Accessed April 25, 2026.